Supporting information

Hydrogen Bond Enhanced Halogen Bonding Organocatalyst with C(sp^3)-Br and Sulfoxide Moieties

Yuheng Zhang, Haimeng Zhu, Boyuan Zhang, Huijuan Yang, Choon-Hong Tan, Chao Wang,* Jin Wen,* and Lili Zong*

Fujian Provincial Key Laboratory of Innovative Drug Target Research, School of Pharmaceutical Sciences, Xiamen University, Xiamen 361102, China;
State Key Laboratory for Modification of Chemical Fibers and Polymer Materials, College of Materials Science and Engineering, Donghua University, Shanghai 201620, China;
Division of Chemistry and Biological Chemistry, School of Physical and Mathematical Science, Nanyang Technological University, 21 Nanyang link, Singapore 637371, Singapore;
Institute of Advanced Synthesis, School of Chemistry and Molecular Engineering, Nanjing Tech University, Nanjing 211816, China.

Corresponding Author Email: Lili.Zong@xmu.edu.cn; iasc.wang@njtech.edu.cn; jinwen@dhu.edu.cn

Contents

1. **General information** S3

2. **Synthesis of catalysts** S3

3. **Screening of reaction conditions**..... S16

4. **Synthesis of substrates** S17

5. **Reduction of quinolines and imines** S21

6. **NMR studies** S28

7. **Reaction monitoring by ^{1}H NMR**..... S45

8. **X-ray analysis**..... S47

9. **Computational details**..... S51

10. **References**..... S60

11. **HPLC trace**..... S62

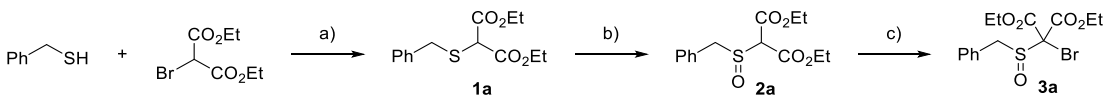
12. **NMR spectra** S64

1. General information

The NMR spectra were recorded on Bruker AV600 MHz and AV500 MHz spectrometer. Chemical shifts are recorded as δ in units of parts per million (ppm). The residual solvent peak was used as an internal reference. High resolution mass spectra (HRMS) were obtained on Q-Exactive using electrospray ionization (ESI). HRMS were reported in units of mass of charge ratio (*m*/*z*). X-ray crystallography analysis was performed on XtaLAB Synergy-R. Enantiomeric excess values were determined by HPLC analysis on Shimadzu LC-20AT HPLC workstations. Analytical thin-layer chromatography (TLC) was performed on 60 GF254 silica gel plates (Leyan). Flash chromatography separations were performed on 300-400 mesh silica gel (Leyan). Visualization was performed using a UV lamp or potassium permanganate stain. IR spectra were recorded with neat solid/liquid samples on a Bruker Vertex 70V spectrometer and reported in wave numbers (cm^{-1})

The mercaptans and quinoline derivatives were purchased from commercial suppliers or synthesized according to the literature procedures. Other reagents and solvents are commercial grade and they were used as supplied without further purification, unless otherwise stated. Anhydrous solvents such as dichloromethane (DCM), tetrahydrofuran (THF) was purchased from J&K Scientific. The deuterated chloroform used in NMR titrations was treated with 5 Å molecular sieves. Experiments involving moisture and/or air sensitive components were performed under a positive pressure of nitrogen in oven-dried glassware equipped with a rubber septum inlet. Catalysts **3f**[1] and **3g**[2] was prepared according to the reported procedures.

2. Synthesis of catalysts

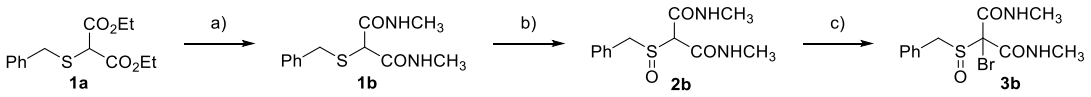


Scheme S1. Synthesis of 3a. Conditions: a) NaH (1.1 equiv.), THF, 0 ℃ ~ rt, 4 h, 72% yield; b) Na_2MoO_4 (0.05 equiv.), 35 wt% H_2O_2 (1.05 equiv.), CF_3CH_2OH, rt, 1 h, 58% yield; c) **NBS** (1.05 equiv.), $TsOH•H_2O$ (1.5 equiv.), CH_3CN, reflux, 2 h, 55% yield.

Step 1: To a solution of phenylmethanethiol (701 μL, 1.0 equiv., 6 mmol) in 20 mL of THF at 0℃ was added sodium hydride (264 mg, 1.1 equiv., 6.6 mmol). 10 min later, diethyl bromomalonate (1.13 mL, 1.1 equiv., 6.6 mmol) was added and the reaction mixture was stirred at room temperature for 4 hours. Then the reaction was quenched by adding water and extracted with DCM (15 mL × 3). The combined organic layer was washed with brine and dried over Na_2SO_4, filtered and concentrated. Sulfide **1a** was obtained by silica gel flash column chromatography (petroleum ether : ethyl acetate = 20:1), as colorless oil, 1.211 g, 72% yield.

Step 2: To a solution of **1a** (1.21 g, 1.0 equiv., 4.29 mmol) in 7 mL of CF_3CH_2OH at rt was added Na_2MoO_4 (43.3 mg, 0.05 equiv., 0.21 mmol), and 35wt% H_2O_2 (385 μL, 1.05 equiv., 4.5 mmol). After stirring for 1 hour, the reaction was quenched with saturated sodium sulfite solution and extracted with DCM (15 mL × 3). The combined organic layer was washed with brine and dried over Na_2SO_4, filtered and concentrated. Sulfoxide **2a** was obtained by silica gel flash column chromatography (petroleum ether : ethyl acetate = 5:1), as colorless oil, 734 mg, 58% yield.

Step 3: To a solution of **2a** (298.4 mg, 1.0 equiv., 1 mmol) and TsOH•H_2O (285.3 mg, 1.5 equiv., 1.5 mmol) in 50 mL of CH_3CN at rt was added *N*-bromosuccinimide (**NBS**) (187 mg, 1.05 equiv., 1.05 mmol). The reaction mixture was stirred under reflux for 2 hours and then cooled to rt. After concentrated under vacuum, the residue was purified by silica gel flash column chromatography (petroleum ether : ethyl acetate = 5:1) to afford **3a** (light yellow solid, 207 mg, 55% yield).



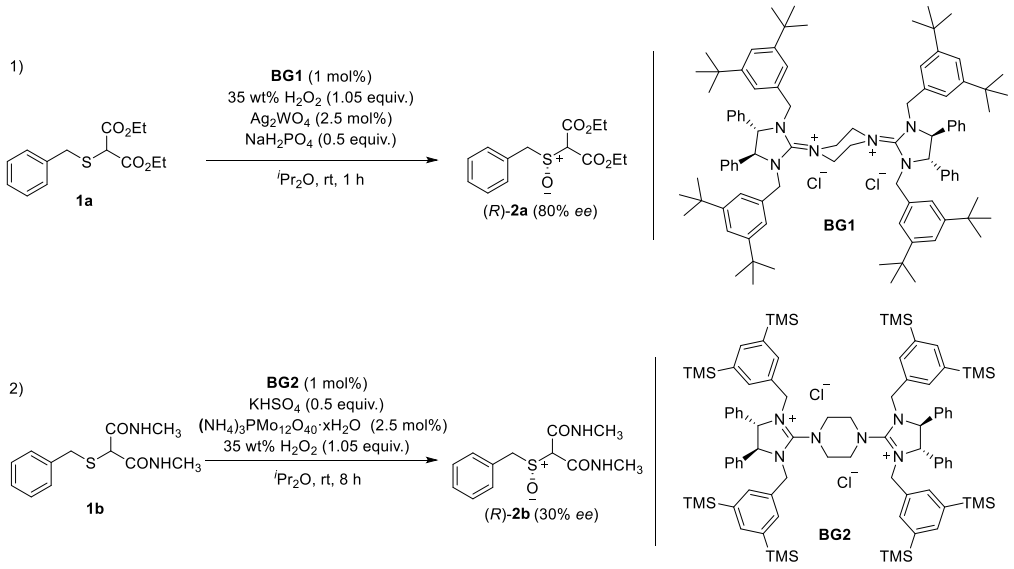
Scheme S2. Synthesis of 3b. Conditions: (a) 27 wt% CH_3NH_2 in MeOH, 60℃, 5 h, 54% yield; (b) Na_2MoO_4 (0.05 equiv.), 35 wt% H_2O_2 (1.05 equiv.), CF_3CH_2OH, rt, 8.5 h, 90% yield; (c) Br_2 (1.0 equiv.), Et_3N (1.0 equiv.), DCM, rt, 3 h, 74% yield.

Step 1: A 25 mL RBF was charged with sulfide **1a** (846 mg, 1.0 equiv., 3.0 mmol) and a solution of 27 wt% CH_3NH_2 in methanol (7 mL). The reaction mixture was stirred at

60℃ for 5 hours. Then solvent was removed under vacuum. The residue was purified by silica gel flash column chromatography (DCM : MeOH = 50: 1) to afford sulfide **1b** (white solid, 407 mg, 54% yield).

Step 2: By following the same procedure to **2a**, **2b** (white solid, 90% yield) was obtained by oxidation of sulfide **1b.**

Step 3: To a solution of **2b** (207 mg, 1.0 equiv., 1 mmol) and Et_3N (108 μL, 1.0 equiv., 1 mmol) in 10 mL of DCM was added Br_2 (40 μL mL, 1.0 equiv., 1 mmol) slowly at 0℃. Then the reaction mixture was stirred at rt for 3 hours. When the reaction was completed, solvent was removed under vacuum. the residue was purified by silica gel flash column chromatography (DCM: MeOH = 50: 1) to give **3b** (white solid, 198 mg, 74% yield).



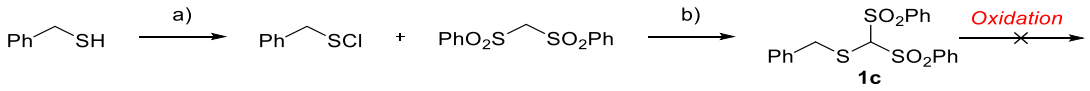
Scheme S3. Attempts to enantioenriched 2a and 2b by asymmetric sulfoxidation

To a solution of **1a** (70.5 mg, 1 equiv., 0.25 mmol), **BG1**[3] (3.6 mg, 1 mol%, 0.0025 mmol), Ag_2WO_4 (2.9 mg, 2.5 mmol%, 0.00625 mmol) and NaH_2PO_4 (15 mg, 0.5 equiv., 0.125 mmol) in 5 mL of isopropyl ether was added 35 wt% H_2O_2 (22 μL, 1.05 equiv., 0.2625 mmol), and the resulting mixture was stirred at rt for 1 h. After quenching by adding saturated sodium sulfite solution, the mixture was extracted by DCM (5 mL × 3). The combined organic layer was washed with brine and dried over Na_2SO_4, filtered

and concentrated. The crude product was purified by silica gel flash column chromatography (petroleum ether:ethyl acetate = 5:1) to afford (*R*)-**2a** (46 mg, 58% yield). HPLC analysis: Chiralcel AD-H (Hexane: Isopropyl alcohol = 90:10, 1.0 mL/min, 254 nm, 25℃) 14.9 min (major), 31.6 min (minor), 83% *ee*. (*R*)-**2a** was then converted to (*R*)-**3a** by bromination. HPLC analysis: Chiralcel AD-H (Hexane: Isopropyl alcohol = 90:10, 1.0 mL/min, 254 nm, 25℃), 12.4 min (major), 15.4 min (minor), 80% *ee*.

2b (62 mg, 93% yield) was also obtained by using **BG2**[4] as catalyst. HPLC analysis: Chiralcel AS-H (Hexane: Isopropyl alcohol = 90:10, 1.0 mL/min, 254 nm, 25℃), 30.6 min (minor), 41.9 min (major), 33% *ee*. (*R*)-**2b** was then converted to (*R*)-**3b** by bromination. HPLC analysis: Chiralcel AS- H (Hexane: Isopropyl alcohol = 90:10, 1.0 mL/min, 254 nm, 25℃), 15.1min (major), 22.1 min (minor), 30% *ee*.

Optimizations of the asymmetric oxidation condition were also conducted, but no further improvement of enantioselectivity was observed.

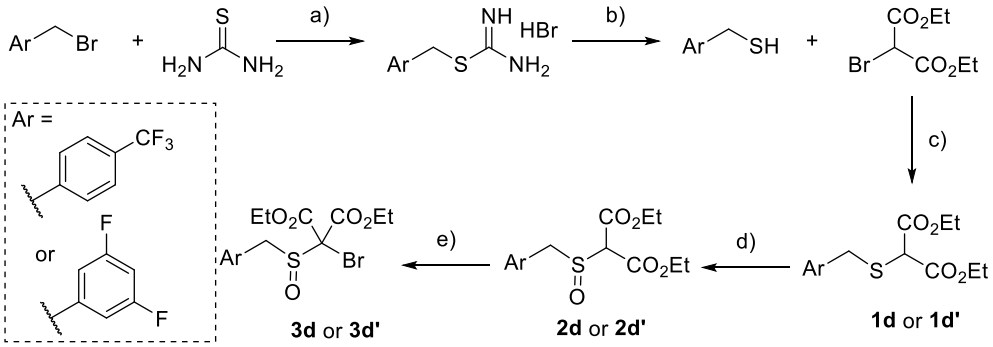


Scheme S4. Synthesis of 1c. Conditions: (a) NCS (1.2 equiv.), DCM, N_2, rt, 1 h; (b) NaH (0.9 equiv.), THF, N_2, 0 ℃~rt, overnight, 24% yield.

Step 1: To a solution of *N*-chlorosuccinimide (**NCS**) (241 mg, 1.2 equiv., 1.8 mmol) in 1.5 mL of DCM was slowly added phenylmethanethiol (176 μL, 1.0 equiv., 1.5 mmol). The reaction mixture was stirred at rt for 1 hour, and then filtered with a pad of celite. The filtrate was concentrated and the residue was used in the next step without further purification.

Step 2: To a solution of bis(phenylsulfonyl)methane (356 mg, 0.8 equiv., 1.2 mmol) in 7 mL of THF was added sodium hydride (54 mg, 0.9 equiv., 1.35 mmol) and stirred at 0 ℃ for 10 min. Then a solution of benzyl hypochlorothioite (237 mg, 1.0 equiv., 1.5 mmol) in 1.5 mL of THF was added slowly. After the reaction was completed, water

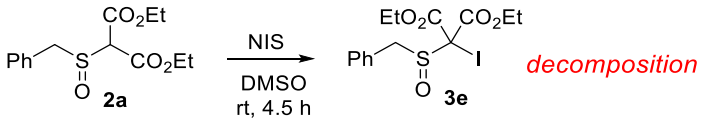
was added to quench the reaction. This mixture was extracted with DCM (5 mL × 3). The combined organic layer was washed with brine and dried over Na_2SO_4, filtered and concentrated. The residue was purified by silica gel flash column chromatography (petroleum ether:ethyl acetate = 5:1) to **1c** (white solid, 74.9 mg, 24% yield). Further oxidation of **1c** failed by using oxidants such as 35 wt% H_2O_2, $NaIO_4$, *m*CPBA or $KMnO_4$.



Scheme S5. Synthesis of 3d and 3d'. Conditions: (a) CH_3OH, reflux, overnight; (b) NaOH (1.1 equiv.), H_2O, reflux, 2 h; (c) NaH (1.1 equiv.), THF, 0°C ~ rt, 4 h; (d) Na_2MoO_4 (0.05 equiv.), 35 wt% H_2O_2 (1.05 equiv.), CF_3CH_2OH, rt, 1 h; (e) **NBS** (1.05 equiv.), TsOH•H_2O (1.5 equiv.), CH_3CN, reflux, 2 h. 62% yield (**3d**), 75% yield (**3d'**).

3d and **3d'** were synthesized by following the same procedure to **3a**. The corresponding mercaptans were prepared by following the reported literature. Take (3,5-difluorophenyl)methanethiol as an example: To a solution of 1-(bromomethyl)-3,5-difluorobenzene (1.29 mL, 1.0 equiv., 10 mmol) in 4 mL of methanol was added thiourea (852 mg, 1.1 equiv., 11 mmol). The reaction mixture was stirred under reflux overnight and cooled to rt. Then methanol was removed under vacuum and the residue was washed with DCM. The resulting 3,5-difluorobenzyl carbamimidothioate hydrobromide (2.087 g, 74% yield) was used directly in next step without purification. To a solution of 3,5-difluorobenzyl carbamimidothioate hydrobromide (1.318 g, 1.0 equiv., 4.67 mmol) in 8 mL of H_2O was added aqueous sodium hydroxide (4 M, 1.0 equiv., 4.67 mmol). The reaction mixture was stirred under reflux for 2 hours and then cooled to rt. After quenching by adding concentrated hydrochloric acid (12 M, 1.1 equiv., 5.14 mmol), the mixture was extracted with DCM (15 mL × 3). The combined organic layer was washed with brine and dried over Na_2SO_4, filtered and concentrated

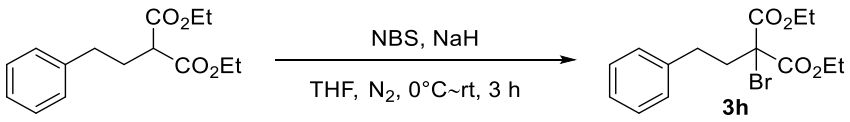
to afford (3,5-difluorophenyl)methanethiol (colorless oil, 608 mg, 95% yield), which was used in next step without purification.



Scheme S6. Synthesis of 3e

To a solution of **2a** (150 mg, 1.0 equiv., 0.5 mmol) in 5 mL of DMSO was added *N*-iodosuccinimide (**NIS**) (118 mg, 1.05 equiv., 0.525 mmol) and the reaction mixture was stirred at rt for 4.5 hours. Then saturated ammonium chloride solution (5 mL) was added and the mixture was extracted with ethyl acetate (5 mL × 3). The combined organic layer was washed with brine and dried over Na_2SO_4, filtered and concentrated. The residue was purified by silica gel flash column chromatography (petroleum ether:ethyl acetate = 5:1) to afford **3e** (light yellow solid). However, **3e** is unstable under ambient atmosphere and only ^{1}H NMR spectrum was obtained in the solution of DMSO-d_6. Its color change of from light yellow to purple indicated the cleavage of carbon-iodine bond.

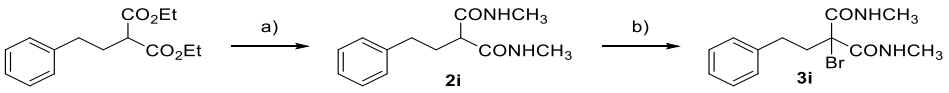
3f and **3g** were prepared by following the reported literature.



Scheme S7. Synthesis of 3h

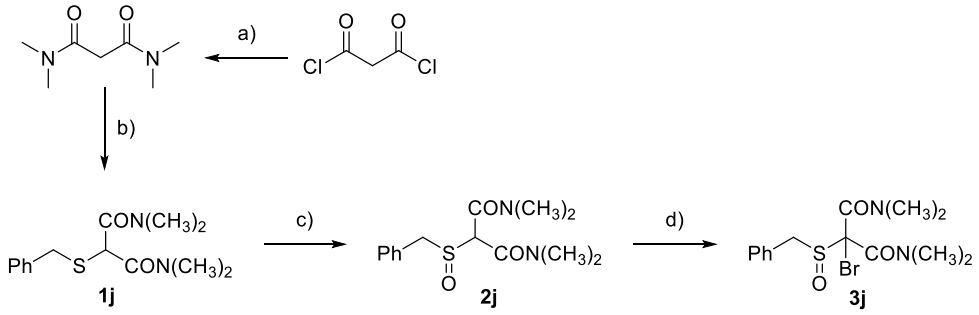
To a solution of diethyl 2-phenethylmalonate (123 μL, 1.0 equiv., 0.5 mmol) in 1 mL of THF was added sodium hydride (24 mg, 1.2 equiv., 0.6 mmol). The mixture was stirred at 0°C for 10 min, following the addition of *N*-bromosuccinimide (107 mg, 1.2 equiv., 0.6 mmol). The reaction mixture was stirred at rt for 3 hours. Then water was added to quench the reaction. The mixture was extracted by DCM (15 mL × 3). The combined organic layer was washed with brine and dried over Na_2SO_4, filtered and

concentrated. The crude product was purified by silica gel flash column chromatography (petroleum ether:ethyl acetate = 30:1) to afford **3h** (93 mg, 54% yield).



Scheme S8. Synthesis of 3i. Conditions: (a) 27 wt% CH_3NH_2 in MeOH, 60℃, 5 h, 99% yield; (b) Br_2 (1.05 equiv.), AcOH, rt, 2 days, 65% yield.

By following the synthetic preproduce for **1b**, N^1,N^3-dimethyl-2-phenethylmalonamide(**2i**) was obtained in 99% yield. Then to a solution **2i** (94 mg, 1.0 equiv., 0.4 mmol) in 6 mL of acetic acid was slowly added Br_2 (22 μL, 1.05 equiv., 0.42 mmol) in 0.335 mL of acetic acid. The reaction mixture was stirred at rt for 2 days and then acetic acid was removed under vacuum. The crude product was purified by silica gel flash column chromatography (DCM: MeOH = 60: 1) to give **3i** (white solid, 40 mg, 65% yield).



Scheme S9. Synthesis of 3j. Conditions: (a) $(CH_3)_2NH$ (2.0 equiv.), Et_3N (2.0 equiv.), THF, N_2, reflux, 4 h, 43% yield; (b) $PhCH_2SCl$ (1.0 equiv.), NaH (1.0 equiv.), THF, N_2, rt, overnight, 16% yield; (c) 35% H_2O_2 (1.05 equiv.), Na_2MoO_4 (2.5 mol%), CF_3CH_2OH, 5 ℃, 51% yield; (d) NBS (1.1 equiv.), NaH (1.1 equiv.), THF, N_2, rt, 91% yield.

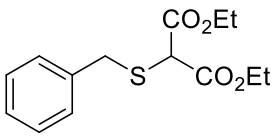
Step 1[5]: A 100 mL RBF was charged with Et_3N (6.98 mL, 2.0 equiv., 50 mmol) and a solution of $(CH_3)_2NH$ in THF (2 M, 25 mL, 2.0 equiv., 50 mmol) at 0 ℃ under nitrogen. A solution of malonylchloride (2.43mL, 25 mmol) in 40 mL of THF was added to the mixture dropwise. Then the reaction mixture was stirred at 70℃ for 4 hours. The resulting white precipitate was removed by filtration and washed with 50 mL of THF, and the solvent was removed under vacuum. The residue was purified by silica gel flash

column chromatography (DCM: MeOH = 100: 1) to afford N^1,N^1,N^3,N^3-tetramethylmalonamide (yellow oil, 543 mg, 43% yield).

Step 2: To a solution of N^1,N^1,N^3,N^3-tetramethylmalonamide (543 mg, 1.0 equiv., 5.3 mmol) in 15 mL THF was added sodium hydride (213 mg, 1.0 equiv., 5.3 mmol) and the resulting mixture was stirred at 0℃ for 10 min. Then a solution of benzyl hypochlorothioite (838 mg, 1.0 equiv., 5.3 mmol) in 5 mL THF was added slowly. After the reaction was completed, water was added to quench the reaction. This mixture was extracted with DCM (15 mL × 3). The combined organic layer was washed with brine and dried over Na_2SO_4, filtered and concentrated. The residue was purified by silica gel flash column chromatography (DCM : MeOH = 400 : 1) to give crude product **1j** (yellow oil, 230 mg, 16% yield).

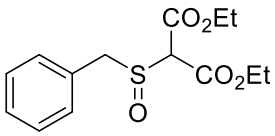
Step 3: By following the same procedure to **2a**, **2j** (yellow oil, 51% yield;) was obtained by oxidation of sulfide **1j.**

Step 4: To a solution of **2j** in 2.5 mL THF was added sodium hydride (5.2 mg, 1.1 equiv., 0.13 mmol) at 0 ℃ under nitrogen. 10 min later, NBS (23 mg, 1.1 equiv., 0.13 mmol) was added and the reaction mixture was stirred at room temperature for 1 hour. Then the reaction was quenched by adding water and extracted with DCM (5 mL × 3). The combined organic layer was washed with brine and dried over Na_2SO_4, filtered and concentrated. **3j** was obtained after silica gel column chromatography (DCM : MeOH = 100:1), as a white solid, 41 mg, 91% yield.



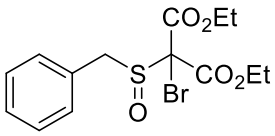
diethyl 2-(benzylthio)malonate (1a)

Colorless oil. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 7.27 – 7.20 (m, 5H), 4.11 (q, *J* = 7.1 Hz, 4H), 3.94 (s, 1H), 3.81 (s, 2H), 1.18 (t, *J* = 7.1 Hz, 6H); ^{13}C NMR (151 MHz, CDCl3): δ = 166.91, 136.64, 129.35, 128.73, 127.61, 62.40, 50.24, 35.94, 14.11; IR: 2982, 1726, 1239, 1141, 1028, 702 cm^{-1}; HRMS (ESI): *m*/*z* calcd. for $C_{14}H_{19}O_4S$ $[M+H]^+$ 283.0999, found 283.0996.



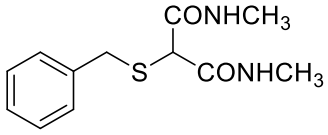
diethyl 2-(benzylsulfinyl)malonate (2a)

Colorless oil. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 7.41 – 7.34 (m, 5H), 4.40 (d, *J* = 13.1 Hz, 1H), 4.37 (s, 1H), 4.34 – 4.25 (m, 4H), 4.17 (d, *J* = 13.0 Hz, 1H), 1.32 (t, *J* = 7.1 Hz, 6H); ^{13}C NMR (151 MHz, $CDCl_3$): δ = 163.46, 162.77, 130.56, 129.35, 129.22, 128.89, 68.08, 63.15, 63.05, 57.57, 14.17, 14.12; IR: 2983, 2929, 1728, 1311, 1258, 1173, 1139, 1029, 734, 702 cm^{-1}; HRMS (ESI): *m*/*z* calcd. for $C_{14}H_{19}O_5S$ $[M+H]^+$ 299.0948, found 299.0944.



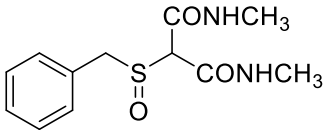
diethyl 2-(benzylsulfinyl)-2-bromomalonate (3a)

Light yellow solid. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 7.40 – 7.34 (m, 5H), 4.43 – 4.40 (m, 1H), 4.40 – 4.29 (m, 4H), 3.91 (d, *J* = 12.7 Hz, 1H), 1.37 – 1.32 (m, 6H); ^{13}C NMR (151 MHz, $CDCl_3$): δ = 163.14, 162.47, 130.74, 129.93, 129.21, 128.90, 76.21, 64.77, 64.44, 59.43, 14.09, 13.98; IR: 2990, 1762, 1725, 1260, 1199, 1022, 698 cm^{-1}; HRMS (ESI): *m*/*z* calcd. for $C_{14}H_{18}BrO_5S$ $[M+H]^+$ 377.0053, found 377.0047.



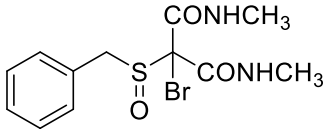
2-(benzylthio)-*N*^{1},*N*^{3}-dimethylmalonamide (1b)

White solid. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 7.30 (h, J = 5.8 Hz, 4H), 7.27 – 7.23 (m, 1H), 6.98 (s, 2H), 3.85 (s, 1H), 3.79 (s, 2H), 2.75 (d, J = 4.8 Hz, 6H); ^{13}C NMR (151 MHz, $CDCl_3$): δ = 167.91, 136.91, 129.18, 128.81, 127.63, 53.09, 36.65, 26.78; IR: 3298, 1674, 1639, 1567, 706 cm^{-1}; HRMS (ESI): *m*/*z* calcd. For $C_{12}H_{17}N_2O_2S$ $[M+H]^+$ 253.1005, found 253.1003.



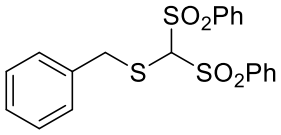
2-(benzylsulfinyl)-*N*^{1},*N*^{3}-dimethylmalonamide (2b)

White solid. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 7.60 (s, 1H), 7.45 (s, 1H), 7.37 (td, *J* = 8.2, 7.1, 4.5 Hz, 5H), 4.21 (d, *J* = 13.2 Hz, 1H), 4.18 (s, 1H), 4.07 (d, *J* = 13.1 Hz, 1H), 2.84 (dd, *J* = 9.3, 4.8 Hz, 6H); ^{13}C NMR (151 MHz, $CDCl_3$): δ = 164.35, 163.83, 130.94, 129.13, 128.93 (d, *J* = 3.4 Hz), 66.79, 56.67, 26.74 (d, *J* = 2.1 Hz); IR: 3304, 1658, 1558, 1074, 1064, 695 cm^{-1}; HRMS (ESI): *m*/*z* calcd. For $C_{12}H_{17}N_2O_3S$ $[M+H]^+$ 269.0954, found 269.0951.



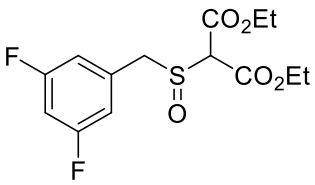
2-(benzylsulfinyl)-2-bromo-*N*^{1},*N*^{3}-dimethylmalonamide (3b)

White solid. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 8.01 (s, 1H), 7.62 (s, 1H), 7.40 – 7.34 (m, 3H), 7.31 (dd, *J* = 7.6, 1.9 Hz, 2H), 4.27 (d, *J* = 12.8 Hz, 1H), 3.90 (d, *J* = 12.8 Hz, 1H), 2.91 (d, *J* = 4.8 Hz, 3H), 2.84 (d, *J* = 4.8 Hz, 3H); ^{13}C NMR (151 MHz, $CDCl_3$): δ = 164.63, 164.24, 130.71, 129.43, 129.16, 128.98, 72.89, 57.91, 27.36, 27.22; IR: 3341, 3301, 3260, 1681, 1663, 1510, 1496, 1039, 759 cm^{-1}; HRMS (ESI): *m*/*z* calcd. For $C_{12}H_{16}BrN_2O_3S$ $[M+H]^+$ 347.0060, found 347.0055.



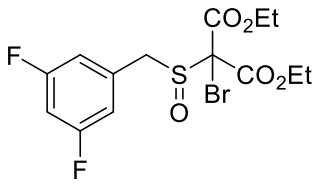
benzyl(bis(phenylsulfonyl)methyl)sulfane (1c)

Yellow solid. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 7.93 – 7.85 (m, 4H), 7.69 (td, *J* = 7.5, 1.3 Hz, 2H), 7.52 (dd, *J* = 8.4, 7.4 Hz, 4H), 7.22 – 7.17 (m, 1H), 7.13 (t, *J* = 7.6 Hz, 2H), 6.98 – 6.93 (m, 2H), 4.64 (s, 1H), 3.95 (s, 2H); ^{13}C NMR (151 MHz, $CDCl_3$): δ = 137.24, 134.78, 133.89, 130.28, 129.45, 129.08, 128.97, 128.02, 83.49, 38.12; IR: 2917, 1335, 1163 cm^{-1}; HRMS (ESI): *m*/*z* calcd. For $C_{20}H_{18}O_4S_3Na$ $[M+Na]^+$ 441.0259, found 441.0270.



diethyl 2-((3,5-difluorobenzyl)sulfinyl)malonate (2d)

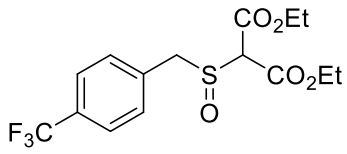
Colorless oil. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 6.93 – 6.89 (m, 2H), 6.83 (tt, *J* = 8.9, 2.4 Hz, 1H), 4.42 (s, 1H), 4.37 – 4.30 (m, 5H), 4.13 (d, *J* = 13.1 Hz, 1H), 1.33 (q, *J* = 7.1 Hz, 6H); ^{13}C NMR (151 MHz, $CDCl_3$): δ = 164.04 (d, *J* = 13.0 Hz), 163.18, 162.67, 162.39 (d, *J* = 13.0 Hz), 133.18 (t, *J* = 9.6 Hz), 113.60 (dd, *J* = 20.3, 5.5 Hz), 104.57 (t, *J* = 25.1 Hz), 68.52, 63.31 (d, *J* = 7.2 Hz), 56.79, 14.12; ^{19}F NMR (565 MHz, $CDCl_3$): δ = –108.27; IR: 2985, 1728, 1025, 866 cm^{-1}; HRMS (ESI): *m*/*z* calcd. for $C_{14}H_{17}F_2O_5S$ $[M+H]^+$ 335.0759, found 335.0754.



diethyl 2-bromo-2-((3,5-difluorobenzyl)sulfinyl)malonate (3d)

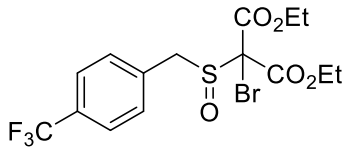
Light yellow solid. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 6.93 (d, *J* = 5.6 Hz, 2H), 6.82 (tt, *J* = 8.9, 2.3 Hz, 1H), 4.45 – 4.34 (m, 5H), 3.87 (d, *J* = 12.7 Hz, 1H), 1.36 (td, *J* = 7.1, 2.0 Hz, 6H); ^{13}C NMR (151 MHz,

$CDCl_3$): δ = 164.10 (d, *J* = 12.6 Hz), 163.15, 162.44 (d, *J* = 12.7 Hz), 162.17, 133.62 (t, *J* = 9.8 Hz), 113.82 (d, *J* = 5.7 Hz), 113.69 (d, *J* = 5.5 Hz), 104.60 (t, *J* = 25.1 Hz), 75.86, 65.01, 64.64, 58.99, 14.09, 13.98; ^{19}F NMR (565 MHz, $CDCl_3$): δ = –108.42; IR: 2980, 1749, 866 cm^{-1}; HRMS (ESI): *m*/*z* calcd. for $C_{14}H_{16}BrF_2O_5S$ $[M+H]^+$ 412.9864, found 412.9861.



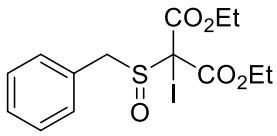
diethyl 2-((4-(trifluoromethyl)benzyl)sulfinyl)malonate (2d')

Colorless oil. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 7.65 (d, *J* = 8.0 Hz, 2H), 7.49 (d, *J* = 8.0 Hz, 2H), 4.43 (d, *J* = 13.1 Hz, 1H), 4.41 (s, 1H), 4.35 – 4.28 (m, 4H), 4.22 (d, *J* = 13.0 Hz, 1H), 1.33 (dt, *J* = 9.6, 7.1 Hz, 6H); ^{13}C NMR (151 MHz, $CDCl_3$): δ =163.21, 162.73, 133.66, 131.01, 126.09 (q, *J* = 3.7 Hz), 124.91, 123.10, 68.52, 63.31, 63.25, 56.92, 14.14; ^{19}F NMR (565 MHz, $CDCl_3$): δ = –62.74; IR: 2982, 2934, 1749, 1116, 1025 cm^{-1}; HRMS (ESI): *m*/*z* calcd. for $C_{15}H_{18}F_3O_5S$ $[M+H]^+$ 367.0822, found 367.0818.



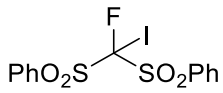
diethyl 2-bromo-2-((4-(trifluoromethyl)benzyl)sulfinyl)malonate (3d')

Light yellow solid. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 7.65 (d, *J* = 8.0 Hz, 2H), 7.51 (d, *J* = 7.9 Hz, 2H), 4.45 – 4.33 (m, 5H), 3.98 (d, *J* = 12.6 Hz, 1H), 1.35 (t, *J* = 7.1 Hz, 6H); ^{13}C NMR (151 MHz, $CDCl_3$): δ = 163.14, 162.23, 134.10, 131.17, 126.09 (q, *J* = 3.8 Hz), 124.91, 123.11, 76.04, 64.96, 64.60, 58.93, 14.07, 13.96; ^{19}F NMR (565 MHz, $CDCl_3$): δ = –62.74; IR: 2983, 1749, 1246, 1227, 1117 cm^{-1}; HRMS (ESI): *m*/*z* calcd. For $C_{15}H_{17}BrF_3O_5S$ $[M+H]^+$ 444.9927, found 444.9923.



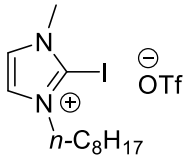
diethyl 2-(benzylsulfinyl)-2-iodomalonate (3e)

Light yellow solid. ^{1}H NMR (600 MHz, DMSO-d_6): δ = 7.44 – 7.37 (m, 3H), 7.35 – 7.33 (m, 2H), 4.34 (d, *J* = 12.6 Hz, 1H), 4.32 – 4.24 (m, 4H), 3.71 (d, *J* = 12.6 Hz, 1H), 1.22 (td, *J* = 7.1, 2.1 Hz, 6H).



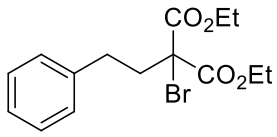
(fluoroiodomethylenedisulfonyl)dibenzene (3f)[1]

White solid. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 8.06 – 8.00 (m, 4H), 7.79 – 7.75 (m, 2H), 7.60 (t, *J* = 8.0 Hz, 4H). The ^{1}H NMR spectrum is consistent with the reported data.



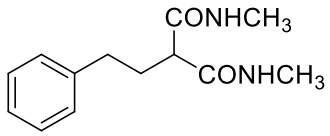
2-iodo-1-methyl-3-octyl-1H-imidazol-3-ium trifluoromethanesulfonate (3g)[2]

White solid. ^{1}H NMR (600 MHz, $CDCl_3$): δ 7.80 (s, 1H), 7.64 (s, 1H), 4.17 (t, *J* = 7.5 Hz, 2H), 3.96 (s, 3H), 1.84 (p, *J* = 7.7, 7.3 Hz, 2H), 1.39 – 1.23 (m, 10H), 0.88 (t, *J* = 6.8 Hz, 3H). ^{1}H NMR spectrum is consistent with the reported data.



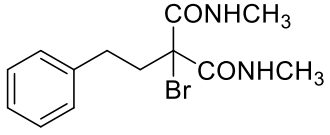
diethyl 2-bromo-2-phenethylmalonate (3h)

Colorless oil. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 7.29 (t, *J* = 7.6 Hz, 2H), 7.20 (d, *J* = 8.1 Hz, 3H), 4.27 (q, *J* = 7.6, 7.1 Hz, 4H), 2.79 – 2.74 (m, 2H), 2.61 – 2.56 (m, 2H), 1.30 (t, *J* = 7.1 Hz, 6H); ^{13}C NMR (151 MHz, $CDCl_3$): δ = 166.84, 140.27, 128.64, 128.62, 126.44, 63.20, 63.12, 39.94, 31.82, 14.00; IR: 2982, 1739, 1248, 1022, 698 cm^{-1}; HRMS (ESI): *m*/*z* calcd. For $C_{15}H_{20}BrO_4$ $[M+H]^+$ 343.0539, found 343.0537.



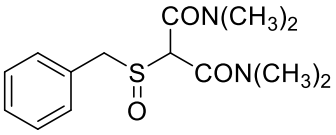
***N*^{1},*N*^{3}-dimethyl-2-phenethylmalonamide (2i)**

White solid. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 7.08 (t, *J* = 7.6 Hz, 2H), 6.99 (dd, *J* = 15.9, 7.0 Hz, 3H), 6.93 (d, *J* = 5.0 Hz, 2H), 2.85 (t, *J* = 7.5 Hz, 1H), 2.62 (d, *J* = 4.8 Hz, 6H), 2.46 – 2.40 (m, 2H), 1.98 (q, *J* = 7.7 Hz, 2H); ^{13}C NMR (151 MHz, $CDCl_3$): δ = 171.59, 140.67, 128.55, 128.52, 126.29, 54.35, 34.26, 33.66, 26.46; IR: 3282, 1663, 1560, 696 cm^{-1}; HRMS (ESI): *m*/*z* calcd. For $C_{13}H_{19}N_2O_2$ $[M+H]^+$ 235.1441, found 235.1448.



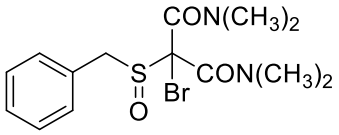
2-bromo-*N*^{1},*N*^{3}-dimethyl-2-phenethylmalonamide (3i)

White solid. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 7.26 (t, *J* = 7.6 Hz, 2H), 7.23 (s, 2H), 7.17 (d, *J* = 8.2 Hz, 3H), 2.85 (d, *J* = 4.8 Hz, 6H), 2.74 – 2.68 (m, 2H), 2.63 – 2.57 (m, 2H); ^{13}C NMR (151 MHz, $CDCl_3$): δ = 167.82, 140.14, 128.60, 128.59, 126.39, 71.80, 43.26, 32.54, 27.56; IR: 3340, 3261, 1657, 1553, 1531, 702 cm^{-1}; HRMS (ESI): *m*/*z* calcd. For $C_{13}H_{18}BrN_2O_2$ $[M+H]^+$ 313.0546, found 313.0543.



2-(benzylsulfinyl)-*N*^{1},*N*^{1},*N*^{3},*N*^{3}-tetramethylmalonamide (2j)

Yellow oil. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 7.39 – 7.33 (m, 5H), 4.91 (s, 1H), 4.56 (d, *J* = 13.2 Hz, 1H), 3.94 (d, *J* = 13.2 Hz, 1H), 3.04 (s, 3H), 2.98 (d, *J* = 8.2 Hz, 6H), 2.85 (s, 3H); ^{13}C NMR (151 MHz, $CDCl_3$): δ = 164.03, 163.56, 130.88, 130.59, 128.79, 128.51, 64.97, 58.13, 37.71, 37.44, 36.16, 36.08; IR: 2976, 1638, 1036 cm^{-1}; HRMS (ESI): *m*/*z* calcd. For $C_{14}H_{21}N_2O_3S$ $[M+H]^+$ 297.1267, found 297.1272.

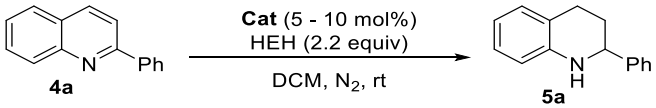


2-(benzylsulfinyl)-2-bromo-*N*^{1},*N*^{1},*N*^{3},*N*^{3}-tetramethylmalonamide (3j)

White solid. ^{1}H NMR (500 MHz, $CDCl_3$): δ = 7.40 (d, *J* = 6.9 Hz, 2H), 7.35 (dt, *J* = 9.9, 6.7 Hz, 3H), 4.68 (d, *J* = 12.8 Hz, 1H), 3.69 (d, *J* = 12.9 Hz, 1H), 3.11 (d, *J* = 5.5 Hz, 6H), 3.08 (s, 3H), 3.04 (s, 3H); ^{13}C NMR (126 MHz, $CDCl_3$): δ = 163.56, 161.52, 131.25, 130.67, 129.03, 128.51, 59.58, 38.84 – 38.13 (m), 37.54. IR: 2926, 1638, 1036, 619 cm^{-1}; HRMS (ESI): *m*/*z* calcd. For $C_{14}H_{20}BrN_2O_3S$ $[M+H]^+$ 375.0373, found 375.0377.

3. Screening of reaction conditions

Table S1. Screening reaction conditions [a]



Entry	Cat (mol%)	Time (h)	Solvent	Yield (%)[b]
1	**3d** (10)	9	CH_2Cl_2	99
2	**3d'** (10)	9	CH_2Cl_2	99
3	**3b** (5)	6.5	$ClCH_2CH_2Cl$	99
4	**3b** (5)	6.5	Et_2O	89
5	**3b** (5)	6.5	CH_3CN	63
6	**3b** (5)	6.5	Toluene	Trace
7	**3b** (5)	6.5	MeOH	45
8	**3b** (5)	6.5	THF	48
10	**3j** (5)	6.5	CH_2Cl_2	3

[a]Unless otherwise indicated, all reactions were carried out with **4a** (0.05 mmol, 1.0 equiv.) and Hantzsch esters (0.11 mmol, 2.2 equiv.) in 0.5 mL of solvent at rt under N_2 atmosphere. [b]Determined by ^1H NMR analysis.

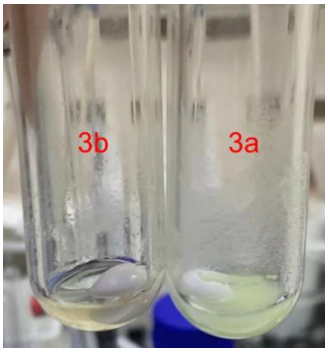
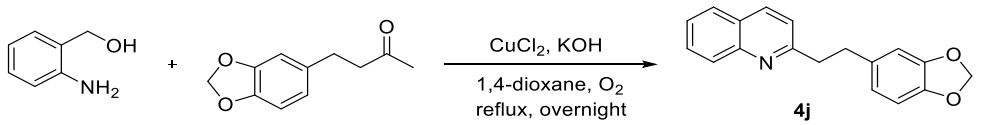


Figure S1. Naked-eye observation of reaction process with 3a and 3b. Reactions were carried out with 0.05 mmol of **4a** in the presence of 10 mol% **3a** (right) or **3b** (left) under standard condition.

Reaction with **3b** as catalyst afforded reduction product **5a** in 89% yield within 3 hours, while 36% yield was obtained with catalyst **3a**. The reaction mixture with **3b** was clearer than that with **3a**, due to the faster consumption of Hantzsch esters with poor solubility in DCM.

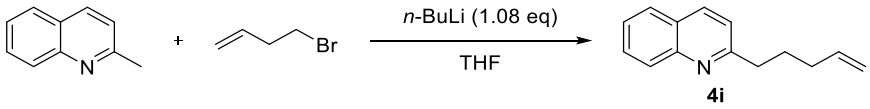
4. Synthesis of substrates

General procedure for the synthesis of 4b, 4c, 4d, 4e, 4f and 4j.[6](Take 4j as an example)



To a solution of (2-aminophenyl)methanol (246 mg, 1.0 equiv., 2 mmol) and 4-(benzo[d][1,3]dioxol-5-yl)butan-2-one (461 mg, 1.2 equiv., 2.4 mmol) in 4 mL of 1,4-dioxane was added $CuCl_2$ (5.4 mg, 0.02 equiv., 0.04 mmol) and KOH (336 mg, 3.0 equiv., 6 mmol) at rt. Then the mixture was stirred under reflux overnight at an oxygen atmosphere. After the reaction was cooled down to room temperature, the solvent was removed under vacuum. Then 5 mL of ethyl acetate was added. The mixture was filtered and the residue was washed with ethyl acetate (5 mL × 3). The combined organic layer was washed with brine and dried over Na_2SO_4, filtered and concentrated. The residue was purified by silica gel flash column chromatography (petroleum ether:ethyl acetate = 20:1) to afford **4j** (yellow solid, 114 mg, 21% yield). Substrates **4b, 4c, 4d, 4e,** and **4f** were prepared in the same way as **4j**.

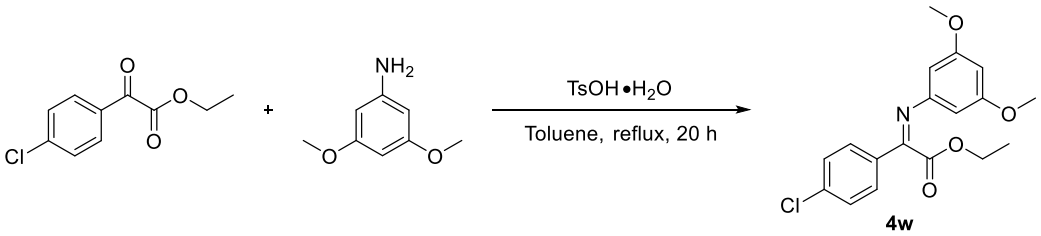
Synthesis of 4i.[7]



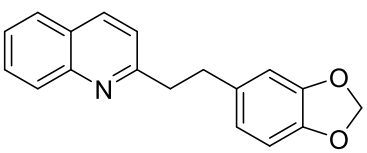
To a solution of 2-methylquinoline (676 μL, 1.0 equiv., 5 mmol) in 20 mL of THF was added 1.6 M *n*-butyllithium in hexanes (3.4 mL, 5.4 mmol, 1.08 equiv.) at 0 ℃, the mixture was stirred at 0 ℃ for 10 min and then stirred at rt for 3 hours. 4-bromobut-1-ene (605 μL, 1.2 equiv., 6 mmol) was added to the above mixture which was cooled down below –30 ℃. The resulting mixture was stirred at –30 ℃ for 20 min and stirred at rt for 12 h. After the reaction was completed, a solution of saturated ammonium chloride (10 mL) was added to quench the reaction at 0 ℃. Then the mixture was warmed to room temperature and extracted with DCM (15 mL × 3). The combined

organic layer was washed with brine and dried over Na_2SO_4, filtered and concentrated. The residue was purified by silica gel flash column chromatography (petroleum ether:ethyl acetate = 20:1) to afford **4i** (yellow liquid, 749 mg, 76% yield).

Synthesis of 4w.[8]

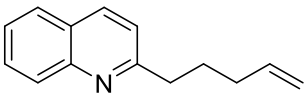


To a solution of ethyl 2-(4-chlorophenyl)-2-oxoacetate (848 mg, 1.0 equiv., 4 mmol) and 3,5-dimethoxyaniline (643 mg, 1.05 equiv., 4.2 mmol) in 6 mL of toluene was added p-toluenesulfonic acid monohydrate (38 mg, 0.05 equiv., 0.2 mmol). And the mixture was stirred under reflux using a Dean-stark apparatus for 20 h. Removing the solvent under vacuum and purified by flash chromatography to give **4w**.



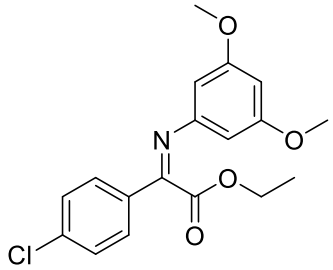
2-(2-(benzo[d][1,3]dioxol-5-yl)ethyl)quinoline (4j)[7]

White solid. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 8.09 (d, *J* = 8.5 Hz, 1H), 8.06 (d, *J* = 8.4 Hz, 1H), 7.78 (d, *J* = 8.0 Hz, 1H), 7.70 (ddd, *J* = 8.4, 6.9, 1.5 Hz, 1H), 7.50 (ddd, *J* = 8.1, 6.8, 1.2 Hz, 1H), 7.23 (d, *J* = 8.4 Hz, 1H), 6.81 – 6.64 (m, 3H), 5.91 (s, 2H), 3.30 – 3.22 (m, 2H), 3.08 (dd, *J* = 9.5, 6.7 Hz, 2H). The ^{1}H NMR spectrum is consistent with the reported data.



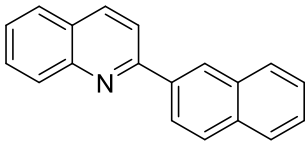
2-(pent-4-en-1-yl)quinolone (4i)[7]

Yellow oil**.** ^{1}H NMR (600 MHz, $CDCl_3$): δ = 8.06 (t, *J* = 9.1 Hz, 2H), 7.77 (d, *J* = 8.1 Hz, 1H), 7.68 (ddd, *J* = 8.4, 6.8, 1.5 Hz, 1H), 7.48 (t, *J* = 7.5 Hz, 1H), 7.30 (d, *J* = 8.3 Hz, 1H), 5.86 (ddt, *J* = 16.9, 10.2, 6.7 Hz, 1H), 5.05 (dq, *J* = 17.1, 1.7 Hz, 1H), 4.99 (ddd, *J* = 10.2, 2.2, 1.2 Hz, 1H), 3.01 – 2.97 (m, 2H), 2.21 – 2.15 (m, 2H), 1.96 – 1.89 (m, 2H). The ^{1}H NMR spectrum is consistent with the reported data.



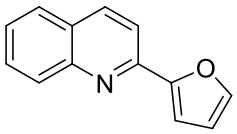
ethyl-2-(4-chlorophenyl)-2-((3,5-dimethoxyphenyl)imino)acetate (4w)

Yellow solid. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 7.86 – 7.81 (m, 2H), 7.48 – 7.43 (m, 2H), 6.28 (t, *J* = 2.3 Hz, 1H), 6.15 (d, *J* = 2.3 Hz, 2H), 4.20 (q, *J* = 7.2 Hz, 2H), 3.78 (s, 6H), 1.09 (t, *J* = 7.1 Hz, 3H); ^{13}C NMR (151 MHz, $CDCl_3$): δ = 164.63, 161.13, 159.17, 151.83, 132.30, 131.55, 129.44, 129.13, 97.92, 97.62, 61.79, 55.51, 13.92.



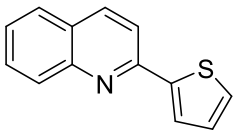
2-(naphthalen-2-yl)quinoline (4b)[9]

White solid. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 8.65 (s, 1H), 8.37 (dd, *J* = 8.6, 1.8 Hz, 1H), 8.31 (d, *J* = 8.6 Hz, 2H), 8.06 (d, *J* = 8.6 Hz, 1H), 8.01 (d, *J* = 8.6 Hz, 2H), 7.91 (dd, *J* = 6.0, 3.4 Hz, 1H), 7.87 (d, *J* = 8.1 Hz, 1H), 7.77 (t, *J* = 7.6 Hz, 1H), 7.59 – 7.53 (m, 3H). The ^{1}H NMR spectrum is consistent with the reported data.



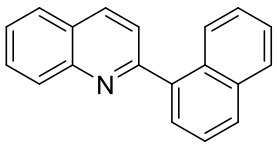
2-(furan-2-yl)quinoline (4c)[10]

Yellow solid. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 8.18 (t, *J* = 9.3 Hz, 2H), 7.84 (d, *J* = 8.6 Hz, 1H), 7.79 (d, *J* = 9.4 Hz, 1H), 7.74 – 7.69 (m, 1H), 7.64 (d, *J* = 1.7 Hz, 1H), 7.53 – 7.49 (m, 1H), 7.26 (s, 1H), 6.60 (dd, *J* = 3.4, 1.7 Hz, 1H). The ^{1}H NMR spectrum is consistent with the reported data.



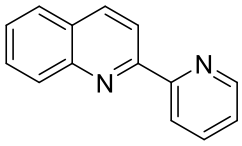
2-(thiophen-2-yl)quinoline (4d)[10]

White solid. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 8.12 (dd, *J* = 22.9, 8.5 Hz, 2H), 7.87 – 7.67 (m, 4H), 7.57 – 7.45 (m, 2H), 7.17 (dd, *J* = 5.0, 3.6 Hz, 1H); The ^{1}H NMR spectrum is consistent with the reported data.



2-(naphthalen-1-yl)quinoline (4e)[11]

Yellow solid. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 8.30 (d, *J* = 8.4 Hz, 1H), 8.24 (d, *J* = 8.4 Hz, 1H), 8.13 (d, *J* = 8.4 Hz, 1H), 8.01 – 7.90 (m, 3H), 7.79 (ddd, *J* = 8.4, 6.8, 1.4 Hz, 1H), 7.76 – 7.70 (m, 2H), 7.65 – 7.58 (m, 2H), 7.55 – 7.50 (m, 1H), 7.50 – 7.46 (m, 1H). The ^{1}H NMR spectrum is consistent with the reported data.



2-(pyridin-2-yl)quinoline (4f)[10]

Yellow solid. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 8.75 (d, *J* = 4.8 Hz, 1H), 8.67 (d, *J* = 7.9 Hz, 1H), 8.57 (d, *J* = 8.6 Hz, 1H), 8.30 (d, *J* = 8.5 Hz, 1H), 8.20 (d, *J* = 8.5 Hz, 1H), 7.92 – 7.85 (m, 2H), 7.78 – 7.72 (m, 1H), 7.60 – 7.54 (m, 1H), 7.40 – 7.35 (m, 1H). The ^{1}H NMR spectrum is consistent with the reported data.

5. Reduction of quinolines and imines

General procedure for the reduction of quinolines (Take 4b as an example)

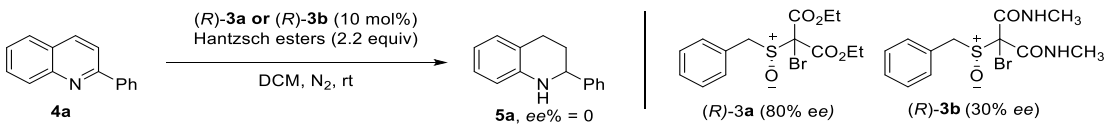
To a solution of **4b** (64 mg**,** 1.0 equiv., 0.25 mmol), **3b** (4.3 mg, 5 mol %, 0.0125 mmol) and Hantzsch esters (140 mg, 2.2 equiv., 0.55 mmol) was added 2.5 mL of DCM under nitrogen atmosphere and the mixture was stirred at room temperature for 24 h**.** After the reaction was completed, solvent was removed under vacuum. The residue was purified by silica gel flash column chromatography (petroleum ether:ethyl acetate = 80:1) to give target product **5b** (white solid, 64 mg, 99% yield).

General procedure for the reduction of 4a in controlled experiments

The reactions were performed on 0.05 mmol scale with TEMPO (1.95 mg, 0.25 equiv., 0.0125 mmol), BHT (5.5 mg, 0.5 equiv., 0.025 mmol), 3,5-lutidine (0.3 μL, 0.05 equiv., 0.0025 mmol), 2,6-di-tert-butylpyridine (0.56 μL, 0.05 equiv., 0.0025 mmol) and **3j** (1.87 mg, 0.1 equiv., 0.005 mmol). The reaction solvent (0.5 mL of DCM) was quickly removed by vacuum pump at ambient temperature at certain time and then $CDCl_3$ (0.5 mL) was added to the residue. The resulting mixture was immediately transferred to NMR tube to run ^1H NMR analysis and the reaction yield was then determined accordingly.

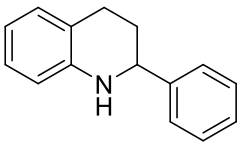
General procedure for the reduction of imines (Take 4s as an example)

To a solution of **4s** (24.9 mg, 1.0 equiv., 0.1 mmol), **3b** (1.7 mg, 5 mol %, 0.005 mmol) and Hantzsch esters (30.4 mg, 1.2 equiv., 0.12 mmol) was added 1 mL of DCM and the mixture was stirred at room temperature for 24 h. After the reaction was completed, solvent was removed under vacuum. The residue was purified by silica gel flash column chromatography (petroleum ether : ethyl acetate = 80: 1) to give target product **5s** (colorless oil, 23 mg, 92% yield).



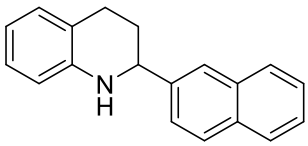
Scheme S10. Transfer hydrogenation of 4a by using enantioenriched 3a and 3b.

No enantioselectivity was observed in the transfer hydrogenation of **4a** using enantioenriched (*R*)-**3a** (80% *ee*) and (*R*)-**3b** (30% *ee*).



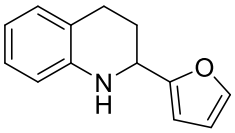
2-phenyl-1,2,3,4-tetrahydroquinoline (5a)[7]

Colorless oil, 52 mg, 99% yield. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 7.43 – 7.39 (m, 2H), 7.36 (t, *J* = 7.6 Hz, 2H), 7.33 – 7.28 (m, 1H), 7.02 (t, *J* = 7.2 Hz, 2H), 6.72 – 6.66 (m, 1H), 6.57 (d, *J* = 8.3 Hz, 1H), 4.45 (dd, *J* = 9.4, 3.3 Hz, 1H), 2.94 (ddd, *J* = 16.2, 10.7, 5.5 Hz, 1H), 2.75 (dt, *J* = 16.3, 4.8 Hz, 1H), 2.19 – 2.10 (m, 1H), 2.10 – 1.98 (m, 1H). The ^{1}H NMR spectrum is consistent with the reported data.



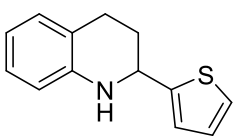
2-(naphthalen-2-yl)-1,2,3,4-tetrahydroquinoline (5b)[7]

White solid, 64 mg, 99% yield. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 7.85 (ddd, *J* = 9.7, 5.4, 2.3 Hz, 4H), 7.56 – 7.46 (m, 3H), 7.04 (t, *J* = 6.9 Hz, 2H), 6.70 (td, *J* = 7.4, 1.2 Hz, 1H), 6.61 (dd, *J* = 8.2, 1.2 Hz, 1H), 4.62 (dd, *J* = 9.4, 3.3 Hz, 1H), 2.98 (ddd, *J* = 16.2, 10.7, 5.4 Hz, 1H), 2.79 (dt, *J* = 16.3, 4.7 Hz, 1H), 2.21 (ddd, *J* = 13.2, 8.3, 5.1 Hz, 1H), 2.11 (ddt, *J* = 18.2, 9.4, 5.0 Hz, 1H). The ^{1}H NMR spectrum is consistent with the reported data.



2-(furan-2-yl)-1,2,3,4-tetrahydroquinoline (5c)[7]

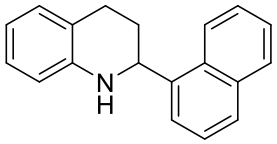
Colorless oil, 49 mg, 99% yield. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 7.38 (d, *J* = 1.9 Hz, 1H), 7.09 – 6.97 (m, 2H), 6.68 (td, *J* = 7.4, 1.1 Hz, 1H), 6.58 (d, *J* = 7.9 Hz, 1H), 6.34 (dd, *J* = 3.2, 1.9 Hz, 1H), 6.22 (d, *J* = 3.3 Hz, 1H), 4.55 (dd, *J* = 8.4, 3.5 Hz, 1H), 2.87 (ddd, *J* = 15.5, 9.4, 5.5 Hz, 1H), 2.76 (dt, *J* = 16.3, 5.5 Hz, 1H), 2.22 (dtd, *J* = 12.8, 5.6, 3.4 Hz, 1H), 2.19 – 2.12 (m, 1H). The ^{1}H NMR spectrum is consistent with the reported data.



2-(thiophen-2-yl)-1,2,3,4-tetrahydroquinoline (5d)[11]

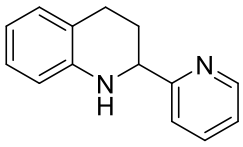
Colorless oil, 56 mg, 99% yield. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 7.24 (dd, *J* = 5.0, 1.3 Hz, 1H), 7.06 – 6.98 (m, 4H), 6.70 (t, *J* = 7.4 Hz, 1H), 6.57 (d, *J* = 7.9 Hz, 1H), 4.77 (dd, *J* = 9.1, 3.2 Hz, 1H), 4.27 (s, 1H), 2.95 (ddd, *J* = 16.2, 10.5, 5.5 Hz, 1H), 2.80 (dt, *J* = 16.4, 5.0 Hz, 1H), 2.24 (dtd, *J* = 13.3, 5.1, 3.2

Hz, 1H), 2.16 – 2.06 (m, 1H). The ^{1}H NMR spectrum is consistent with the reported data.



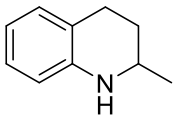
2-(naphthalen-1-yl)-1,2,3,4-tetrahydroquinoline (5e)[11]

White solid, 64 mg, 99% yield. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 8.17 (d, *J* = 8.3 Hz, 1H), 7.92 (dd, *J* = 7.9, 1.6 Hz, 1H), 7.81 (d, *J* = 8.2 Hz, 1H), 7.69 (d, *J* = 7.1 Hz, 1H), 7.61 – 7.45 (m, 3H), 7.14 – 7.05 (m, 2H), 6.72 (td, *J* = 7.4, 1.2 Hz, 1H), 6.64 (d, *J* = 7.8 Hz, 1H), 5.30 (dd, *J* = 8.5, 3.5 Hz, 1H), 4.33 (s, 1H), 3.00 (ddd, *J* = 15.7, 9.9, 5.3 Hz, 1H), 2.77 (dt, *J* = 16.3, 5.2 Hz, 1H), 2.43 – 2.31 (m, 1H), 2.26 – 2.16 (m, 1H). The ^{1}H NMR spectrum is consistent with the reported data.



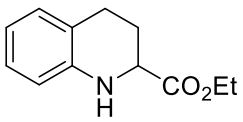
2-(pyridin-2-yl)-1,2,3,4-tetrahydroquinoline (5f)[7]

White solid, 50 mg, 95% yield. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 8.58 (d, *J* = 4.8 Hz, 1H), 7.70 (td, *J* = 7.7, 1.8 Hz, 1H), 7.45 (d, *J* = 7.9 Hz, 1H), 7.21 (ddd, *J* = 7.6, 4.9, 1.2 Hz, 1H), 7.08 – 6.96 (m, 2H), 6.72 – 6.61 (m, 2H), 4.62 (dd, *J* = 8.7, 3.6 Hz, 1H), 2.92 (ddd, *J* = 15.6, 9.8, 5.3 Hz, 1H), 2.69 (dt, *J* = 16.2, 5.3 Hz, 1H), 2.27 (dtd, *J* = 12.8, 5.4, 3.6 Hz, 1H), 2.05 (dddd, *J* = 13.2, 9.8, 8.7, 4.9 Hz, 1H). The ^{1}H NMR spectrum is consistent with the reported data.



2-methyl-1,2,3,4-tetrahydroquinoline (5g)[7]

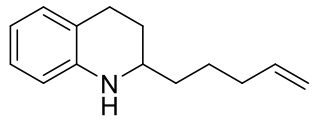
Colorless oil, 36 mg, 98% yield. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 6.98 (t, *J* = 6.6 Hz, 2H), 6.63 (td, *J* = 7.3, 1.2 Hz, 1H), 6.50 (dd, *J* = 8.3, 1.2 Hz, 1H), 3.41 (ddq, *J* = 12.6, 6.2, 3.2 Hz, 1H), 2.85 (ddd, *J* = 17.4, 11.6, 5.7 Hz, 1H), 2.79 – 2.70 (m, 1H), 1.95 (ddt, *J* = 12.3, 6.0, 3.2 Hz, 1H), 1.61 (dddd, *J* = 12.9, 11.7, 9.9, 5.3 Hz, 1H), 1.23 (d, *J* = 6.3 Hz, 3H). The ^{1}H NMR spectrum is consistent with the reported data.



ethyl 1,2,3,4-tetrahydroquinoline-2-carboxylate (5h)[12]

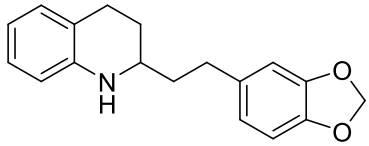
Colorless oil, 49 mg, 97% yield. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 7.06 – 6.95 (m, 2H), 6.70 – 6.59 (m, 2H), 4.28 – 4.20 (m, 2H), 4.03 (dd, *J* = 9.0, 3.7 Hz, 1H), 2.88 – 2.73 (m, 2H), 2.30 (dtd, *J* = 13.0, 5.5, 3.7 Hz, 1H),

2.00 (dtd, *J* = 14.4, 9.3, 5.1 Hz, 1H), 1.30 (t, *J* = 7.1 Hz, 3H). The ^{1}H NMR spectrum is consistent with the reported data.



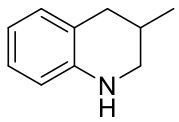
2-(pent-4-en-1-yl)-1,2,3,4-tetrahydroquinoline (5i)[7]

Colorless oil, 49 mg, 99% yield. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 6.97 (t, *J* = 7.7 Hz, 2H), 6.63 (t, *J* = 7.6 Hz, 1H), 6.53 (d, *J* = 7.8 Hz, 1H), 5.83 (ddt, *J* = 16.9, 10.1, 6.6 Hz, 1H), 5.04 (dq, *J* = 17.1, 1.8 Hz, 1H), 4.98 (dt, *J* = 10.2, 1.6 Hz, 1H), 3.26 (ddt, *J* = 9.2, 6.1, 2.8 Hz, 1H), 2.86 – 2.72 (m, 2H), 2.11 (t, *J* = 6.7 Hz, 2H), 1.98 (ddt, *J* = 12.7, 5.5, 3.4 Hz, 1H), 1.63 (dddd, *J* = 12.8, 10.8, 9.5, 5.3 Hz, 1H), 1.57 – 1.50 (m, 4H). The ^{1}H NMR spectrum is consistent with the reported data.



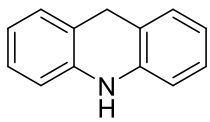
2-(2-(benzo[d][1,3]dioxol-5-yl)ethyl)-1,2,3,4-tetrahydroquinoline (5j)[7]

White solid, 69 mg, 99% yield. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 6.97 (t, *J* = 7.2 Hz, 2H), 6.77 – 6.69 (m, 2H), 6.69 – 6.62 (m, 2H), 6.54 (d, *J* = 8.0 Hz, 1H), 5.93 (s, 2H), 3.30 (dtd, *J* = 9.4, 6.4, 3.0 Hz, 1H), 2.82 (ddd, *J* = 16.3, 10.6, 5.6 Hz, 1H), 2.75 (dt, *J* = 16.4, 4.9 Hz, 1H), 2.67 (t, *J* = 7.9 Hz, 2H), 2.01 (dddd, *J* = 13.0, 5.6, 4.4, 2.9 Hz, 1H), 1.87 – 1.78 (m, 2H), 1.69 (dddd, *J* = 12.9, 10.6, 9.4, 5.4 Hz, 1H). The ^{1}H NMR spectrum is consistent with the reported data.



3-methyl-1,2,3,4-tetrahydroquinoline (5k)[7]

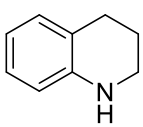
Colorless oil, 36 mg, 97% yield. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 6.98 (dd, *J* = 18.0, 7.4 Hz, 2H), 6.63 (td, *J* = 7.4, 1.2 Hz, 1H), 6.51 (dd, *J* = 7.9, 1.2 Hz, 1H), 3.28 (ddd, *J* = 11.0, 3.7, 2.0 Hz, 1H), 2.91 (dd, *J* = 11.0, 9.7 Hz, 1H), 2.86 – 2.75 (m, 1H), 2.45 (dd, *J* = 16.0, 10.3 Hz, 1H), 2.08 (tddd, *J* = 10.1, 6.8, 4.9, 3.7 Hz, 1H), 1.06 (d, *J* = 6.6 Hz, 3H). The ^{1}H NMR spectrum is consistent with the reported data.



9,10-dihydroacridine (5l)[7]

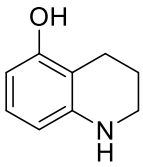
White solid, 43 mg, 94% yield. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 7.12 (ddd, *J* = 17.1, 7.4, 1.4 Hz, 4H), 6.89 (td, *J* = 7.4, 1.2 Hz, 2H),

6.68 (dd, *J* = 7.9, 1.2 Hz, 2H), 5.96 (s, 1H), 4.08 (s, 2H). The ^{1}H NMR spectrum is consistent with the reported data.



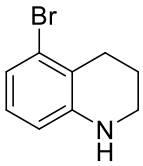
1,2,3,4-tetrahydroquinoline (5m)[7]

Colorless oil, 28 mg, 78% yield. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 6.97 (ddd, *J* = 13.4, 6.9, 1.5 Hz, 2H), 6.62 (td, *J* = 7.4, 1.2 Hz, 1H), 6.50 (d, *J* = 7.9 Hz, 1H), 3.33 – 3.29 (m, 2H), 2.78 (t, *J* = 6.4 Hz, 2H), 1.99 – 1.93 (m, 2H). The ^{1}H NMR spectrum is consistent with the reported data.



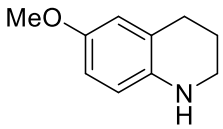
1,2,3,4-tetrahydroquinolin-5-ol (5n)[7]

Black oil, 23 mg, 62% yield. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 6.83 (t, *J* = 7.9 Hz, 1H), 6.12 (dd, *J* = 8.0, 3.2 Hz, 2H), 3.31 – 3.24 (m, 2H), 2.65 (t, *J* = 6.5 Hz, 2H), 1.99 – 1.94 (m, 2H). The ^{1}H NMR spectrum is consistent with the reported data.



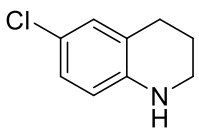
5-bromo-1,2,3,4-tetrahydroquinoline (5o)[7]

Colorless oil, 45 mg, 85% yield. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 6.86 (dd, *J* = 7.9, 1.2 Hz, 1H), 6.81 (t, *J* = 7.9 Hz, 1H), 6.42 (dd, *J* = 8.0, 1.2 Hz, 1H), 3.30 – 3.23 (m, 2H), 2.76 (t, *J* = 6.6 Hz, 2H), 1.99 – 1.93 (m, 2H). The ^{1}H NMR spectrum is consistent with the reported data.



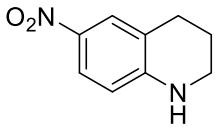
6-methoxy-1,2,3,4-tetrahydroquinoline (5p)[13]

Colorless oil, 31 mg, 78% yield. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 6.62 – 6.56 (m, 2H), 6.49 (d, *J* = 8.6 Hz, 1H), 3.73 (s, 3H), 3.29 – 3.23 (m, 2H), 2.76 (t, *J* = 6.5 Hz, 2H), 2.00 – 1.89 (m, 2H). The ^{1}H NMR spectrum is consistent with the reported data.



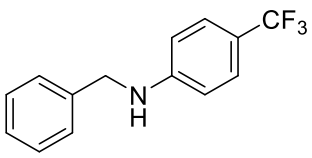
6-chloro-1,2,3,4-tetrahydroquinoline (5q)[7]

Colorless oil, 33 mg, 79% yield. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 6.97 – 6.86 (m, 2H), 6.39 (d, *J* = 8.3 Hz, 1H), 3.32 – 3.26 (m, 2H), 2.72 (t, *J* = 6.4 Hz, 2H), 1.96 – 1.88 (m, 2H). The ^{1}H NMR spectrum is consistent with the reported data.



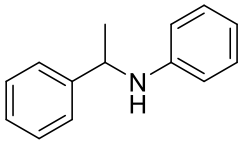
6-nitro-1,2,3,4-tetrahydroquinoline (5r)[13]

Yellow solid, 22 mg, 50% yield. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 7.92 – 7.87 (m, 2H), 6.41 (d, *J* = 9.1 Hz, 1H), 3.42 (dd, *J* = 6.6, 4.8 Hz, 2H), 2.80 (t, *J* = 6.3 Hz, 2H), 1.99 – 1.94 (m, 2H). The ^{1}H NMR spectrum is consistent with the reported data.



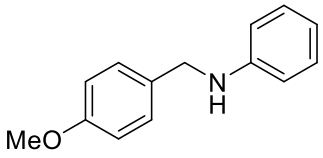
_N_-benzyl-4-(trifluoromethyl)aniline (5s)[14]

Colorless oil, 23 mg, 92% yield. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 7.41 (d, *J* = 8.4 Hz, 2H), 7.36 (d, *J* = 5.3 Hz, 4H), 7.33 – 7.28 (m, 1H), 6.66 (d, *J* = 8.4 Hz, 2H), 4.37 (s, 2H). The ^{1}H NMR spectrum is consistent with the reported data.



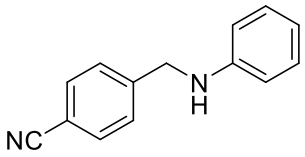
_N_-(1-phenylethyl)aniline (5t)[15]

Colorless oil, 18 mg, 91% yield. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 7.40 – 7.36 (m, 2H), 7.32 (t, *J* = 7.7 Hz, 2H), 7.26 – 7.21 (m, 1H), 7.12 – 7.08 (m, 2H), 6.67 (tt, *J* = 7.3, 1.1 Hz, 1H), 6.58 – 6.52 (m, 2H), 4.49 (q, *J* = 6.7 Hz, 1H), 1.54 (d, *J* = 6.7 Hz, 3H). The ^{1}H NMR spectrum is consistent with the reported data.



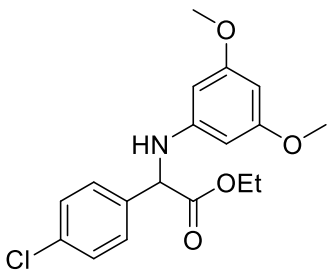
_N_-(4-methoxybenzyl)aniline (5u)[16]

Colorless oil, 21 mg, 99% yield. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 7.32 – 7.28 (m, 2H), 7.22 – 7.16 (m, 2H), 6.91 – 6.87 (m, 2H), 6.74 (td, *J* = 7.3, 1.1 Hz, 1H), 6.69 – 6.65 (m, 2H), 4.26 (s, 2H), 3.81 (s, 3H). The ^{1}H NMR spectrum is consistent with the reported data.



4-((phenylamino)methyl)benzonitrile (5v)[17]

Colorless oil, 21 mg, 99% yield. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 7.66 – 7.60 (m, 2H), 7.48 (d, *J* = 8.0 Hz, 2H), 7.21 – 7.16 (m, 2H), 6.77 (tt, *J* = 7.4, 1.1 Hz, 1H), 6.64 – 6.59 (m, 2H), 4.43 (s, 2H). The ^{1}H NMR spectrum is consistent with the reported data.



ethyl 2-(4-chlorophenyl)-2-((3,5-dimethoxyphenyl)amino)acetate (5w)

Yellow oil,15 mg, 43% yield. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 7.45 – 7.40 (m, 2H), 7.34 – 7.29 (m, 2H), 5.87 (s, 1H), 5.73 (s, 2H), 5.00 (s, 1H), 4.23 (dq, *J* = 10.7, 7.1 Hz, 1H),

4.14 (dq, *J* = 10.8, 7.1 Hz, 1H), 3.69 (s, 6H), 1.21 (t, *J* = 7.1 Hz, 3H). ^{13}C NMR (151 MHz, $CDCl_3$): δ = 171.25, 161.73, 147.39, 136.23, 134.23, 129.16, 128.66, 92.72, 90.77, 62.27, 60.36, 55.26, 14.15. IR: 3396, 2935, 1735, 1616, 1597, 1203, 1153, 811 cm^{-1}; HRMS (ESI): *m/z* calcd. For $C_{18}H_{21}ClNO_4$ $[M+H]^+$ 350.1154, found 350.1160.

6. NMR studies

^{1}H NMR spectra of 4a in the presence of catalysts

General procedure: A solution of 2-phenylquinoline (0.01 mmol, 1.0 equiv.) in 0.6 mL $CDCl_3$ was added to a NMR tube to give standard solution of 2-phenylquinoline. To six NMR tubes containing standard solution of 2-phenylquinoline was added **3a, 2a, 3h, 3b, 2b**, **3i** (0.01 mmol, 1.0 equiv.) respectively. ^{1}H NMR spectrum of each sample was recorded at room temperature. Chemical shift changes ($\Delta\delta(^1H)$(ppm)) of C_4–H in 2-phenylquinoline[18] was accordingly calculated by comparing with the chemical shift of 2-phenylquinoline solution itself (0.01 mmol in 0.6 mL of $CDCl_3$).

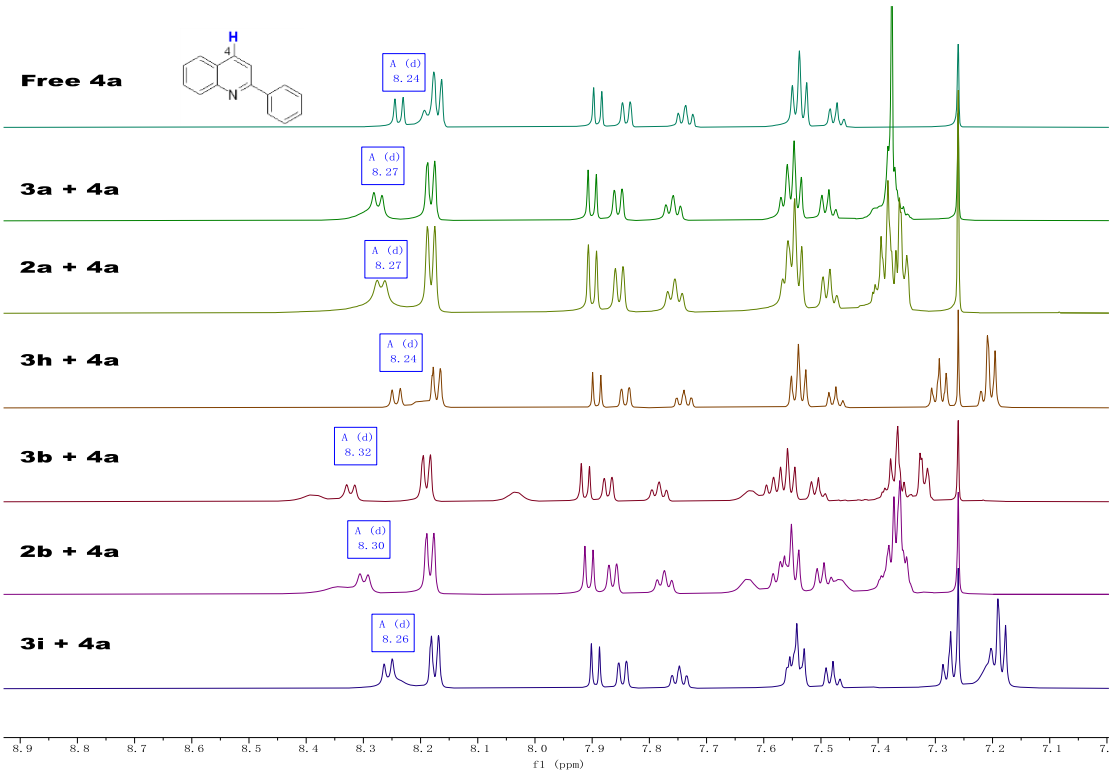


Figure S2. ^{1}H NMR spectra of 4a after binding with catalysts.

Table S2. Chemical shift changes of C_4–H in 4a with catalysts.

Entry	Catalyst	$\Delta\delta(^1H)$(ppm)[a]
1	**3a**	0.03
2	**2a**	0.03
3	**3h**	0.00
4	**3b**	0.08
5	**2b**	0.06
6	**3i**	0.02

[a] $\Delta\delta(^1H) = \delta$ (**4a• Catalyst**) - δ (free **4a**)

^{31}P NMR spectra of TMPO in the presence of catalysts[2]

General procedure: TMPO (0.035 mmol) was dissolved in 0.7 mL of DCM to prepare the TMPO stock solution with concentration of 0.05 M. To six NMR tubes containing **3a, 2a, 3h, 3b, 2b** and **3i** (5 equiv., 0.025 mmol) in mixed solvent of 0.3 mL of DCM and 0.2 mL of CD_2Cl_2 was added aliquots of **TMPO** stock solution (0.1 mL) respectively. The ^{31}P NMR spectrum of each sample was then recorded at room temperature. Chemical shift changes ($\Delta\delta(^{31}P)$(ppm)) of **TMPO** was accordingly calculated by comparing with the chemical shift of **TMPO** itself (8.3 mM in 0.4 mL of DCM and 0.2 mL of CD_2Cl_2).

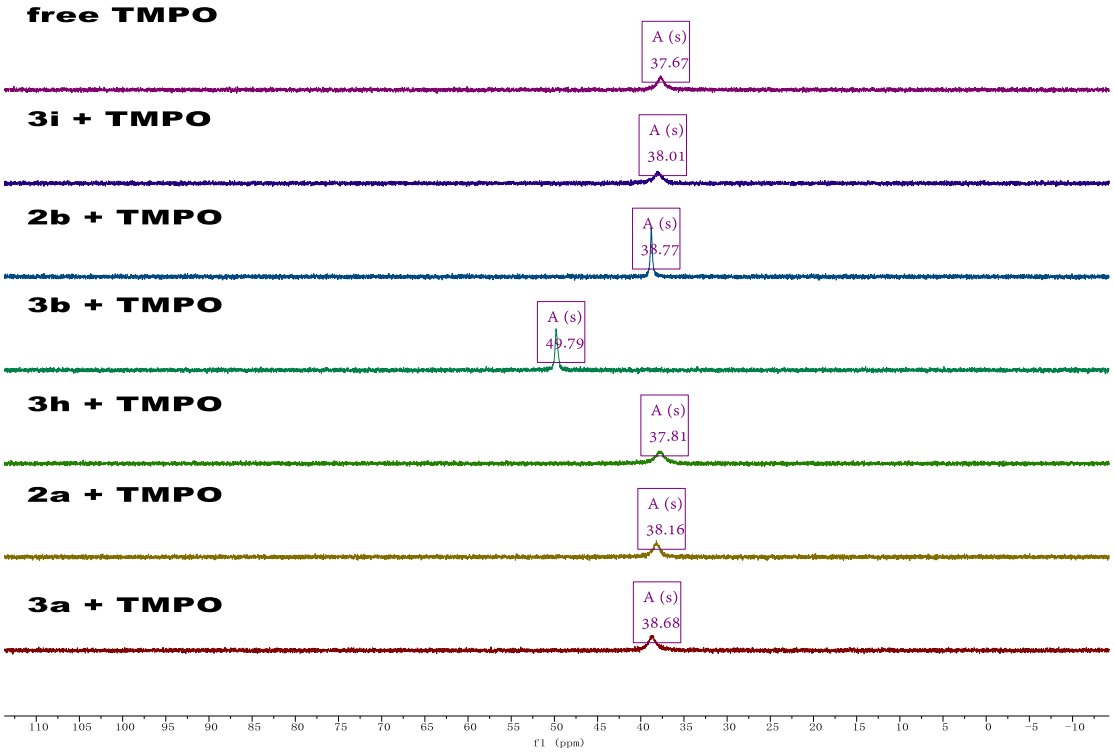


Figure S3. ^{31}P NMR spectra of TMPO after binding with catalysts:

Table S3. Chemical shift changes of TMPO with catalysts.

Entry	Catalyst	$\Delta\delta(^{31}P)$(ppm)[a]
1	**3a**	1.01
2	**2a**	0.49
3	**3h**	0.14
4	**3b**	12.12
5	**2b**	1.1
6	**3i**	0.34

[a] $\Delta\delta(^{31}P) = \delta$(**TMPO• Catalyst**) - δ(free **TMPO**)

^{31}P NMR spectra of TEPO in the presence of catalysts [2]

General procedure: TEPO (0.083 mmol) was dissolved in 1 mL of DCM to prepare the **TEPO** stock solution with concentration of 0.083 M. To four NMR tubes containing **3a, 3b, 3f, 3g** (0.083 mmol, 10 equiv.) in mixed solvent of 0.3 mL of DCM and 0.2 mL of CD_2Cl_2 were added aliquots of **TEPO** stock solution (0.1 mL) respectively. The ^{31}P NMR spectrum of each sample was then recorded at room temperature. Chemical shift changes ($\Delta\delta(^{31}P)$(ppm)) of **TEPO** was accordingly calculated by comparing with the chemical shift of TEPO itself (13.8 mM in 0.4 mL of DCM and 0.2 mL of CD_2Cl_2). The ^{31}P NMR study of **2b** (0.042 mmol, 10 equiv.) and **TEPO** (0.083 M, 0.05 mL, 1.0 equiv.) in 0.35 mL of DCM and 0.2 mL of CD_2Cl_2 was also conducted to exclude the interference of the extra hydrogen bond sites from the amino group.

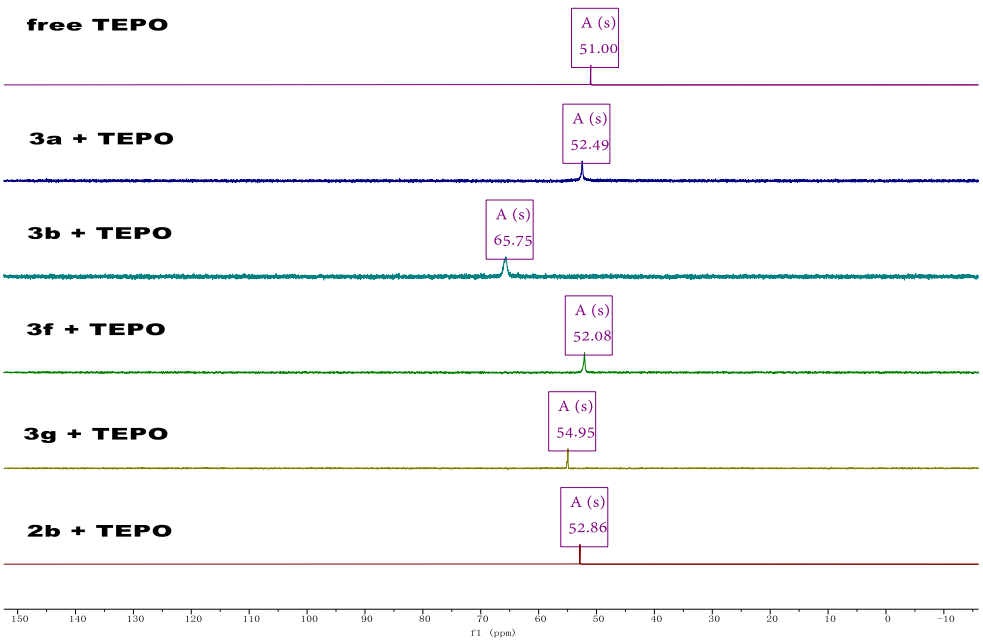


Figure S4. ^{1}H NMR spectra of TEPO after binding with catalyst.

Table S4. Chemical shift changes of TEPO with catalysts.

Entry	Catalyst	$\Delta\delta(^{31}P)$(ppm)[a]
1	**3a**	1.49
2	**3b**	14.75
3	**3f**	1.08
4	**3g**	3.95
5	**2b**	1.95

[a] $\Delta\delta(^{31}P) = \delta$ (**TEPO• Catalyst**) - δ (free **TEPO**)

2D NOESY spectrum of 2b

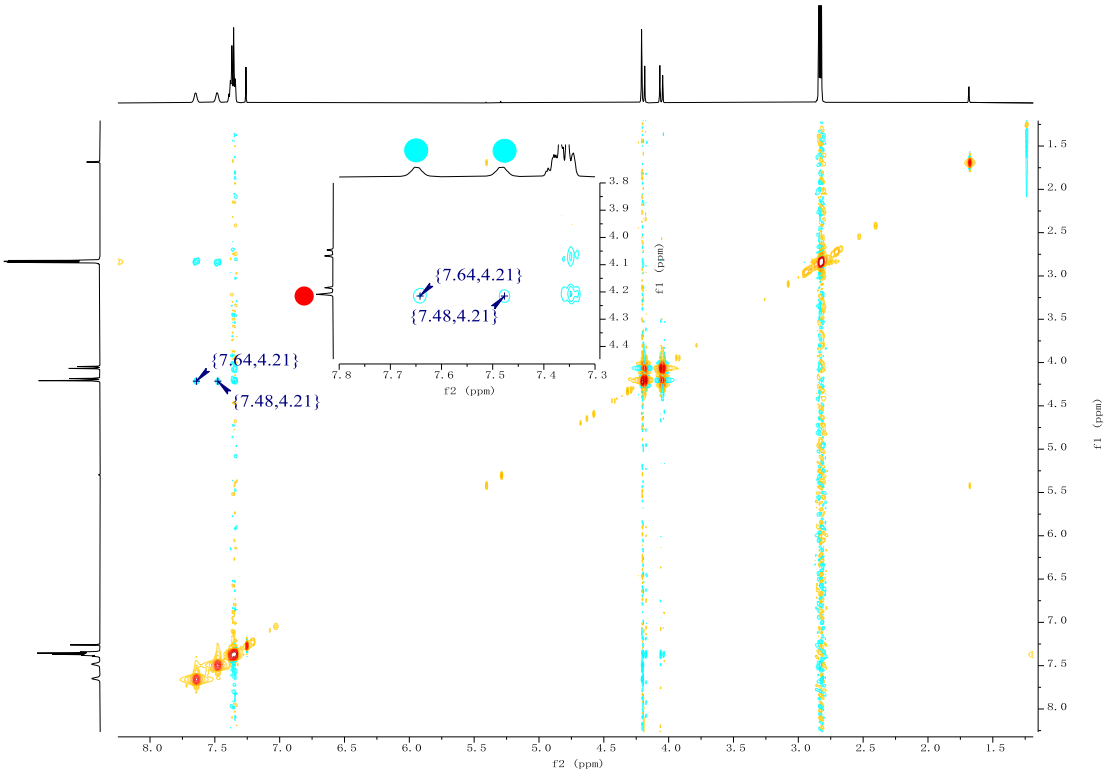


Figure S5. NOESY spectrum of 2b in $CDCl_3$ (0.046 M, 600MHz)

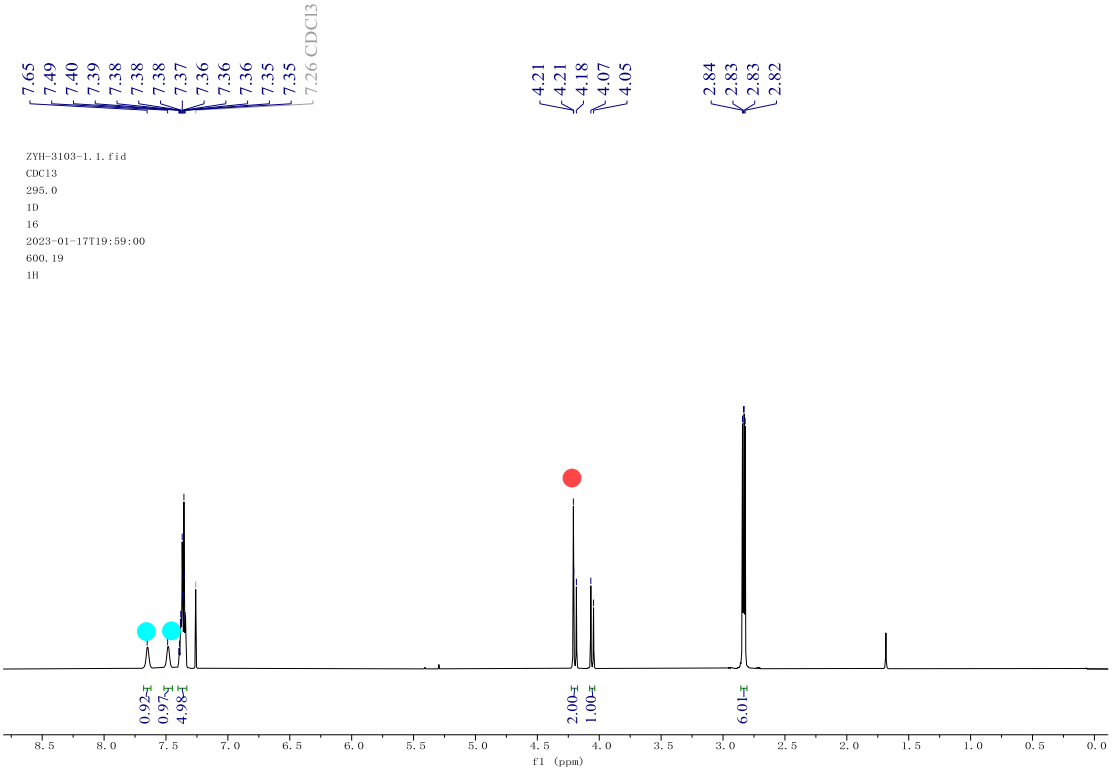


Figure S6. ^{1}H NMR spectrum of 2b in $CDCl_3$ (0.046 M)

For the NMR spectra of **2b**, the resonances of amide protons N–H were marked with blue dots and methine C–H was marked with red dot in Figure S5, S6. NOE signals were detected in the NOESY spectrum of **2b** (Figure S5). Indeed, through-space correlations appeared between the methine proton (C(sp^3)–H, 4.21 ppm) and each of two amide protons (7.64, 7.48 ppm) respectively, but that was not observed between these two amide protons. Therefore, it could be imagined that the methine proton should located in betwixt two amide protons. Based on the NOESY spectrum and structural features of amide, we speculated a possible conformation of **2b** (Figure S7), in which intramolecular hydrogen bonds of N–H···O=C and another N–H···O=S may occur.

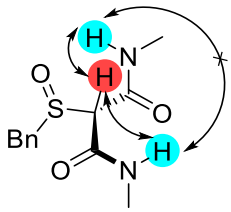


Figure S7. Detected NOE and proposed conformation of 2b

Stacked ^{1}H NMR spectra of 2b and 3b

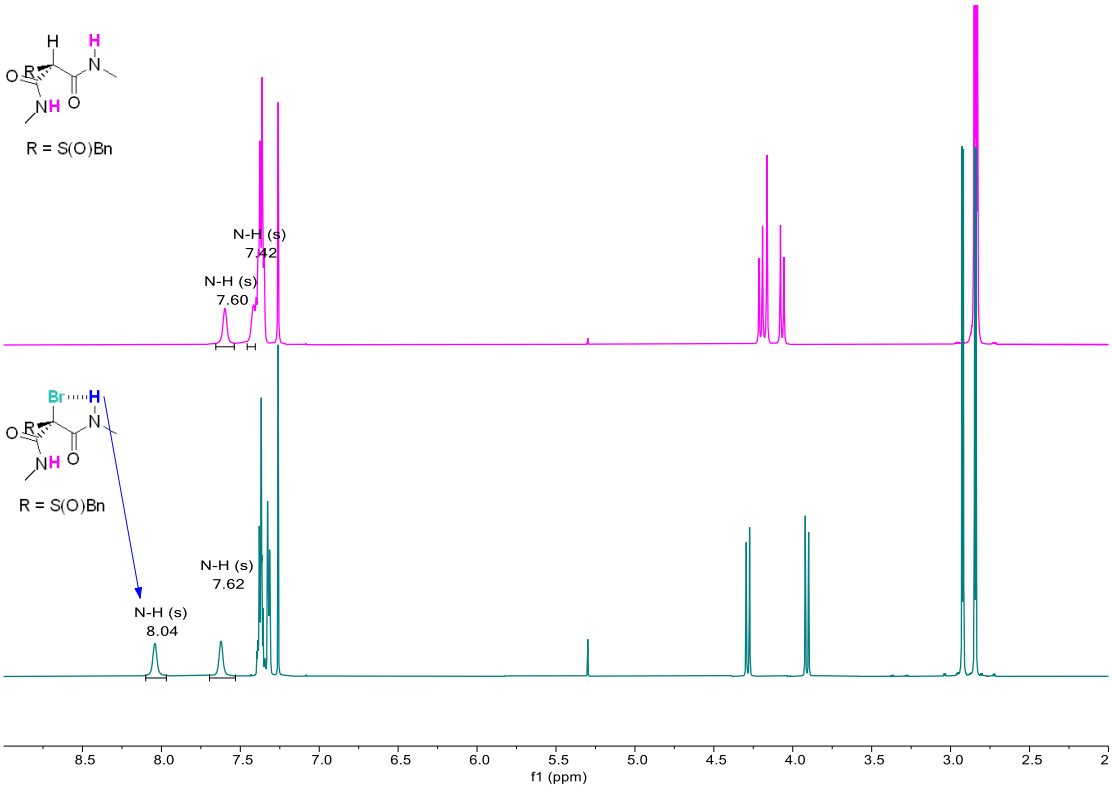


Figure S8. Stacked ^{1}H NMR spectra of 2b and 3b (0.02 M, $CDCl_3$)

^{1}H NMR spectra of 2b and 3b at different concentrations

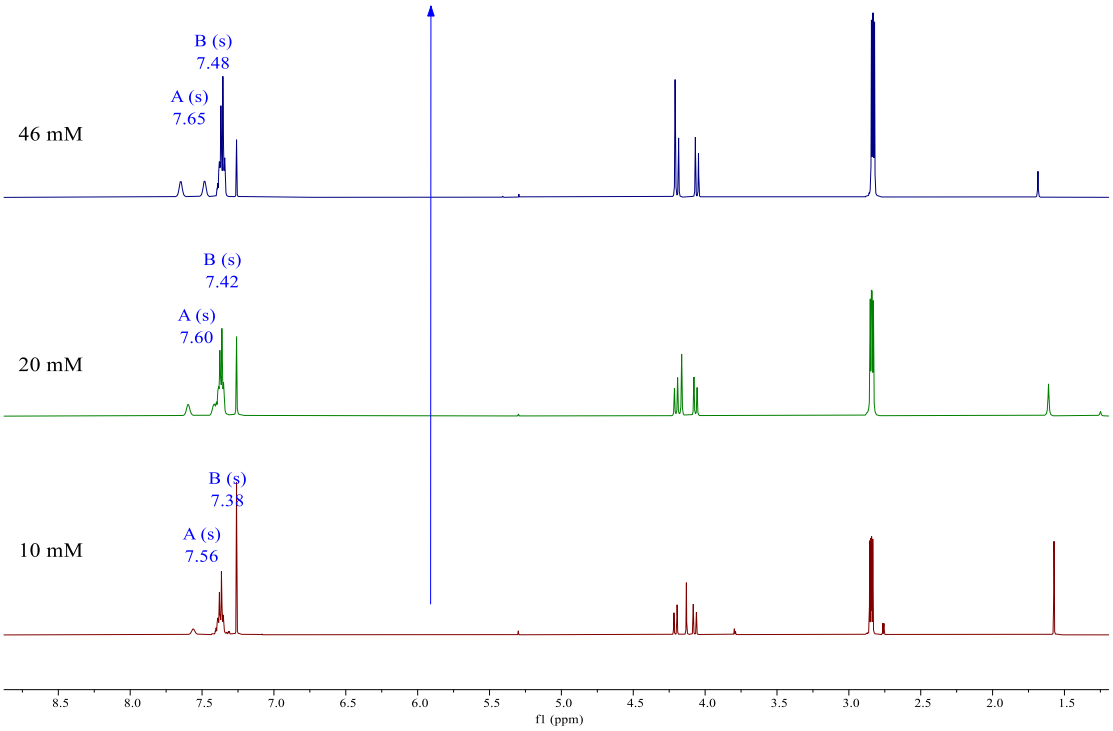


Figure S9. ^{1}H NMR spectra of 2b at different concentrations. From top to bottom, the concentrations of **2b** were 46 mM, 20 Mm and 10 mM in $CDCl_3$. The peaks of amide protons were picked. The large downfield of the NH chemical shift with increasing concentration indicated the intermolecular hydrogen bonds in **2b** with the studied concentrations.

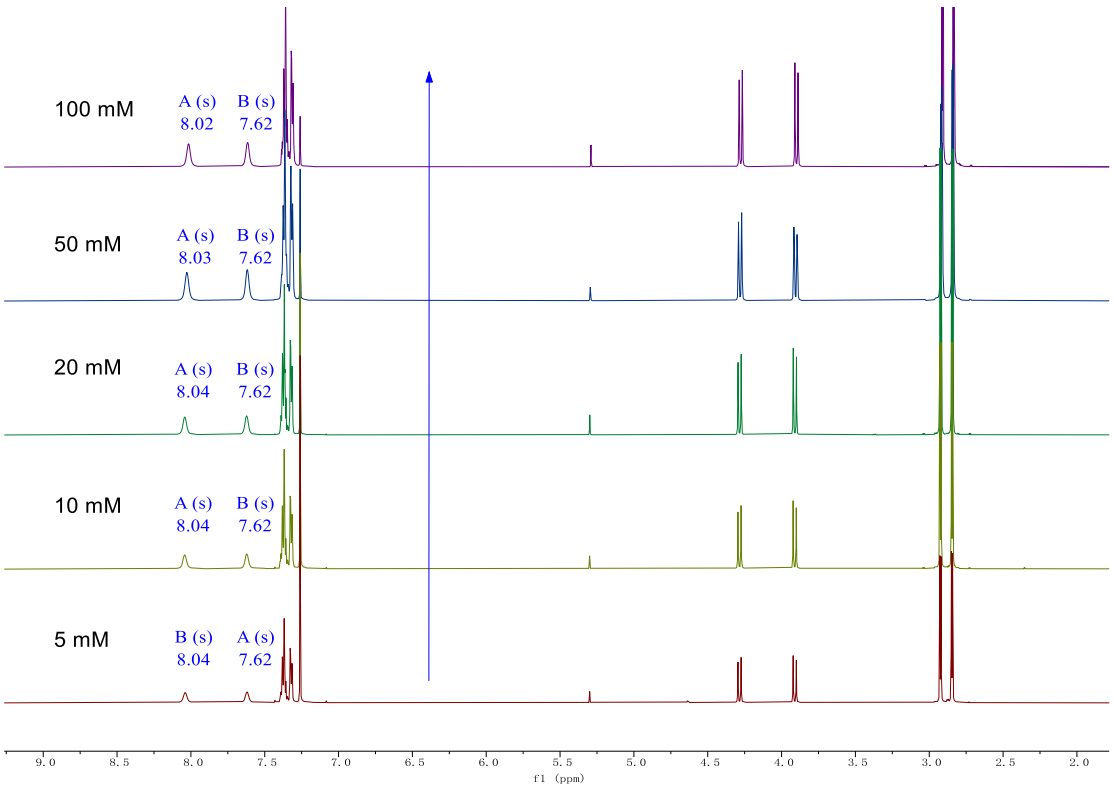


Figure S10 ^{1}H NMR spectra of 3b at different concentrations. From top to bottom, the concentrations of **3b** were 100 mM, 50 Mm, 20 mM, 10 mM and 5 mM in $CDCl_3$. The peaks of amide protons were picked. The independence of the NH chemical shift with increasing concentration indicated that no intermolecular hydrogen bonding in **3b** with the studied concentrations.

NOESY NMR studies of 3b

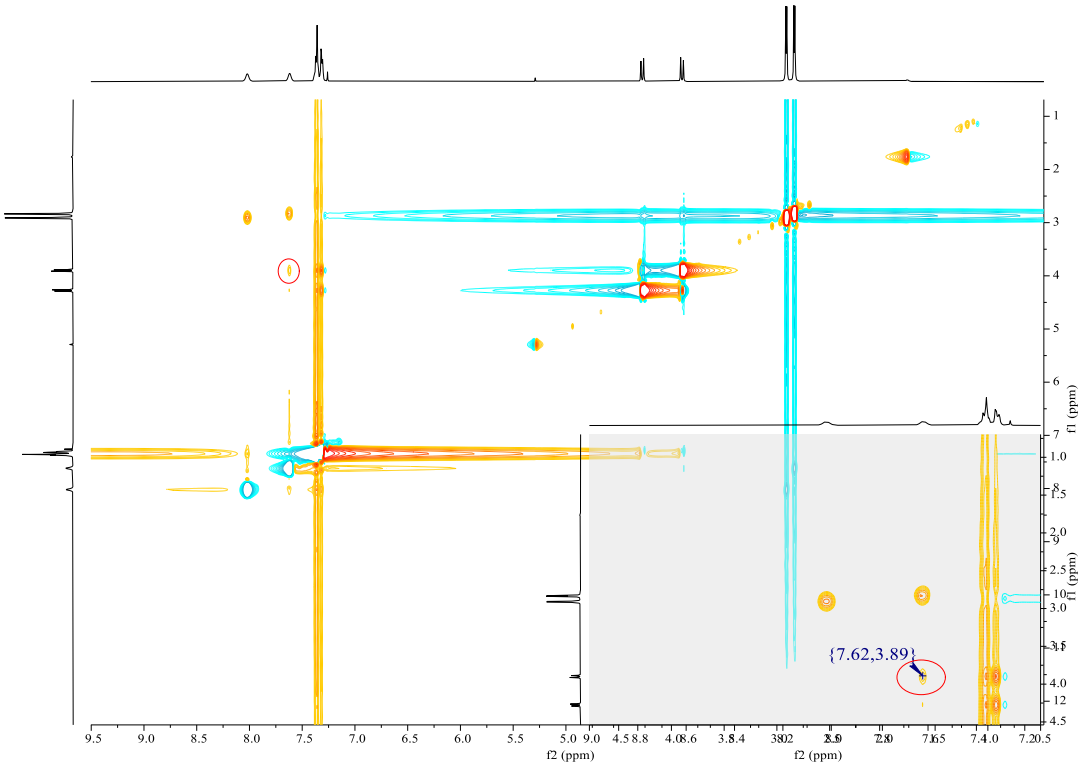


Figure S11 2D NOESY spectra of 3b in $CDCl_3$ (0.1 M, 500 MHz). The NOE signal in red dash cycle was between amide proton (δ = 7.62 ppm) and benzyl methylene proton in **3b**

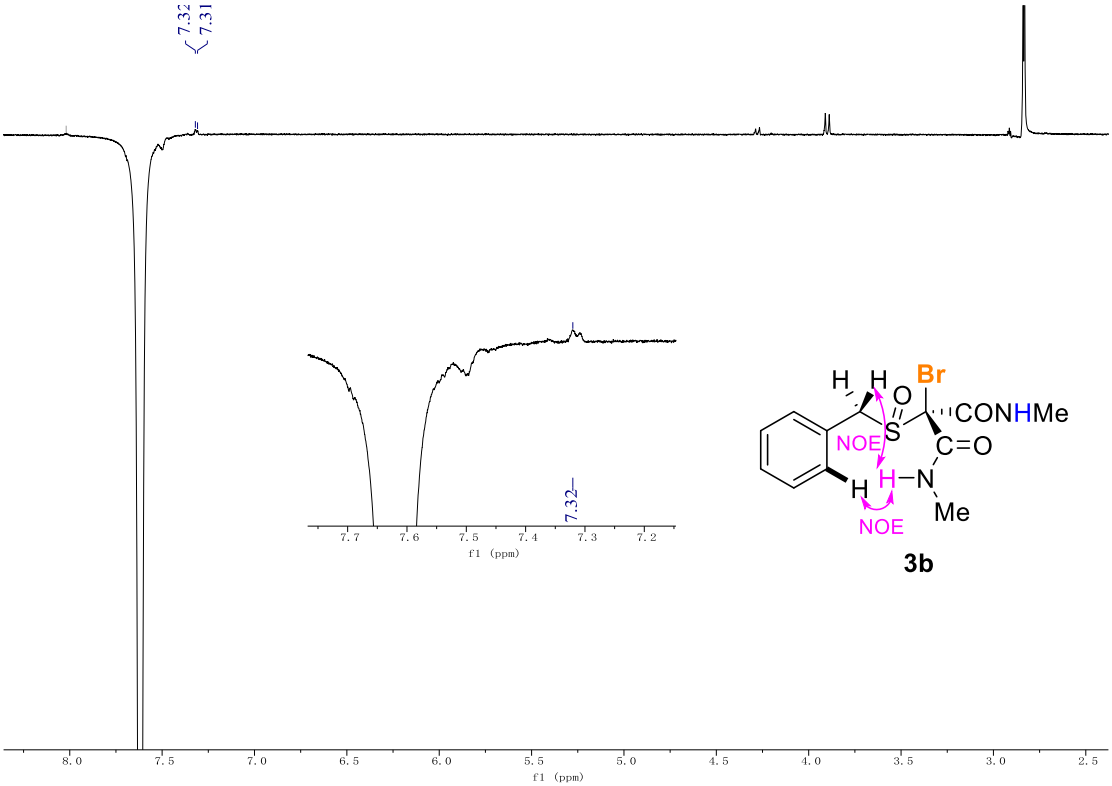


Figure S12. 1D NOESY spectra of 3b in $CDCl_3$ (0.1 M, 600 MHz). The picked NOE signal was between amide proton (δ = 7.62 ppm) and *ortho*-proton of phenyl ring in **3b**.

^{1}H NMR studies of 3b and 2b with adding TMPO

General produce: A solution of **2b** or **3b** (0.01 mmol, 1.0 equiv.) in 0.5 mL $CDCl_3$ was added to a NMR tube to give standard solution of **2b** or **3b**. To NMR tubes containing standard solution was added varying amounts of **TMPO** respectively (the resulting molar ration of **2b** or **3b** : **TMPO** was indicated in the resulting ^{1}H NMR spectra). ^{1}H NMR spectrum of each sample was recorded at room temperature.

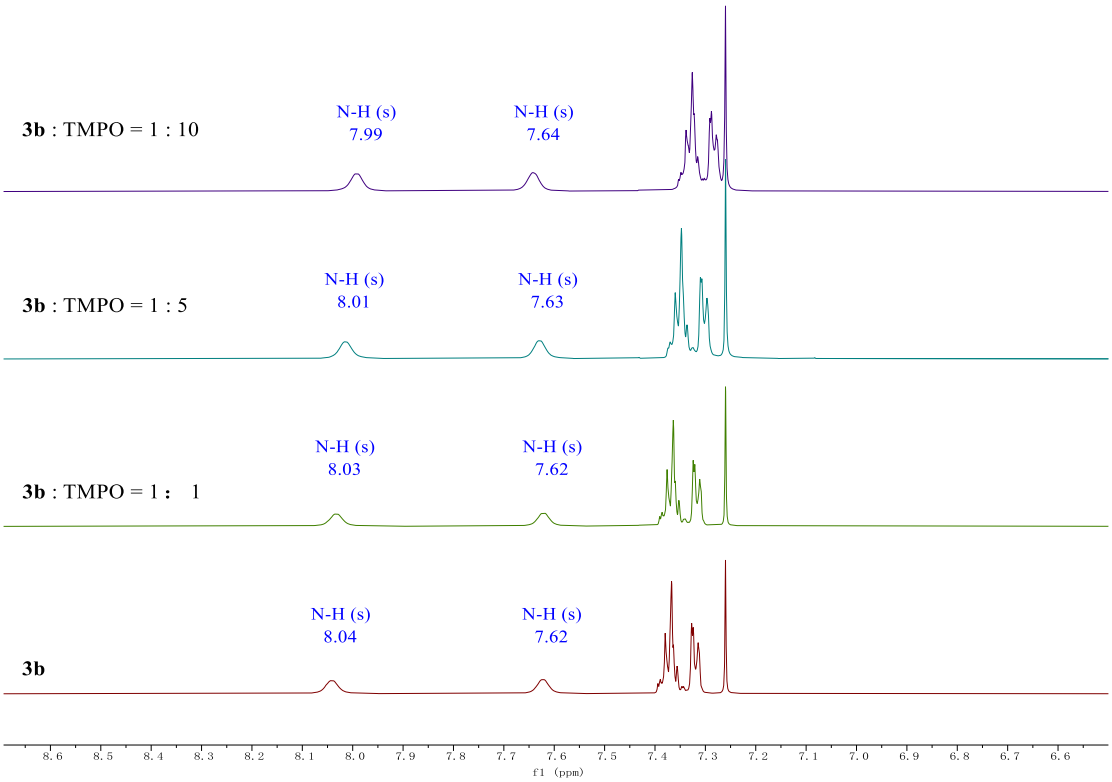


Figure S13. Partial ^{1}H NMR spectra of 3b with different equiv. of TMPO in $CDCl_3$. The peaks of amide protons were picked.

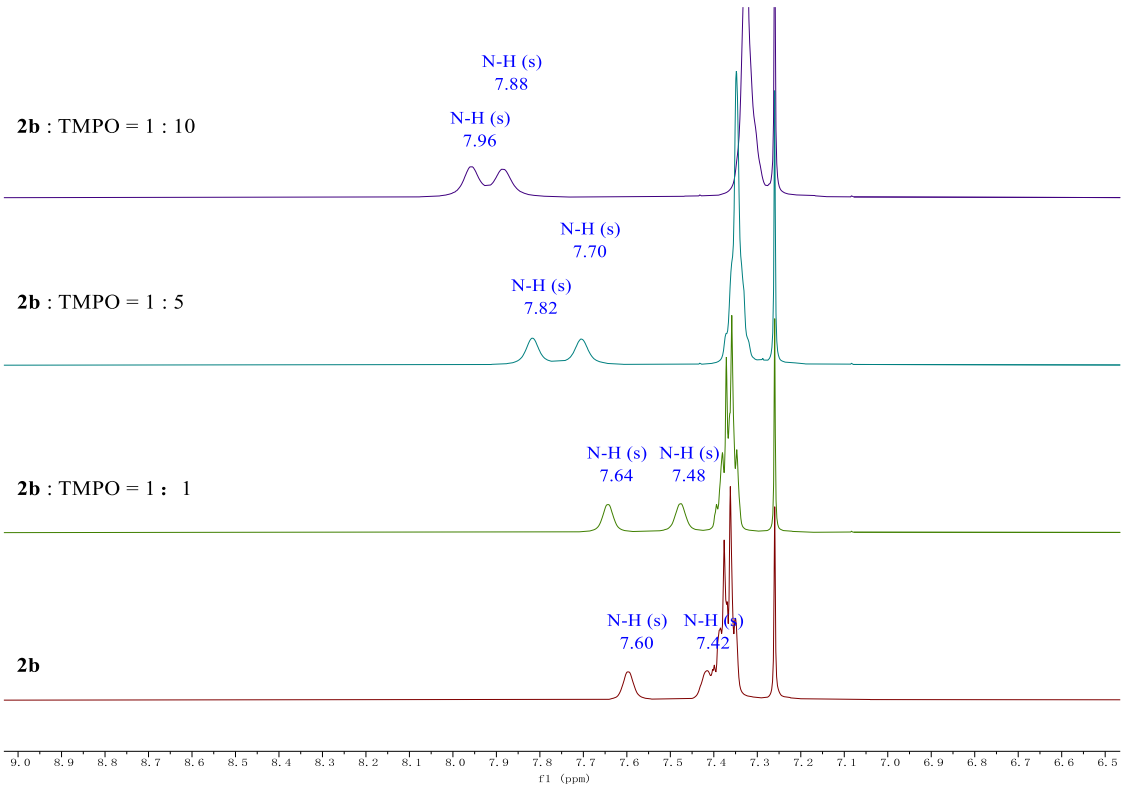


Figure S14. Partial ^{1}H NMR spectra of 2b with different equiv. of TMPO in $CDCl_3$. The peaks of amide protons were picked.

^{1}H NMR titration of 3b in $CDCl_3$ with DMSO-d_6

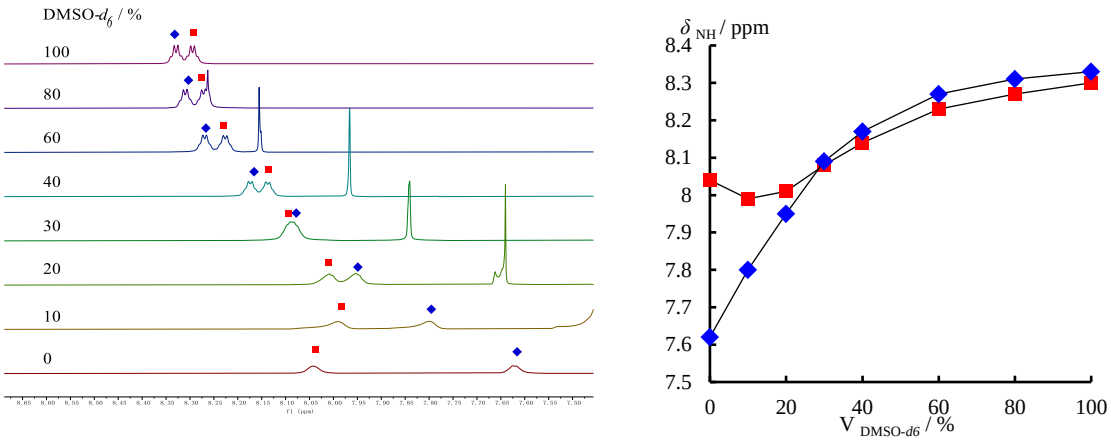


Figure S15. Partial ^{1}H NMR spectra in $CDCl_3$/DMSO-d_6 (0.01 M) of 3b. The amide protons were picked, red square and blue diamond represent amide protons. Large downfield chemical shift indicated the blue diamond marked amide proton was more prone to intermolecular hydrogen bond.

^{13}C NMR titration of 3a with TMPO

A solution of **3a** (0.05 mmol, 1.0 equiv.) in 0.6 mL $CDCl_3$ was added to a NMR tube to give standard solution of **3a**. To two NMR tubes containing standard solution of **3a** was added **TMPO** (0.25 mmol, 5.0 equiv. or 0.5 mmol, 10.0 equiv.) respectively. ^{13}C NMR spectrum of each sample was recorded at room temperature. With the increase of **TMPO**, there was no downfield shift on the peak of bromine-carried carbon in **3a**, but all peaks were very slightly upfield, which may be caused by system errors or other interactions between **3a** and **TMPO**.

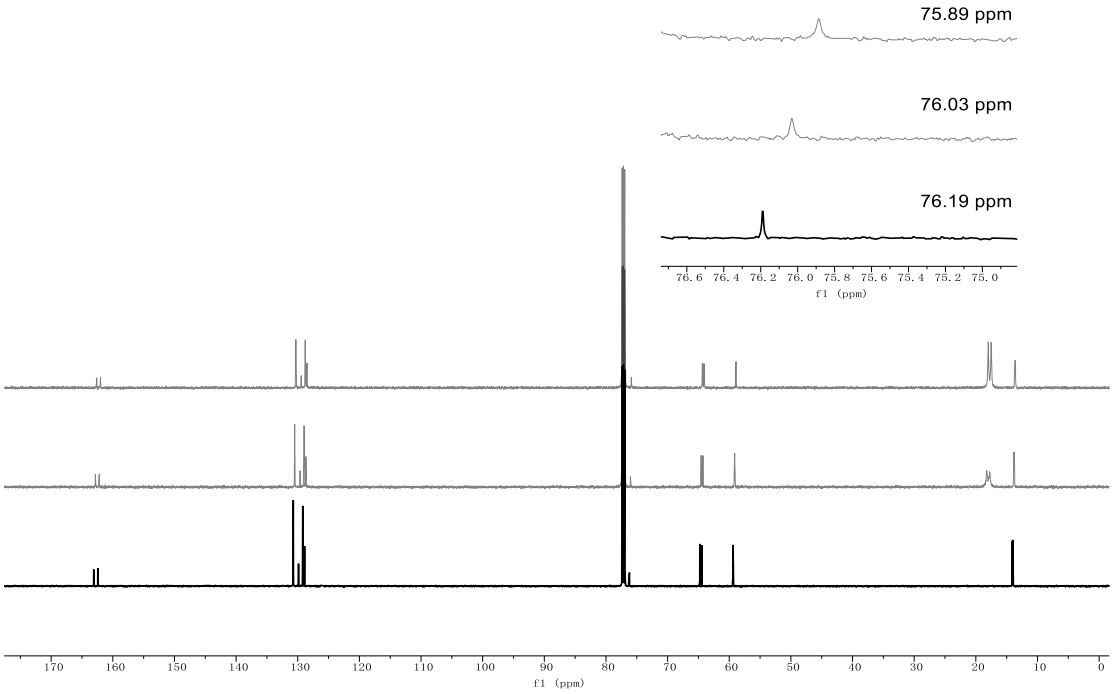


Figure S16. ^{13}C NMR titration spectra of 3a with TMPO. From top to bottom, ^{13}C NMR spectra of **3a** + TMPO (10.0 equiv.), **3a** + TMPO (5.0 equiv.) and **3a** in $CDCl_3$; the peaks of bromine-carried carbon were picked.

^{13}C NMR titration of 3a with 4a

To two NMR tubes containing standard solution of **3a** was added **4a** (0.05 mmol, 1.0 equiv. or 0.25 mmol, 5.0 equiv.) respectively. ^{13}C NMR spectrum of each sample was recorded at room temperature. With the increase of **4a**, there was no downfield shift on the peak of bromine-carried carbon in **3a**.

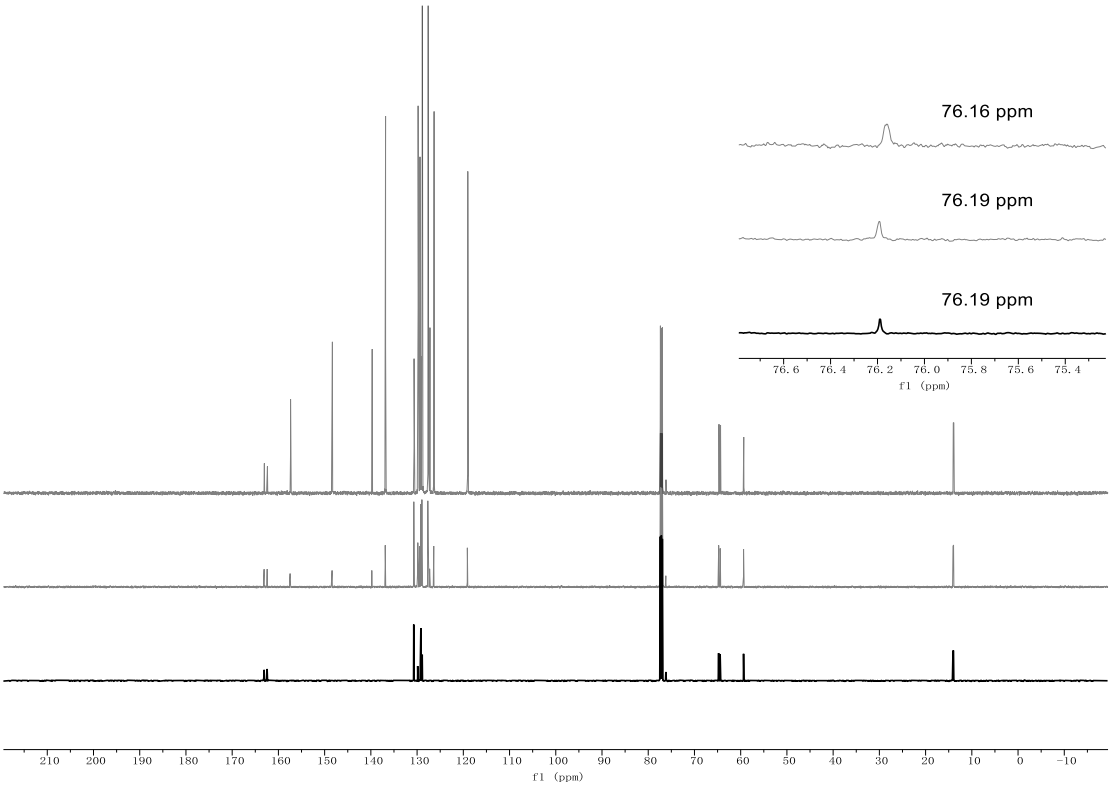


Figure S17. ^{13}C NMR titration spectra of 3a with 4a. From the top to bottom, ^{13}C NMR spectra of **3a** + **4a** (5.0 equiv.), **3a** + **4a** (1.0 equiv.) and **3a** in $CDCl_3$; the peaks of bromine-carried carbon were picked.

^{13}C NMR titration of 3b with TMPO

A solution of **3b** (0.05 mmol, 1.0 equiv.) in 0.6 mL $CDCl_3$ was added to a NMR tube to give standard solution of **3b**. To two NMR tubes containing standard solution of **3b** was added **TMPO** (0.25 mmol, 5.0 equiv. or 0.5 mmol, 10.0 equiv.) respectively. ^{13}C NMR spectrum of each sample was recorded at room temperature. Downfield shift of the peak of bromine-carried carbon in **3b** was observed with the addition of **TMPO**.

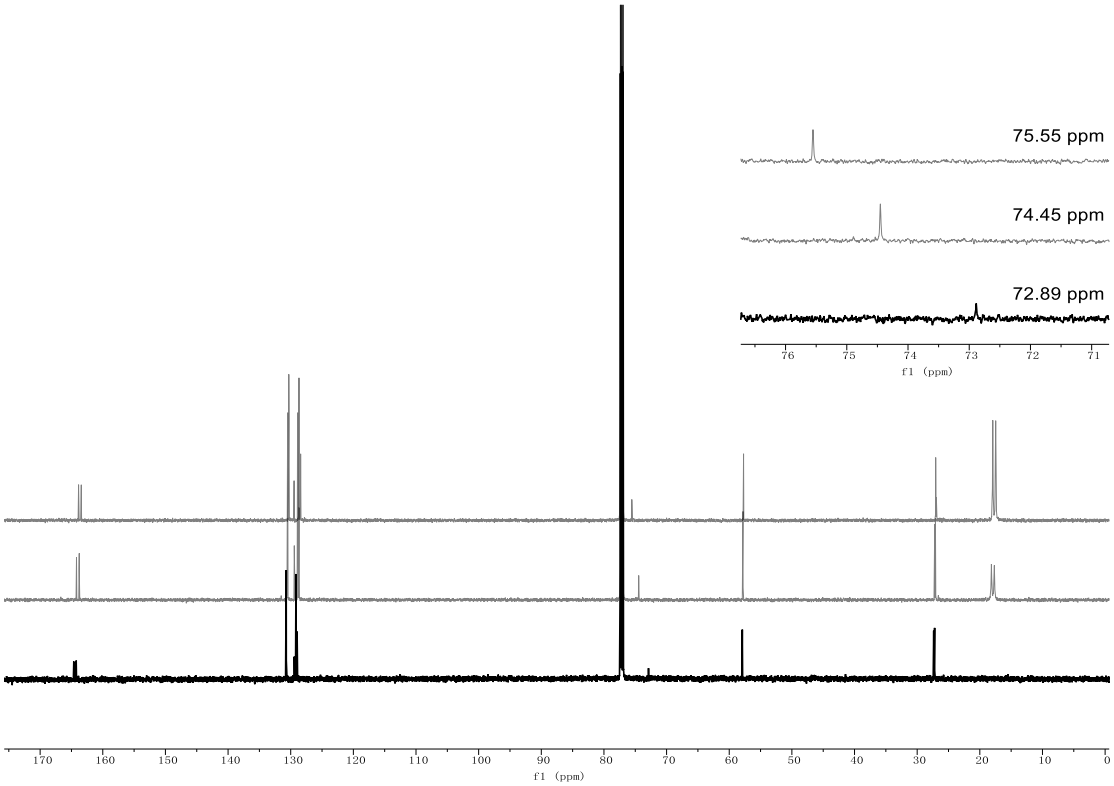


Figure S18. ^{13}C NMR titration of 3b with TMPO. From the top to bottom, ^{13}C NMR spectra of **3b** + TMPO (10.0 equiv.), **3b** + TMPO (5.0 equiv.) and **3b** in $CDCl_3$; the peaks of bromine-carried carbon were picked.

^{13}C NMR titration of 3b with 4a

To two NMR tubes containing standard solution of **3b** was added **4a** (0.05 mmol, 1.0 equiv. or 0.25 mmol, 5.0 equiv.) respectively. ^{13}C NMR spectrum of each sample was recorded at room temperature. Downfield shift of the peak of bromine-carried carbon in **3b** was observed with the addition of **4a**.

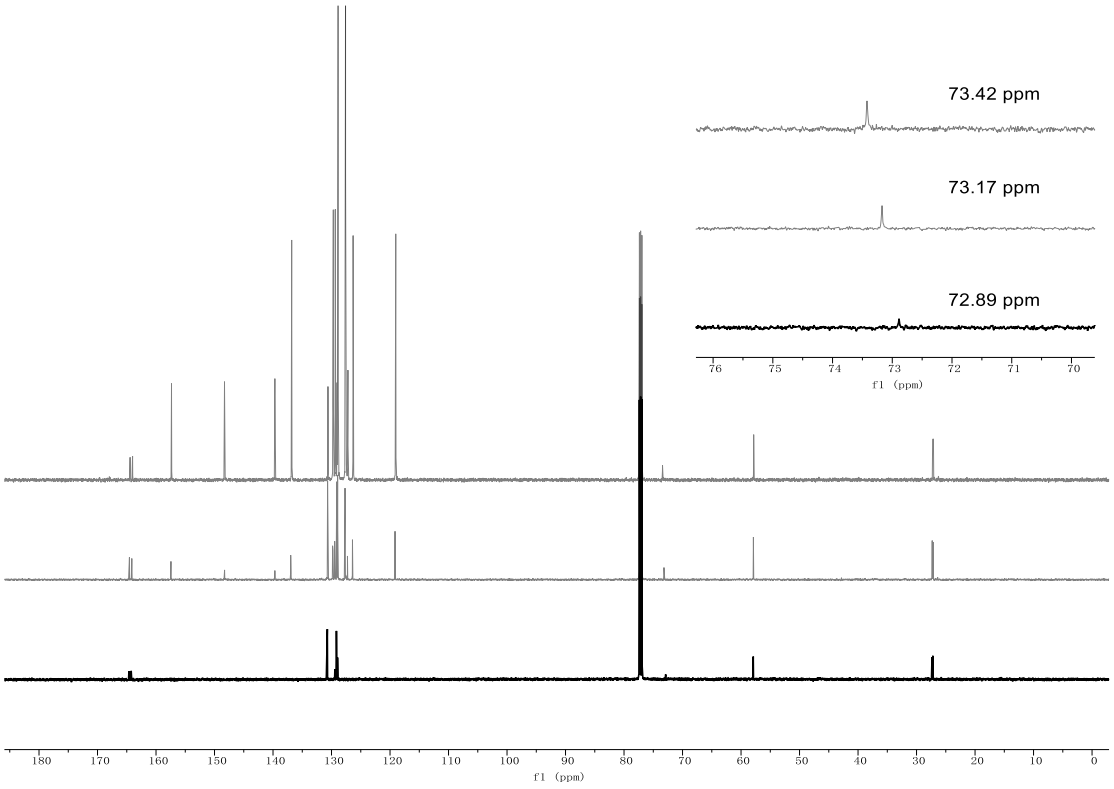


Figure S19. ^{13}C NMR titration of 3b with 4a. From the top to bottom, ^{13}C NMR spectra of **3b** + **4a** (5.0 equiv.), **3b** + **4a** (1.0 equiv.) and **3b** in $CDCl_3$, the peaks of bromine-carried carbon were picked.

^{1}H NMR titration of 3a and 3b with substrate 4a

General procedure: A solution of **3a** or **3b** (0.01 mmol, 1.0 equiv.) in 0.5 mL $CDCl_3$ was added to a NMR tube to give standard host **XBD** solution. To ten NMR tubes containing standard solution was added varying amounts of **4a** respectively (the resulting molar ration of **3a** or **3b** : **4a** was indicated in the footnotes of resulting ^{1}H NMR spectra). ^{1}H NMR spectra of each sample was recorded at room temperature. The chemical shift data were fitted to a 1:1 binding isotherm and calculated binding constant (K_a) using BindFit, freely available at http://supramolecular.org (P. Thordarson, *Chem. Soc. Rev.*, 2011, 40, 1305-1323; D. Brynn Hibbert, Pall Thordarson, *Chem. Commun.*, 2016, 52, 12792-12805).

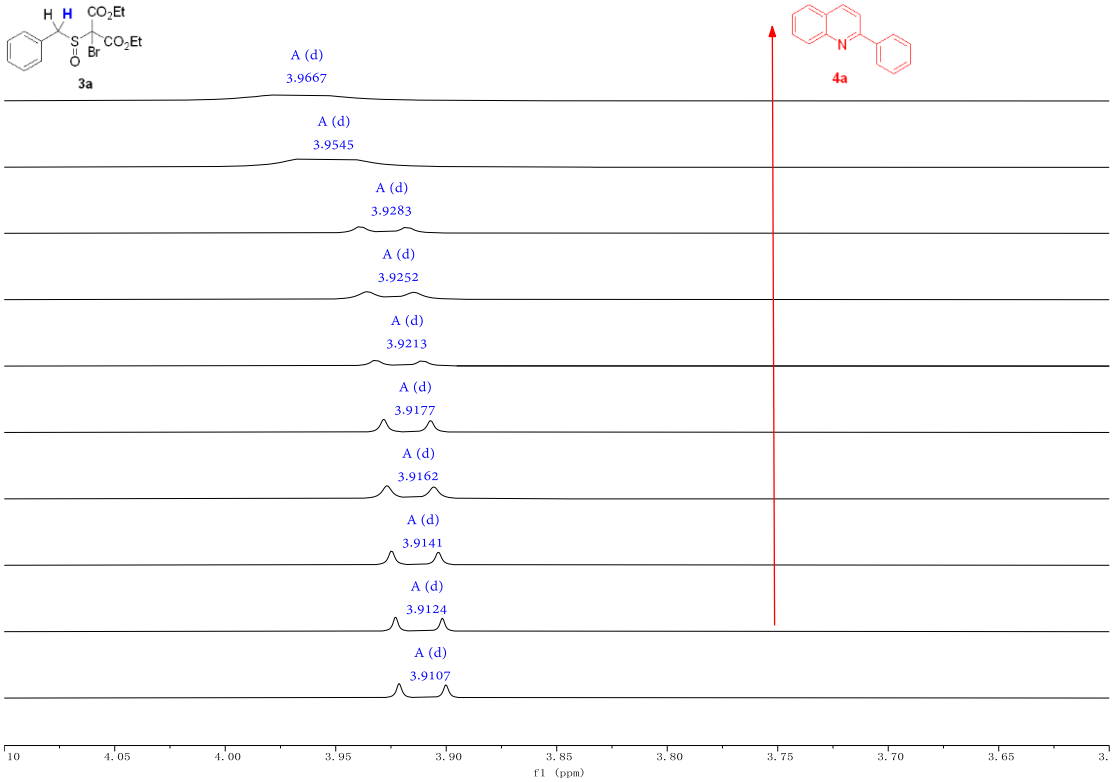


Figure S20. ^{1}H NMR spectra titration of 3a with 4a in $CDCl_3$ at room temperature. Showing one of proton of **3a** benzyl methyl (from bottom to top, the equivalents of **4a** are 0, 1.0, 2.0, 3.0, 4.1, 6.1, 8.0, 10.25, 29.85, 39.95).

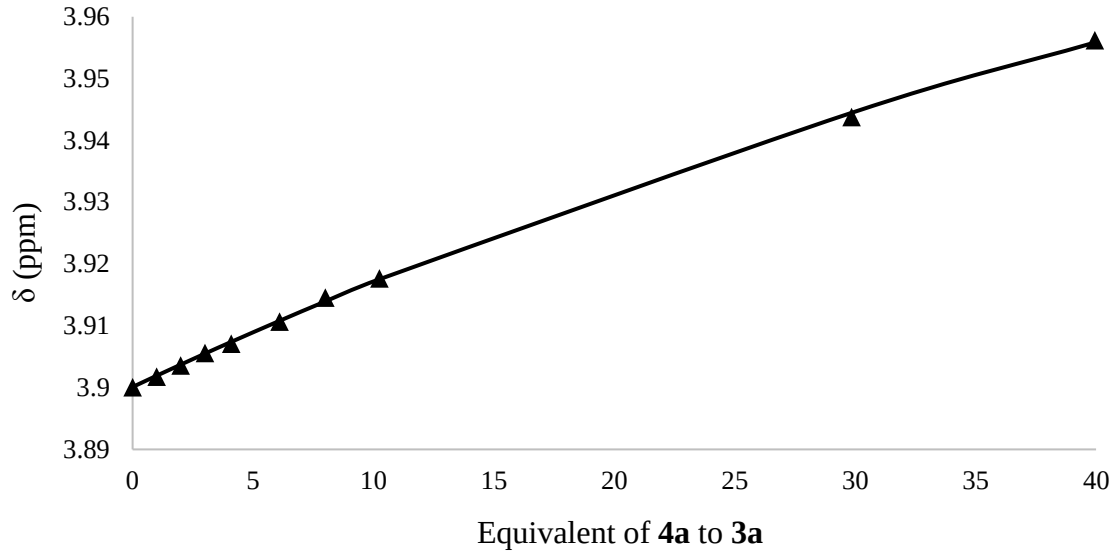


Figure S21. Calculated chemical shift using BindFit to give binding constant (K_a = 0.40 ± 0.006 M^{-1}).

For more details, please see: http://app.supramolecular.org/bindfit/view/3223d34f-6c1b-4780-8c49-b21f6ee644ac.

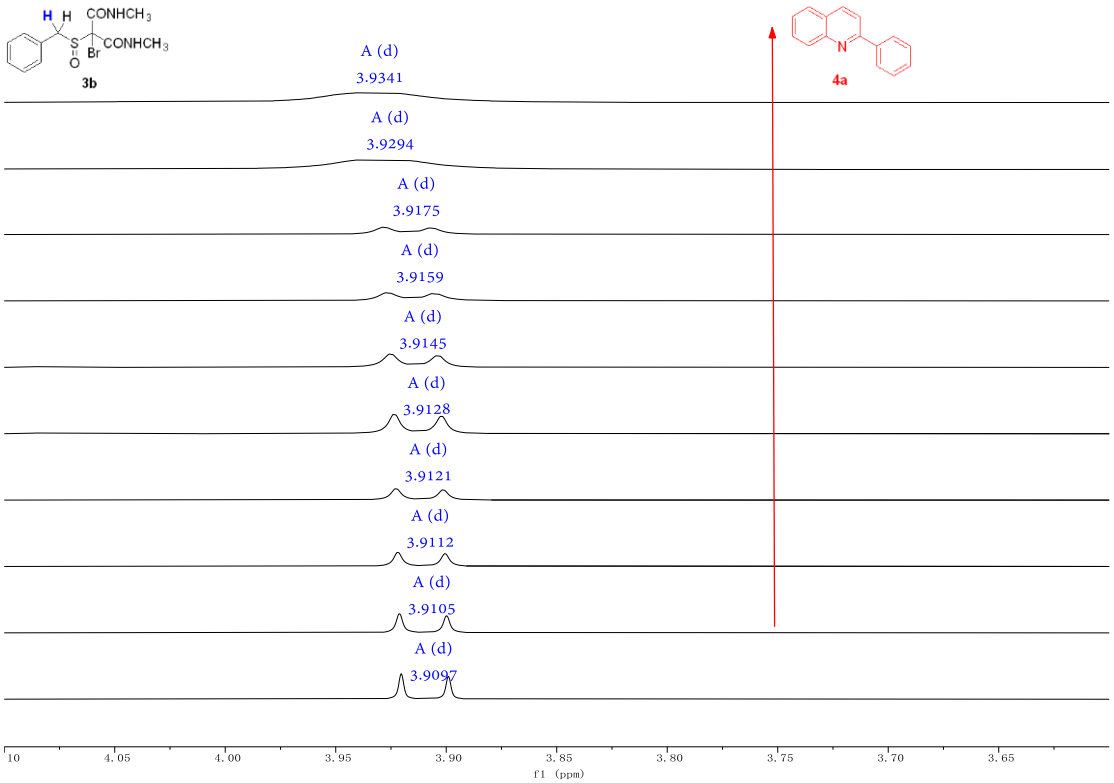


Figure S22. ^{1}H NMR spectra titration of 3b with 4a in $CDCl_3$ at room temperature. Showing one of proton of **3b** benzyl methyl (from bottom to top, the equivalents of **4a** are 0, 1.05, 2.0, 3.05, 4.0, 6.05, 7.95, 10.15, 29.85, 40.0).

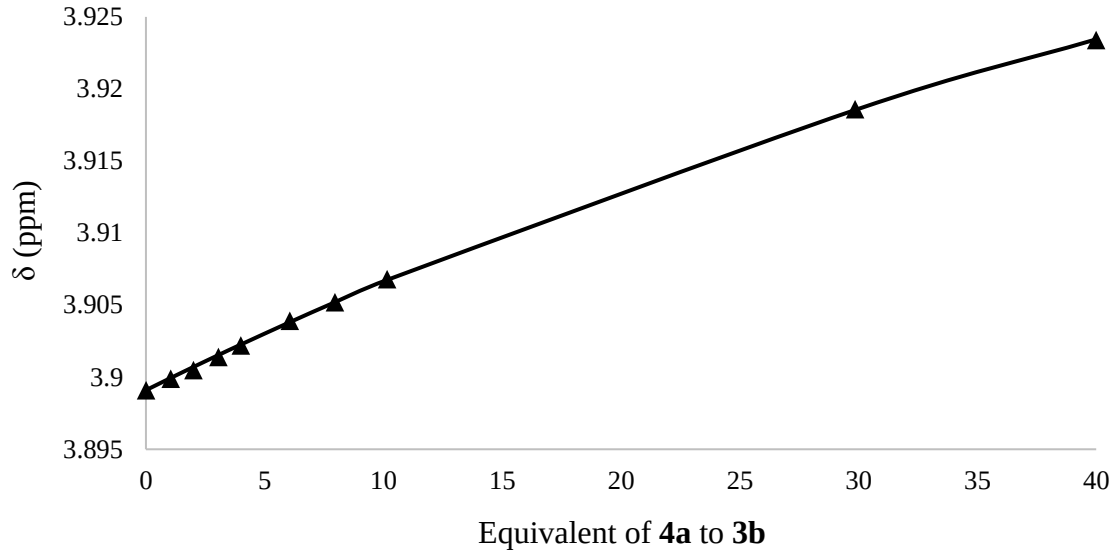
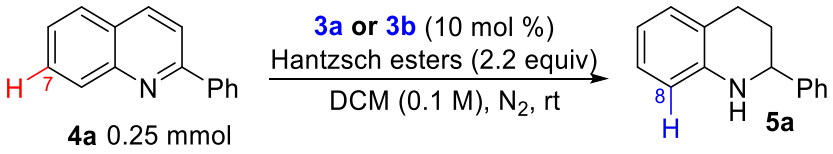


Figure S23. Calculated chemical shift using BindFit to give binding constant (K_a = 0.44 ± 0.004 M^{-1}).

For more details, please see: http://app.supramolecular.org/bindfit/view/ef5e9297-772b-4064-813f-0f6944b627cc.

7. Reaction monitoring by ^{1}H NMR



Reactions were carried out with 0.25 mmol of **4a** in the presence of 10 mol% **3a** or **3b** under standard conditions.

Aliquots of the reaction mixture (100 μL) was taken in appropriate intervals. After a rapid removal of CH_2Cl_2 under vacuum pump, the residue was immediately dissolved in $CDCl_3$(0.6 mL). ^{1}H NMR spectrum of the resulting solution was then recorded. Integrals associated with the protons in substrate **4a** at positions 7 (δ = 7.75 ppm, t, 1H) and product **5a** at position 8 (δ 6.53 ppm; d, 1H) were observed (see Figures S18-S19). The conversion of **4a** was then determined by comparing the integration of the above-mentioned resonances in the crude NMR.

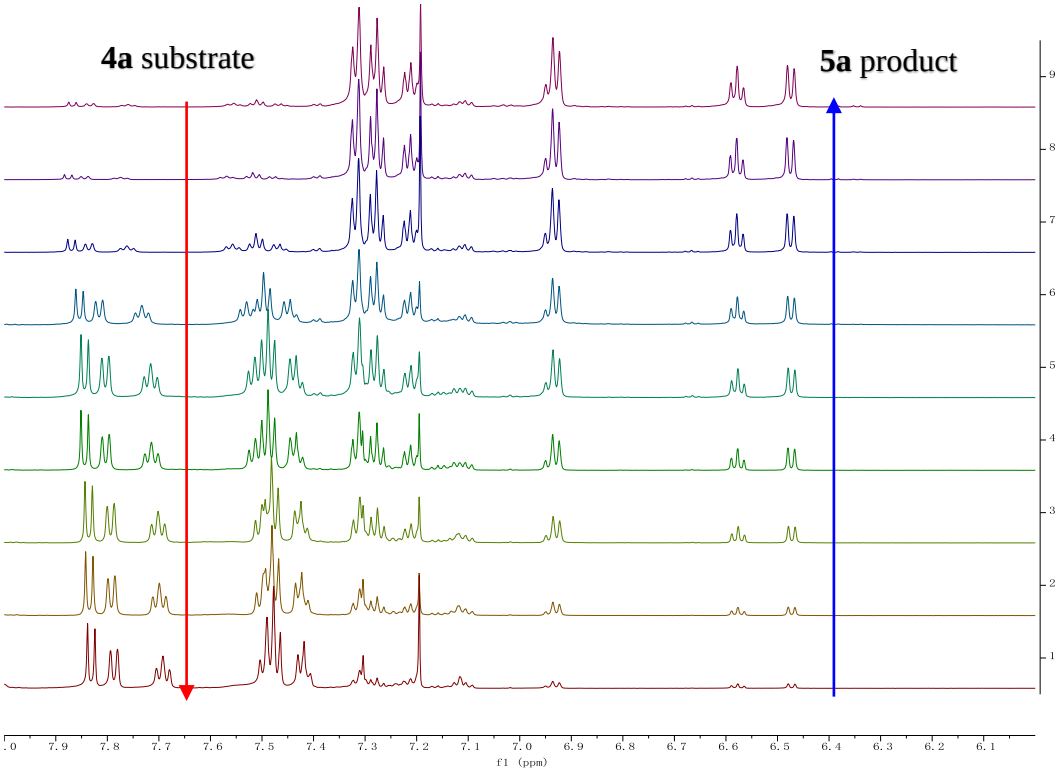


Figure S24. Reaction progress with catalyst 3a by ^{1}H NMR monitoring. Samples (100 μL) were taken in appropriate intervals. Relevant substrate signals for integration are shown with a red arrow, product signals with a blue arrow.

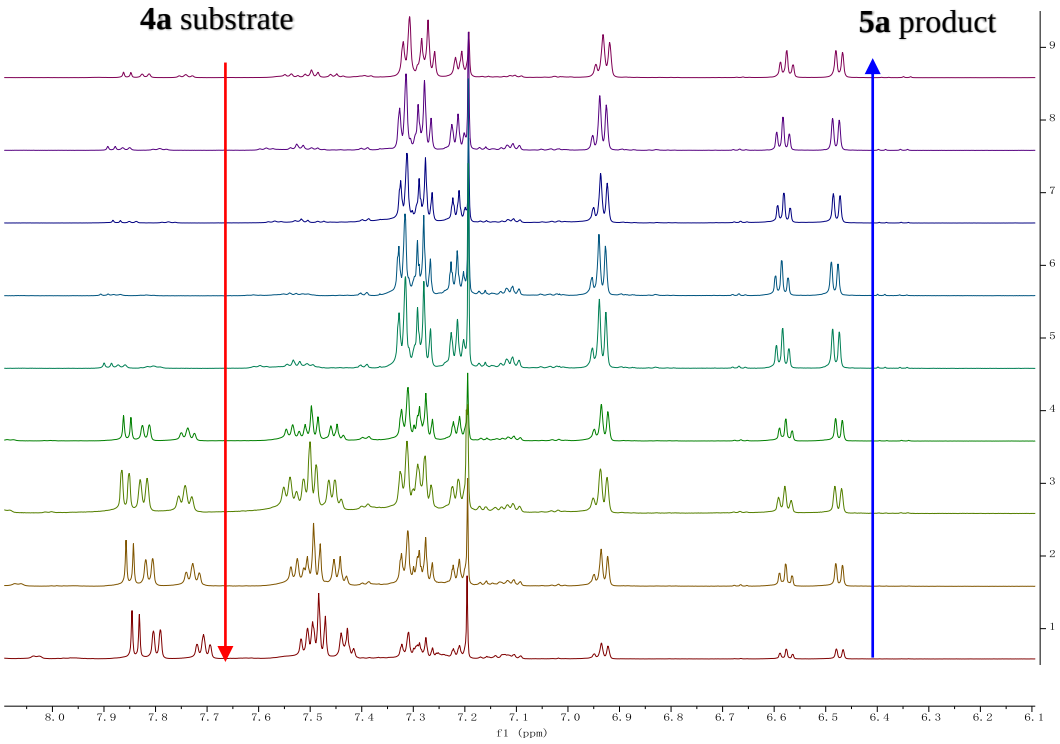


Figure S25. Reaction progress with catalyst 3b by ^{1}H NMR monitoring. Samples (100 µL) were taken in appropriate intervals. Relevant substrate signals for integration are shown with a red arrow, product signals with a blue arrow.

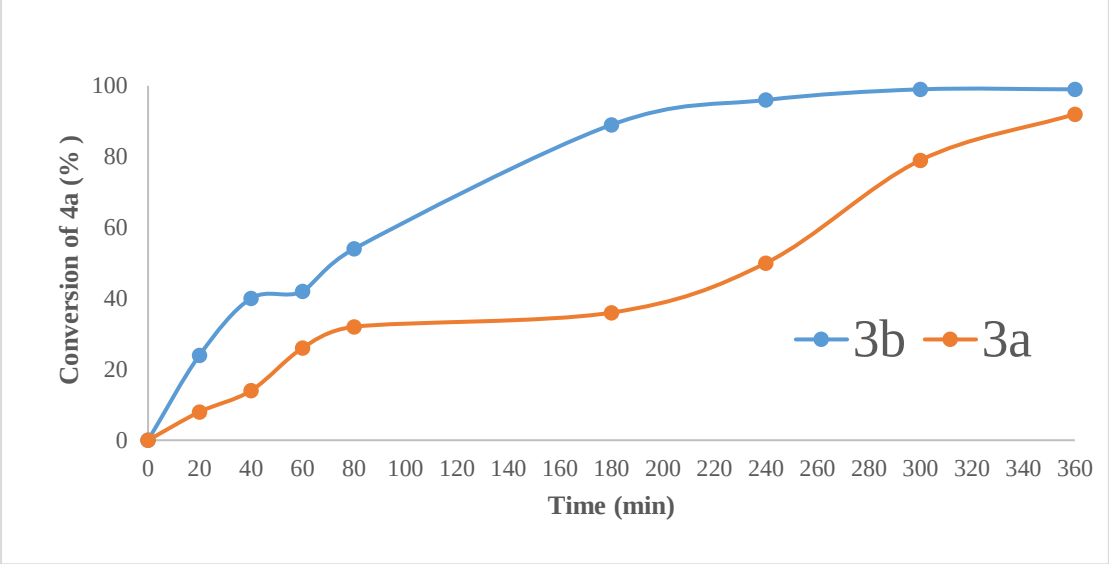
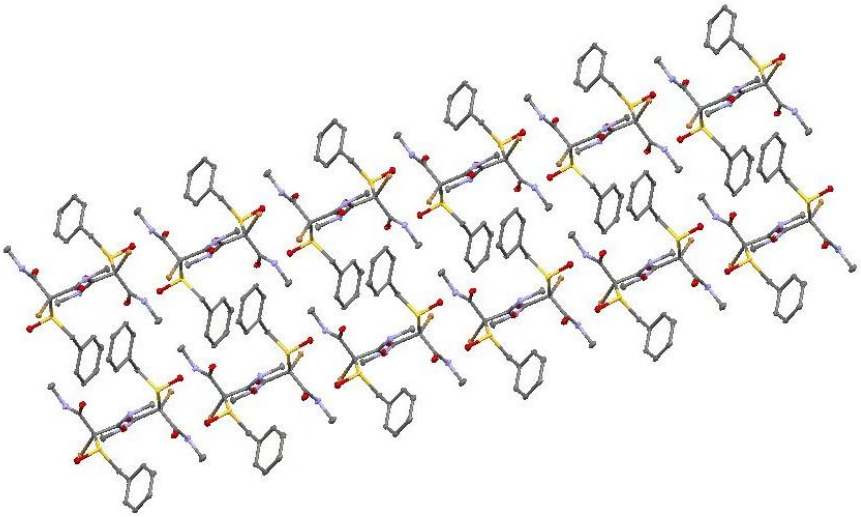


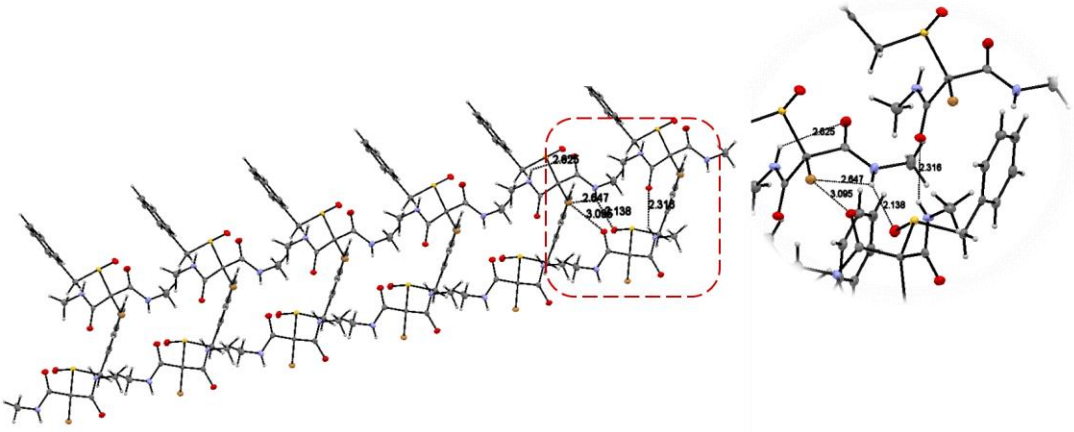
Figure S26. Conversion of 4a with time in the presence of 10 mol% 3a and 3b

8. X-Ray structural analysis

a)



b)



c)

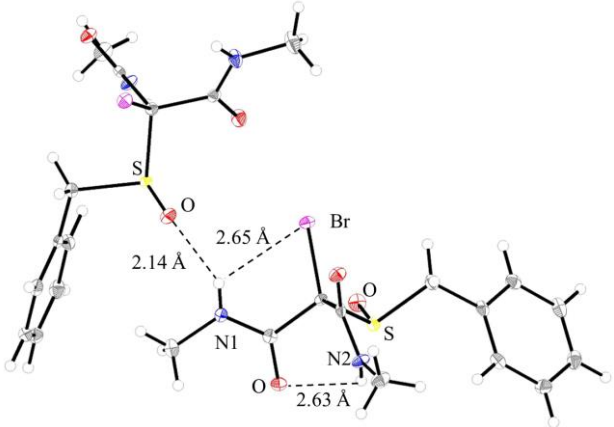


Figure S27. X-ray packing of 3b a) π–π stacking. b) Intramolecular HB, intermolecular HB and XB. c) Intramolecular HB and intermolecular HB

CCDC 2213198 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from the Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.

Crystallographic Methods: The X-ray intensity data of **3b** were measured on a XtaLAB Synergy R, DW system, HyPix diffractometer. The crystal was kept at 99.99(10) K during data collection. Using Olex2 (Dolomanov, O.V., Bourhis, L.J., Gildea, R.J, Howard, J.A.K. & Puschmann, H. (2009), J. Appl. Cryst. 42, 339-341), the structure was solved with the SHELXT (Sheldrick, G.M. (2015). Acta Cryst. A71, 3-8) structure solution program using Intrinsic Phasing and refined with the SHELXL (Sheldrick, G.M. (2015). Acta Cryst. C71, 3-8) refinement package using Least Squares minimisation. (Dolomanov, O.V., Bourhis, L.J., Gildea, R.J, Howard, J.A.K. & Puschmann, H. *J. Appl. Cryst.* **2009**, 42, 339-341. Sheldrick, G.M. *Acta Cryst.* **2015**, *A71*, 3-8. Sheldrick, G.M. *Acta Cryst.* **2015**, *C71*, 3-8).

Table S5. Crystal data and structure refinement for exp_1895_auto

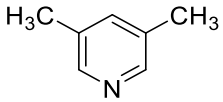
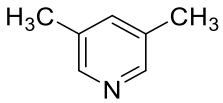
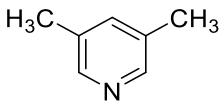
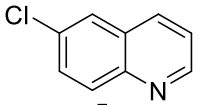
Identification code	exp_1895_autored
Empirical formula	$C_{12}H_{15}BrN_2O_3S$
Formula weight	347.23
Temperature/K	99.99(10)
Crystal system	monoclinic
Space group	$P2_1/c$
a/Å	7.99040(10)
b/Å	10.11790(10)
c/Å	17.12180(10)
α/°	90
β/°	89.4790(10)
γ/°	90
Volume/Å^3	1384.17(2)
Z	4
ρ_{calcd} g/cm^3	1.666
μ/mm^{-1}	5.524

F(000)	704.0
Crystal size/mm^3	0.3 × 0.2 × 0.2
Radiation	Cu Kα (λ = 1.54184)
2Θ range for data collection/°	10.156 to 133.182
Index ranges	-9 ≤ h ≤ 9, -11 ≤ k ≤ 12, -19 ≤ l ≤ 20
Reflections collected	8744
Independent reflections	2445 [R_{int} = 0.0195, R_{sigma} = 0.0156]
Data/restraints/parameters	2445/0/182
Goodness-of-fit on F^2	1.092
Final R indexes [I> =2σ (I)]	R_1 = 0.0240, wR_2 = 0.0595
Final R indexes [all data]	R_1 = 0.0245, wR_2 = 0.0598
Largest diff. peak/hole / e Å^{-3}	0.39/-0.41

Attempt to grow co-crystals of 3b and XB acceptors

We tried to grow the co-crystals of **3b** and XB acceptors by varying the XB acceptor, the equivalent of XB acceptor, the temperature and method for crystal culture, the solvent mixture. Up to now, no cocrystal was observed.

Table S6. Attempt to grow co-crystals.

XB acceptor	Eq.	Temperature	Solvent 1	Solvent 2	Note
4a	2.5	rt	DCE	Et_2O	No crystal
4a	7	-20	DCM	/	Single **3b**
5o	2	-20	DCE	Hexane	Poor quality crystal
	5	-20	DCE	/	Single **3b**
	10	-20	DCE	/	Single **3b**
	100	-20	DCE	/	Single **3b**
 **5q**	1	-20	DCE	Hexane	No crystal
5q	2	-20	DCM	/	Single **3b**
5q	1	RT	DCM	Hexane	Single **3b**
5q	1	-20	DCM	/	Poor quality crystal
TMPO	3	-20	DCM	/	Single **3b**
TMPO	5	-20	DCM	/	No crystal
TMPO	6	RT	DCM	Hexane	Single **TMPO**

9. Computational Details

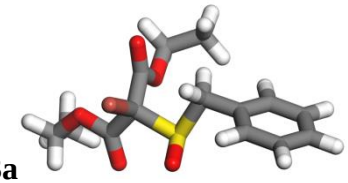
Geometries of **3a** and **3b** were optimized at the B3LYP/def2-TZVPP level to locate their minimal structures, which were confirmed by the following frequency calculations. To identify the reactive sites, the electrostatic potentials (ESPs) were mapped into surfaces using isodensity of 0.001 a.u. as shown in Figure 4. We further obtained the optimized complexes of the reactant with **3a** and **3b** in the gas phase respectively at the B3LYP/def2-TZVPP level with the binding energies calculated based on Eq. S1,

$$E(\text{Binding energy}) = E(\text{AB}) - E(\text{A}) - E(\text{B}) \qquad \text{(S1)}$$

Where E(A) and E(B) were electronic energies of isolated reactant and **3a**/**3b** respectively, and E(AB) was the energy of the complex. All calculations were performed using packages Gaussian 16 in the revision of C.02 and Multiwfn. (Frisch, M. J.; Trucks, G. W.; Schlegel, H. B.; et al. Gaussian 16, revision C.02; Gaussian, Inc., Wallingford CT, **2016**, Lu T.; Chen. F. Quantitative analysis of molecular surface based on improved Marching Tetrahedra algorithm. *J. Mol. Graph. Model.* **2012**, *38*(Complete):314-323, Lu. T.; Chen F. Multiwfn: A multifunctional wavefunction analyzer. *J Comput Chem.* **2012**, *33*(5):580-592).

Table S7. Cartesian coordinates of the optimized minimum in the ground state of 3a at the B3LYP/def2-TZVPP level.

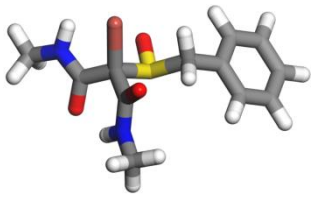
3a



C	-5.751558	0.076638	0.820237
C	-5.144481	-1.128712	0.485806
C	-3.837014	-1.14755	0.019908
C	-3.121512	0.041917	-0.119028
C	-3.734922	1.246149	0.225802
C	-5.043625	1.265576	0.689614
C	-1.702431	0.022777	-0.589597

S	-0.609996	-0.388987	0.846828
O	-0.904102	-1.79403	1.181466
C	1.03657	-0.37396	-0.15009
C	2.155295	-0.448786	0.89582
C	1.129139	0.921339	-0.944984
Br	1.130116	-1.936707	-1.295303
C	1.22602	3.280494	-0.604354
C	-0.164592	3.791843	-0.913977
C	4.507885	-0.309633	1.178749
C	4.755858	1.058959	1.77577
O	1.114323	1.03698	-2.136844
O	1.964266	-0.609679	2.069068
H	-6.771132	0.089303	1.180805
H	-5.689507	-2.05696	0.59045
H	-3.357685	-2.085271	-0.223176
H	-3.185209	2.173405	0.123789
H	-5.509987	2.206781	0.947567
H	-1.397355	0.993477	-0.971998
H	-1.517385	-0.743574	-1.339489
H	1.702795	3.862139	0.180523
H	1.85803	3.274479	-1.489748
H	-0.620352	3.219357	-1.72026
H	-0.108543	4.833975	-1.230672
H	-0.800706	3.737572	-0.030648
H	4.352878	-1.062117	1.948849
H	5.318417	-0.620017	0.524273
H	4.878253	1.805916	0.991628
H	5.668059	1.037207	2.373444
H	3.931074	1.354196	2.42203
O	1.180589	1.934612	-0.06441
O	3.343729	-0.292087	0.312085

Table S8. Cartesian coordinates of the optimized minimum in the ground state of 3b derived from the X-ray structure at the B3LYP/def2-TZVPP level.

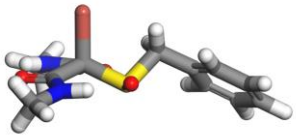


3b derived from X-ray structure

Br	-1.637605	-0.923002	-1.697169
S	0.346589	-0.311564	0.714192
O	-1.081165	1.991375	-1.660857
O	-2.378081	0.595273	2.01575
O	0.338787	-1.782152	0.882094
N	-3.016965	-1.293686	0.973603
H	-2.853004	-1.906252	0.192784
N	-1.19512	2.501382	0.543551
H	-1.510274	2.129349	1.433686
C	2.860365	-0.083737	-0.27998
C	3.463331	-1.340031	-0.212565
H	2.895624	-2.216112	-0.494026
C	4.774269	-1.466567	0.226553
H	5.232146	-2.445247	0.273816
C	5.496492	-0.340657	0.605616
H	6.518681	-0.440262	0.944588
C	4.900847	0.913963	0.54652
H	5.457783	1.793816	0.838928
C	3.58888	1.03995	0.109706
H	3.126139	2.017563	0.063436
C	1.437785	0.051477	-0.723917
H	1.202484	1.053722	-1.074146
H	1.162828	-0.673357	-1.486532
C	-1.336001	0.159457	-0.09625
C	-2.319352	-0.158664	1.046278
C	-3.84115	-1.738701	2.082279
H	-3.227078	-1.993963	2.946777
H	-4.532901	-0.95131	2.375069
H	-4.402705	-2.61384	1.76696
C	-1.231171	1.655562	-0.500978

C	-1.080218	3.931965	0.334909
H	-0.859843	4.405853	1.288263
H	-0.275636	4.139522	-0.367529
H	-1.999683	4.356673	-0.073256

Table S9. Cartesian coordinates of the optimized minimum in the ground state of proposed conformation of 3b at the B3LYP/def2-TZVPP level.

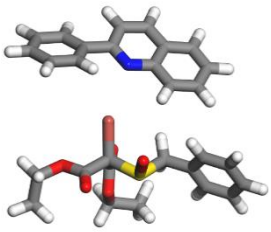


Proposed conformation of **3b**

C	5.611528	0.023150	-0.538103
C	5.038878	-1.053217	0.129339
C	3.716615	-0.989014	0.549444
C	2.956136	0.154888	0.310111
C	3.534452	1.228609	-0.368188
C	4.856207	1.164160	-0.787038
C	1.524306	0.224079	0.738949
S	0.494399	-0.489095	-0.611037
O	0.585371	-1.969878	-0.381197
C	-1.245830	0.053035	-0.088917
C	-2.360948	-0.818022	-0.739231
C	-1.308467	1.563554	-0.421561
Br	-1.546858	-0.165559	1.864367
O	-0.261640	2.190222	-0.541674
O	-3.481536	-0.351854	-0.924847
H	6.641981	-0.026348	-0.863083
H	5.621428	-1.943883	0.322130
H	3.264480	-1.830050	1.057126
H	2.941221	2.111902	-0.565580
H	5.297191	2.003724	-1.307188
H	1.185351	1.239723	0.901105
H	1.310309	-0.407655	1.597075
N	-2.522864	2.102594	-0.533510

N	-2.038577	-2.077738	-1.027844
C	-2.692466	3.509156	-0.840804
C	-3.028564	-2.996328	-1.557551
H	-3.298834	1.449425	-0.581966
H	-1.094598	-2.396799	-0.814210
H	-2.119212	4.117364	-0.143043
H	-3.746805	3.757741	-0.752943
H	-2.350571	3.742664	-1.850791
H	-2.541044	-3.946006	-1.759953
H	-3.461201	-2.608038	-2.478831
H	-3.838617	-3.149857	-0.843587

Table S10. Cartesian coordinates of the optimized minimum in the ground state of 3a•4a at the B3LYP/def2-TZVPP level



3a•4a

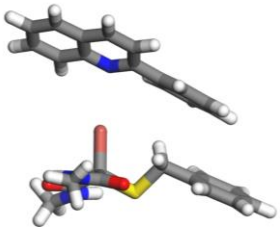
Binding energy: -7.8558 kcal/mol

C	5.652897	-3.104386	-1.237788
C	4.428392	-3.087884	-1.897033
C	3.402837	-2.267173	-1.448754
C	3.589437	-1.448999	-0.332756
C	4.824183	-1.463572	0.314964
C	5.849331	-2.288589	-0.130317
C	2.492167	-0.554528	0.151229
S	2.290313	0.858157	-1.026201
O	1.735360	0.263321	-2.258470
C	0.830297	1.683346	-0.084503
C	0.604860	3.033697	-0.766563
C	1.227822	1.874045	1.371744
Br	-0.754407	0.582010	-0.276553
C	2.850707	3.006143	2.703236

C	3.850746	1.940977	3.100649
C	-0.599145	5.072008	-0.592100
C	0.409970	6.094245	-0.113972
O	0.742618	1.338800	2.327264
O	1.161012	3.383110	-1.771136
H	6.449374	-3.747483	-1.586814
H	4.272353	-3.713900	-2.765267
H	2.459882	-2.235940	-1.974382
H	4.979535	-0.830081	1.179126
H	6.799174	-2.294192	0.387176
H	2.722083	-0.146611	1.131995
H	1.526505	-1.055596	0.182762
H	3.326759	3.974802	2.572918
H	2.045149	3.090277	3.429655
H	3.356945	0.983633	3.259389
H	4.339618	2.227163	4.032720
H	4.616943	1.827023	2.333879
H	-0.633388	5.020090	-1.678230
H	-1.595235	5.274110	-0.206225
H	0.455592	6.111256	0.974861
H	0.118590	7.086788	-0.460339
H	1.400553	5.872085	-0.507018
O	2.271475	2.722064	1.405509
O	-0.283453	3.753973	-0.077940
C	0.041463	-5.002384	0.692804
C	0.398703	-4.067588	-0.302681
C	-0.450906	-3.046690	-0.643024
C	-1.700688	-2.913771	0.007828
C	-2.061557	-3.859435	1.009666
C	-1.166400	-4.902759	1.333396
C	-3.320847	-3.690849	1.626408
C	-4.120323	-2.644550	1.266379
C	-3.670837	-1.729056	0.275729
N	-2.509743	-1.879838	-0.329340
C	-4.490262	-0.561388	-0.122054

C	-5.461066	-0.022430	0.726341
C	-6.203359	1.085593	0.340855
C	-5.990680	1.670755	-0.901275
C	-5.025020	1.143622	-1.752799
C	-4.280077	0.040709	-1.366715
H	0.727438	-5.798871	0.946914
H	1.357310	-4.158125	-0.794033
H	-0.192506	-2.319591	-1.399838
H	-1.448295	-5.618097	2.095700
H	-3.643392	-4.399915	2.378587
H	-5.093279	-2.520036	1.716575
H	-5.622434	-0.448663	1.706258
H	-6.943214	1.495939	1.014862
H	-6.570666	2.532696	-1.202529
H	-4.852066	1.593874	-2.721013
H	-3.523852	-0.369454	-2.018932

Table S11 Cartesian coordinates of the optimized minimum in the ground state of 3b•4a at the B3LYP/def2-TZVPP level.



3b•4a

Binding energy: -10.2811 kcal/mol

Br	0.764965	0.821024	-0.430752
S	-2.357154	1.558416	-0.128940
O	-1.266015	1.448674	2.434760
O	0.633524	4.202410	0.011364
O	-2.546062	1.624072	-1.619634
N	-0.829645	3.703255	-1.624309
H	-1.509775	3.024559	-1.964875
N	0.474861	2.887609	2.317888
H	0.951357	3.522893	1.684865
C	-3.614661	-0.824251	0.206630

C	-4.273665	-1.091147	-0.993655
H	-3.782605	-0.857219	-1.927643
C	-5.545659	-1.646768	-0.987885
H	-6.047249	-1.850906	-1.924388
C	-6.173395	-1.943359	0.217370
H	-7.162297	-2.381684	0.221085
C	-5.526007	-1.672052	1.417347
H	-6.010360	-1.896889	2.358110
C	-4.255535	-1.110431	1.410929
H	-3.750175	-0.894761	2.343175
C	-2.236500	-0.243367	0.208747
H	-1.738792	-0.362258	1.162389
H	-1.623555	-0.630297	-0.600354
C	-0.580351	2.120103	0.238750
C	-0.216090	3.454558	-0.467817
C	-0.493182	4.874978	-2.408801
H	-1.141282	4.901614	-3.280695
H	-0.633924	5.784862	-1.826142
H	0.547813	4.841696	-2.733077
C	-0.487822	2.135660	1.781306
C	0.658511	2.956886	3.753143
H	-0.194079	3.429773	4.244095
H	0.770961	1.956100	4.168151
H	1.556035	3.534170	3.959665
C	6.523770	-0.171076	-0.172070
C	5.711775	0.381654	-1.185731
C	4.431131	-0.066875	-1.378544
C	3.903444	-1.092940	-0.561369
C	4.721318	-1.646880	0.464668
C	6.038034	-1.165819	0.636629
C	4.150747	-2.664658	1.260089
C	2.865681	-3.064873	1.031388
C	2.114352	-2.449637	-0.006476
N	2.629073	-1.506816	-0.771494
C	0.716898	-2.856312	-0.273010

C	-0.104966	-3.350265	0.743418
C	-1.415815	-3.721863	0.477333
C	-1.923231	-3.616108	-0.811201
C	-1.115876	-3.118292	-1.828666
C	0.189851	-2.735086	-1.561773
H	7.531706	0.196149	-0.035465
H	6.107402	1.168831	-1.813188
H	3.793189	0.347846	-2.146042
H	6.654705	-1.593422	1.417026
H	4.742998	-3.124328	2.041394
H	2.424853	-3.858444	1.615554
H	0.268857	-3.416281	1.755666
H	-2.047154	-4.079523	1.278941
H	-2.946879	-3.895717	-1.014906
H	-1.508398	-3.022018	-2.831981
H	0.819205	-2.335504	-2.343014

10. References

(1) Matsuzaki, K.; Uno, H.; Tokunaga, E.; Shibata, N. Fluorobissulfonylmethyl Iodides: An Efficient Scaffold for Halogen Bonding Catalysts with an sp^3-Hybridized Carbon–Iodine Moiety. *ACS Catal.* **2018**, *8*, 6601-6605.

(2) Chang, Y. P.; Tang, T.; Jagannathan, J. R.; Hirbawi, N.; Sun, S.; Brown, J.; Franz, A. K. NMR Quantification of Halogen-Bonding Ability To Evaluate Catalyst Activity. *Org. Lett.* **2020**, *22*, 6647-6652.

(3) Zong, L.; Wang, C.; Moeljadi, A. M.; Ye, X.; Ganguly, R.; Li, Y.; Hirao, H.; Tan, C. H. Bisguanidinium Dinuclear Oxodiperoxomolybdosulfate Ion Pair-Catalyzed Enantioselective Sulfoxidation. *Nat Commun* **2016**, *7*, 13455.

(4) Cao, W.; Tan, D.; Lee, R.; Tan, C.-H. Enantioselective 1,2-Anionotropic Rearrangement of Acylsilane through a Bisguanidinium Silicate Ion Pair. *J. Am. Chem. Soc.* **2018**, *140*, 1952-1955.

(5) Dul, M.-C.; Bourgeois, D.; Maynadié, J.; Meyer, D. Synthesis of fluorinated malonamides and use in L/L extraction of f-elements. *Tetrahedron Lett.* **2013**, *54*, 6271-6274.

(6) Cho, C. S.; Ren, W. X.; Shim, S. C. A Copper(II)-Catalyzed Protocol for Modified Friedländer Quinoline Synthesis. *Tetrahedron Lett.* **2006**, *47*, 6781-6785.

(7) Wang, Y.; Dong, B.; Wang, Z.; Cong, X.; Bi, X. Silver-Catalyzed Reduction of Quinolines in Water. *Org. Lett.* **2019**, *21*, 3631-3634.

(8) Tang, J.; Dong, W.; Chen, F.; Deng, L.; Xian, M. Rhodium catalysts with cofactor mimics for the biomimetic reduction of C=N bonds. *Catal. Sci. Technol.* **2021**, *11*, 5564-5569.

(9) Tobisu, M.; Hyodo, I.; Chatani, N. Nickel-Catalyzed Reaction of Arylzinc Reagents with N-Aromatic Heterocycles: A Straightforward Approach to C−H Bond Arylation of Electron-Deficient Heteroaromatic Compounds. *J. Am. Chem. Soc.* **2009**, *131*, 12070-12071.

(10) Martínez, R.; Ramón, D. J.; Yus, M. Transition-Metal-Free Indirect Friedländer Synthesis of Quinolines from Alcohols. *J. Org. Chem.* **2008**, *73*, 9778-9780.

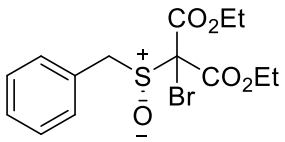
(11) Wang, T.; Zhuo, L.-G.; Li, Z.; Chen, F.; Ding, Z.; He, Y.; Fan, Q.-H.; Xiang, J.; Yu, Z.-X.; Chan, A. S. C. Highly Enantioselective Hydrogenation of Quinolines Using Phosphine-Free Chiral Cationic Ruthenium Catalysts: Scope, Mechanism, and Origin of Enantioselectivity. *J. Am. Chem. Soc.* **2011**, *133*, 9878-9891.

(12) Maj, A. M.; Suisse, I.; Hardouin, C.; Agbossou-Niedercorn, F. Synthesis of new chiral 2-functionalized-1,2,3,4-tetrahydroquinoline derivatives via asymmetric hydrogenation of substituted quinolines. *Tetrahedron* **2013**, *69*, 9322-9328.

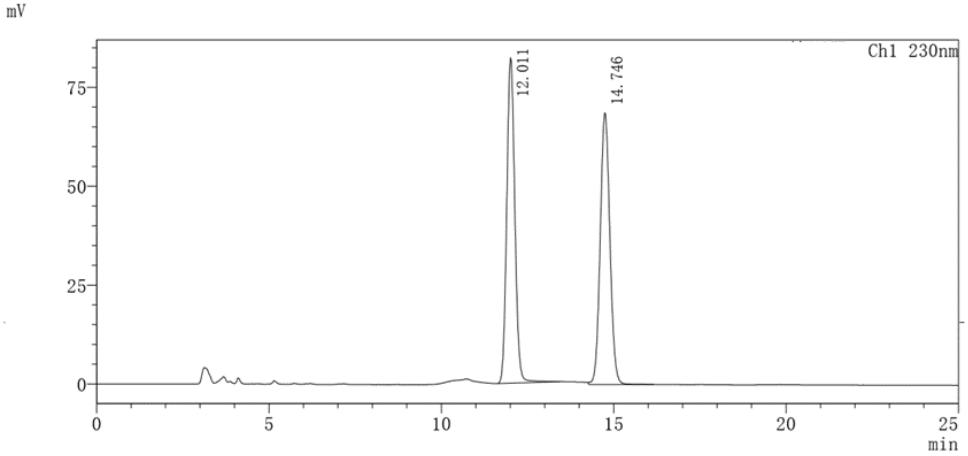
(13) Yang, C.-H.; Chen, X.; Li, H.; Wei, W.; Yang, Z.; Chang, J. Iodine Catalyzed Reduction of Quinolines under Mild Reaction *Conditions. Chem. Commun.* 2018, 54, 8622-8625.

(14) Yang, C. T.; Fu, Y.; Huang, Y. B.; Yi, J.; Guo, Q. X.; Liu, L. Room-Temperature Copper-Catalyzed Carbon-Nitrogen Coupling of Aryl Iodides and Bromides Promoted by Organic Ionic Bases. *Angew. Chem. Int. Ed. Engl.* **2009**, *48*, 7398-7401.
(15) Kato, H.; Shibata, I.; Yasaka, Y.; Tsunoi, S.; Yasuda, M.; Baba, A. The Reductive Amination of Aldehydes and Ketones by Catalytic Use of Dibutylchlorotin Hydride Complex. *Chem. Commun.* **2006**, 4189-4191.
(16) Liao, W.; Chen, Y.; Liu, Y.; Duan, H.; Petersen, J. L.; Shi, X. 1,2,3-Triazole-Boranes: Stable and Efficient Reagents for Ketone and Aldehyde Reductive Amination in Organic Solvents or in Water. *Chem. Commun.* 2009, 6436-6438.
(17) Likhar, P. R.; Arundhathi, R.; Kantam, M. L.; Prathima, P. S. Amination of Alcohols Catalyzed by Copper-Aluminium Hydrotalcite: A Green Synthesis of Amines. *Eur. J. Org. Chem.* **2009**, *2009*, 5383-5389.
(18) Barros, A. I. R. N. A.; Dias, A. F. R.; Silva, A. M. S. Reductive Coupling Reactions of 2-Nitrochalcones and their β-Hydroxy-analogues: New Syntheses of 2-Arylquinoline and 2-Aryl-4-hydroxyquinoline Derivatives. *Monatsh. Chem.* **2007**, *138*, 585-594.

11. HPLC trace

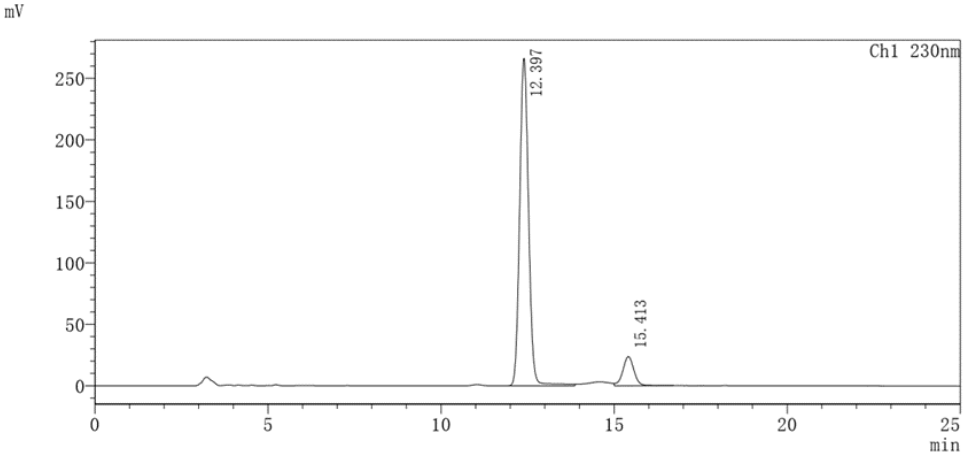


(*R*)-**3a** (80% *ee*)



Detect A 230nm

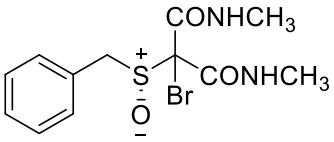
Peak	Ret. Time	Area	Height	Area%
1	12.011	1332657	82160	49.946
2	14.746	1335527	68674	50.054
Total		2668183	150834	100.000



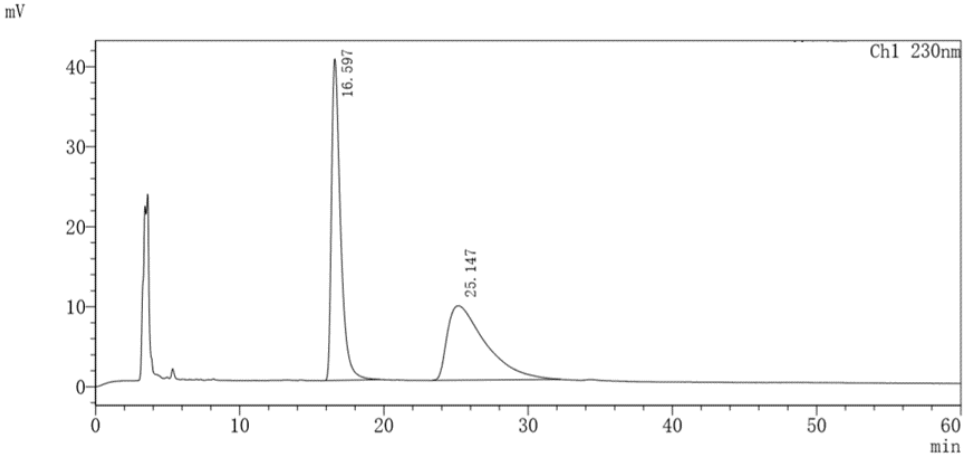
Detect A 230nm

Peak	Ret. Time	Area	Height	Area%
1	12.397	4776564	266288	90.107
2	15.413	524431	23643	9.893
Total		5300995	289931	100.000

Figure S28. HPLC spectra for 3a.

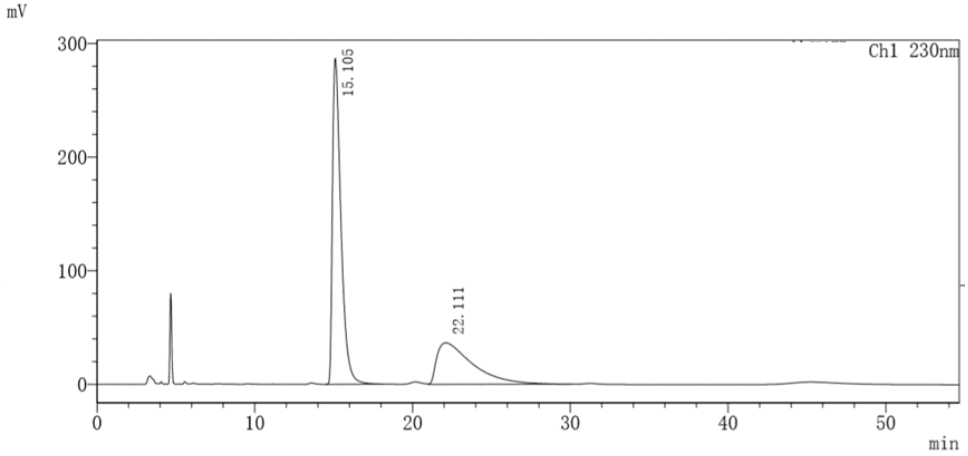


(*R*)-**3b** (30% ee)



Detect A 230nm

Peak	Ret. Time	Area	Height	Area%
1	16.597	1749579	40175	50.560
2	25.147	1710826	9298	49.440
Total		3460405	49474	100.000



Detect A 230nm

Peak	Ret. Time	Area	Height	Area%
1	15.105	10729309	286890	65.066
2	22.111	5760489	36486	34.934
Total		16489798	323376	100.000

Figure S29. HPLC spectra for 3b.

12. NMR spectra

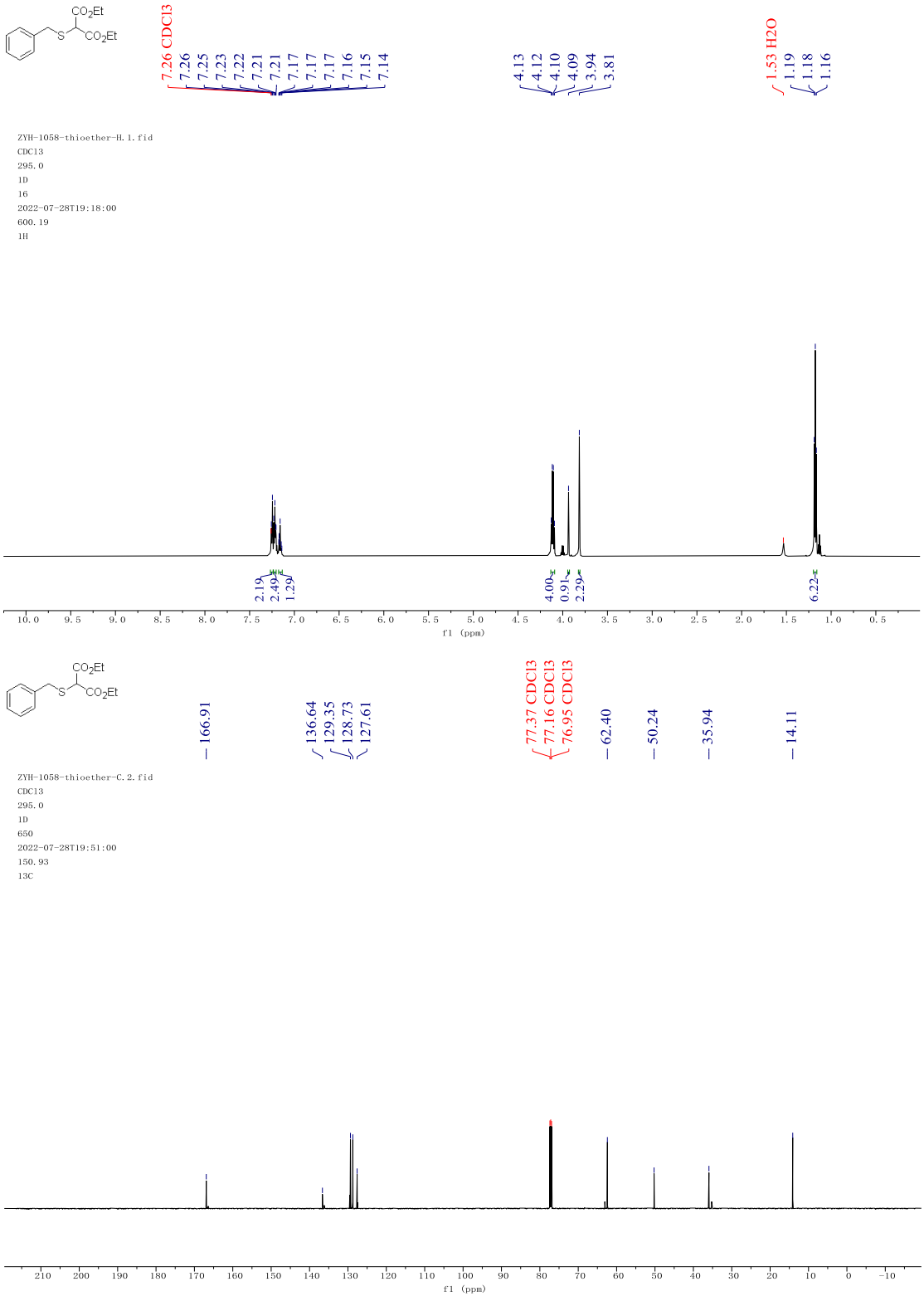


Figure S30. ^{1}H and ^{13}C NMR spectra of 1a in $CDCl_3$.

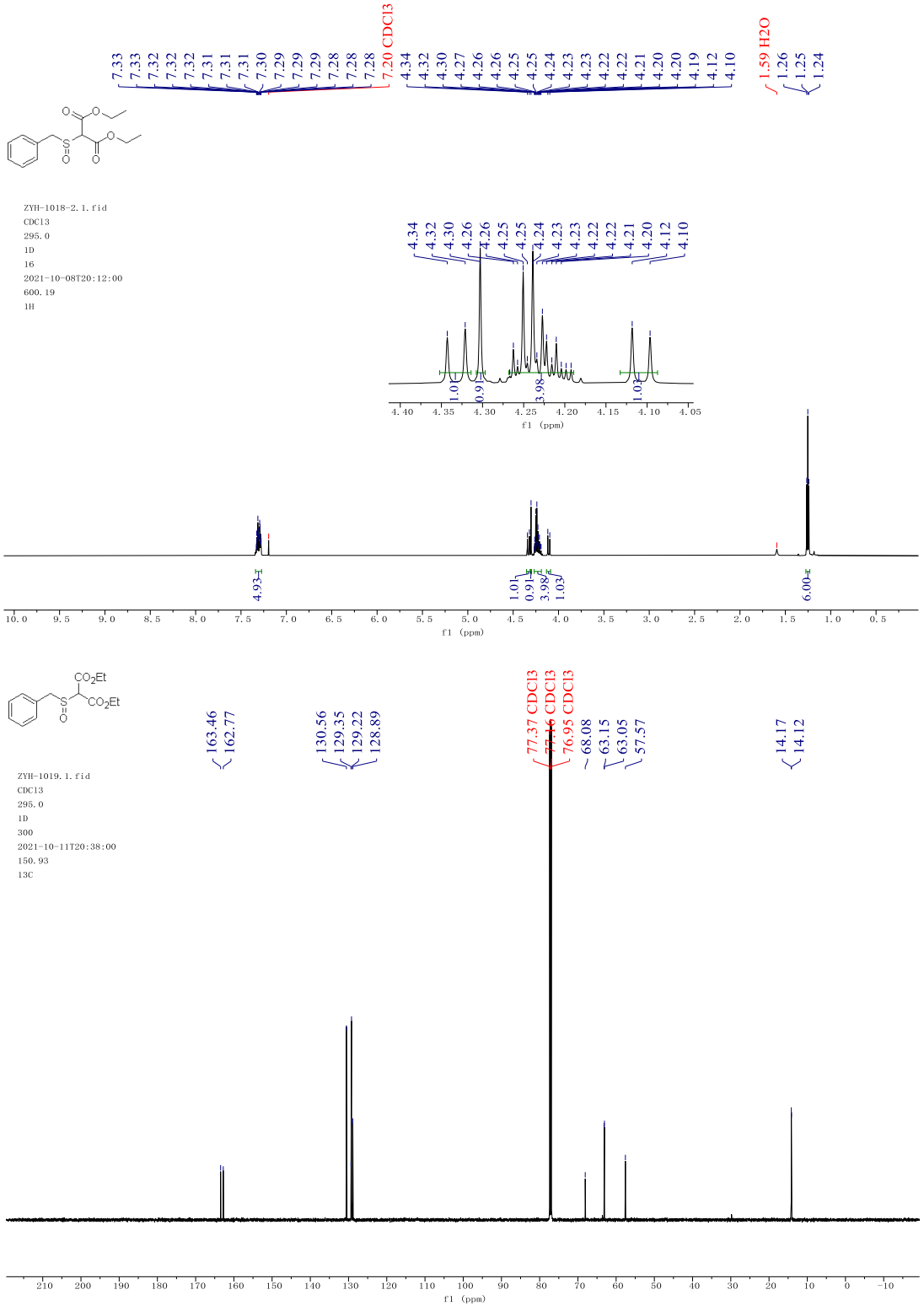


Figure S31. ^{1}H and ^{13}C NMR spectra of 2a in $CDCl_3$.

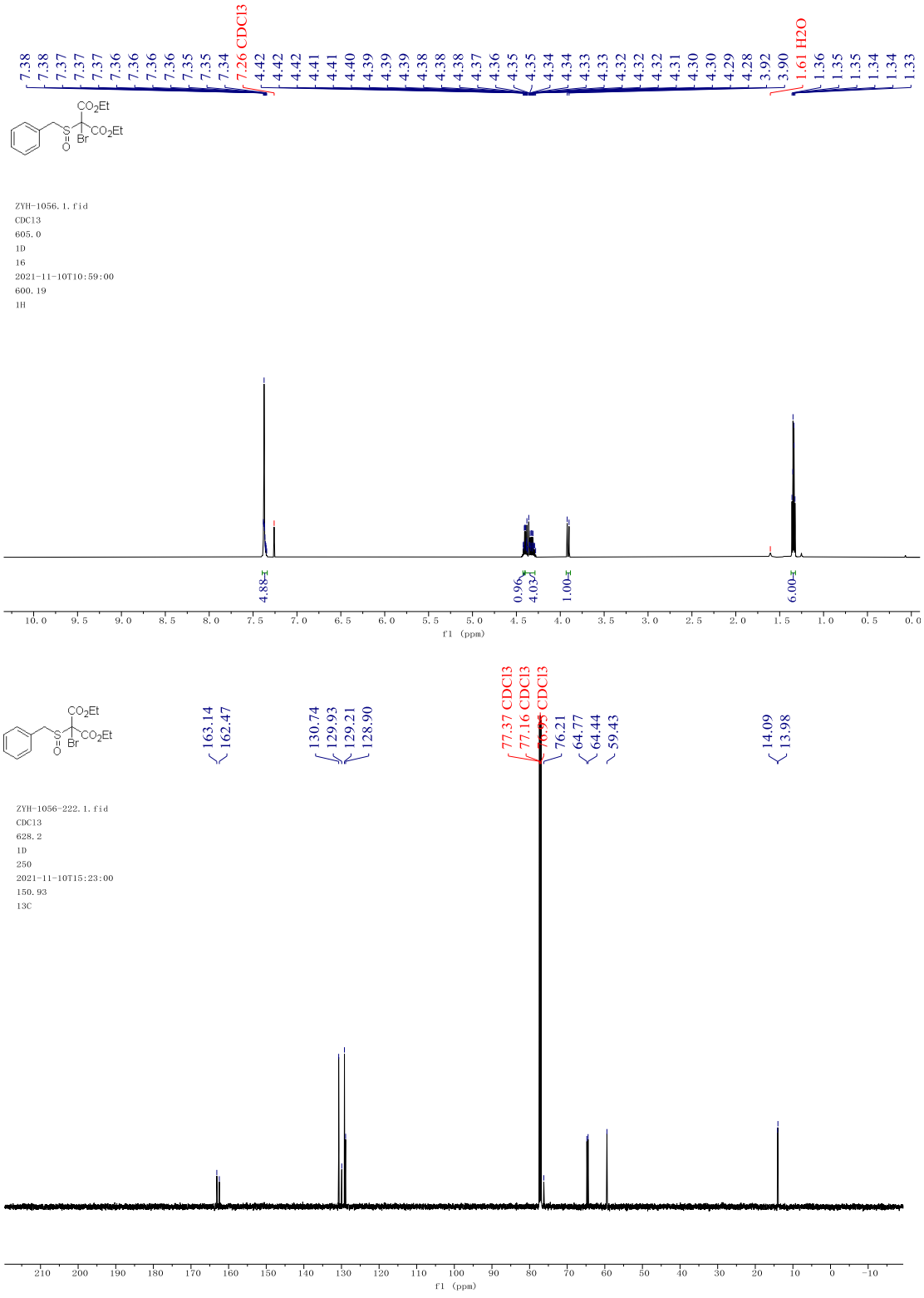


Figure S32. ^{1}H and ^{13}C NMR spectra of 3a in $CDCl_3$.

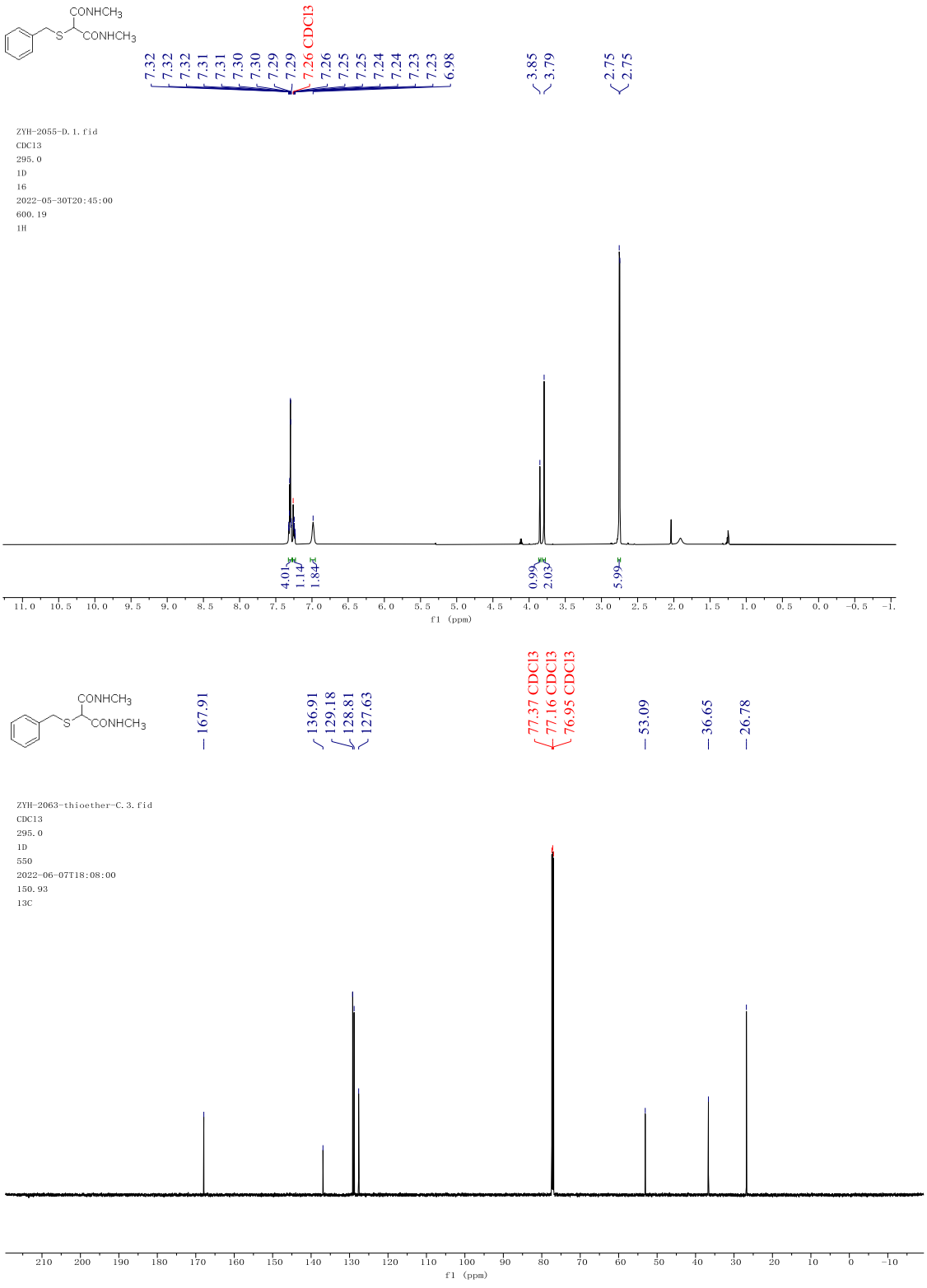


Figure S33. ^{1}H and ^{13}C NMR spectra of 1b in $CDCl_3$.

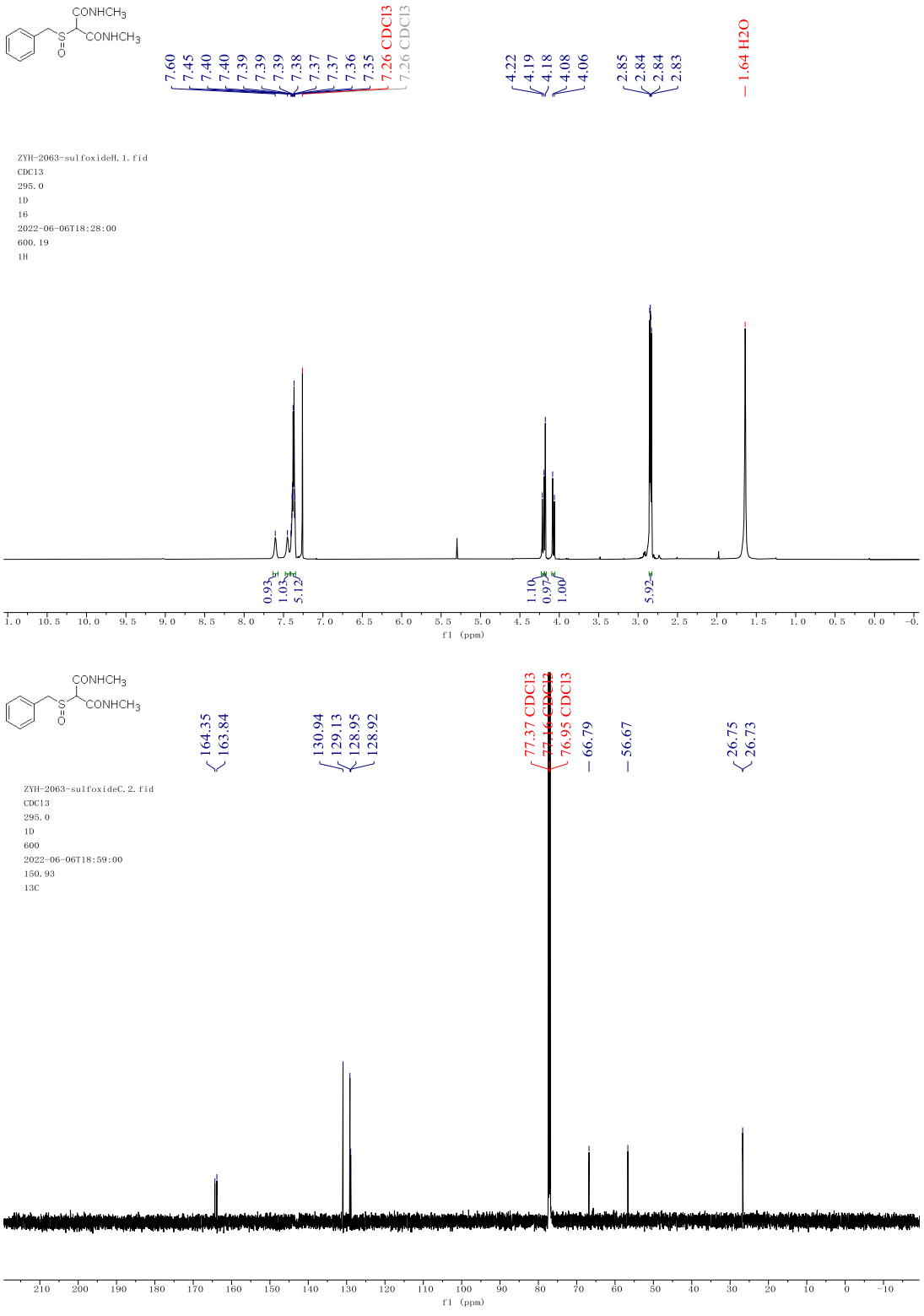


Figure S34. ^{1}H and ^{13}C NMR spectra of 2b in $CDCl_3$.

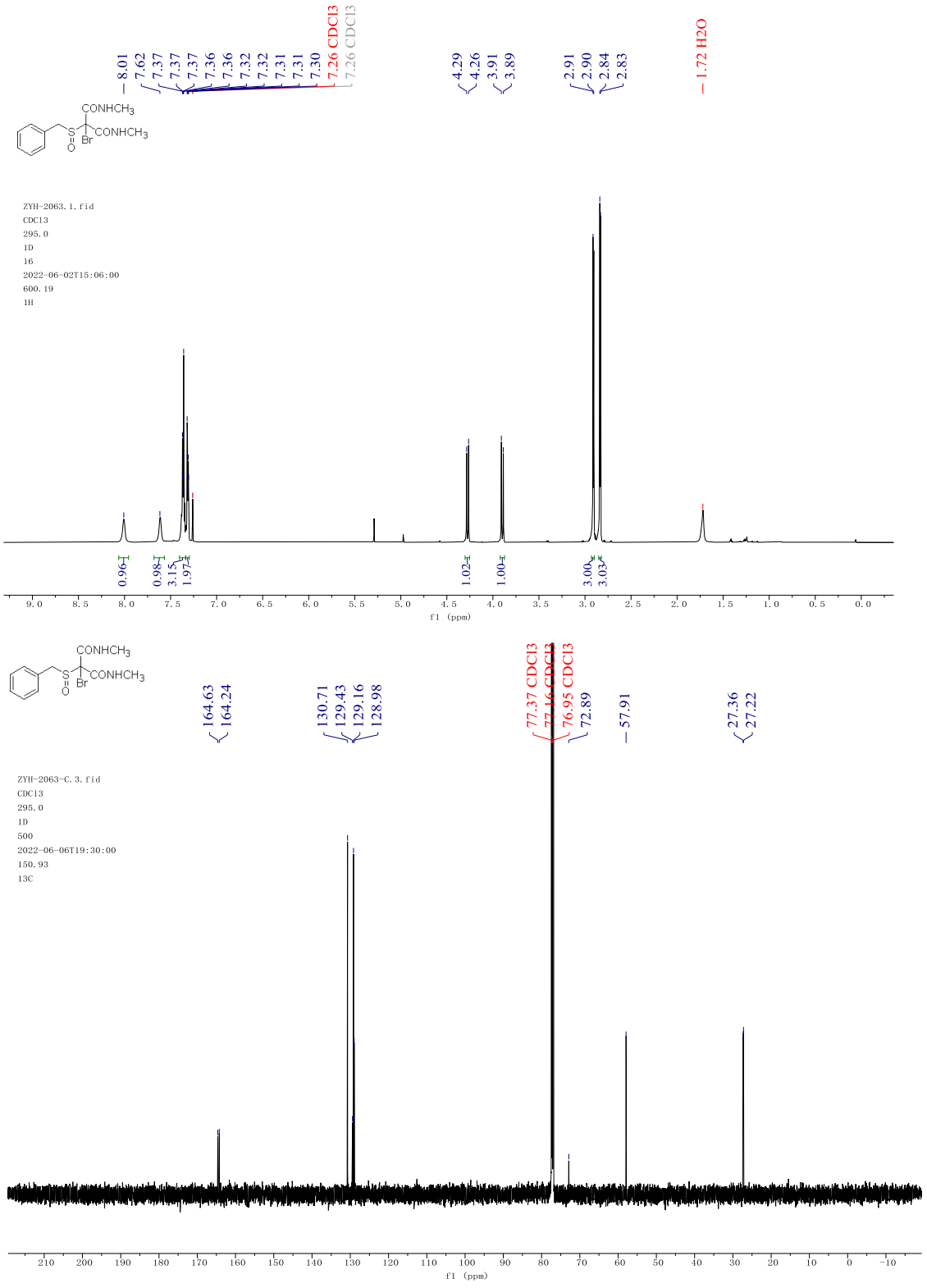


Figure S35. ^{1}H and ^{13}C NMR spectra of 3b in $CDCl_3$.

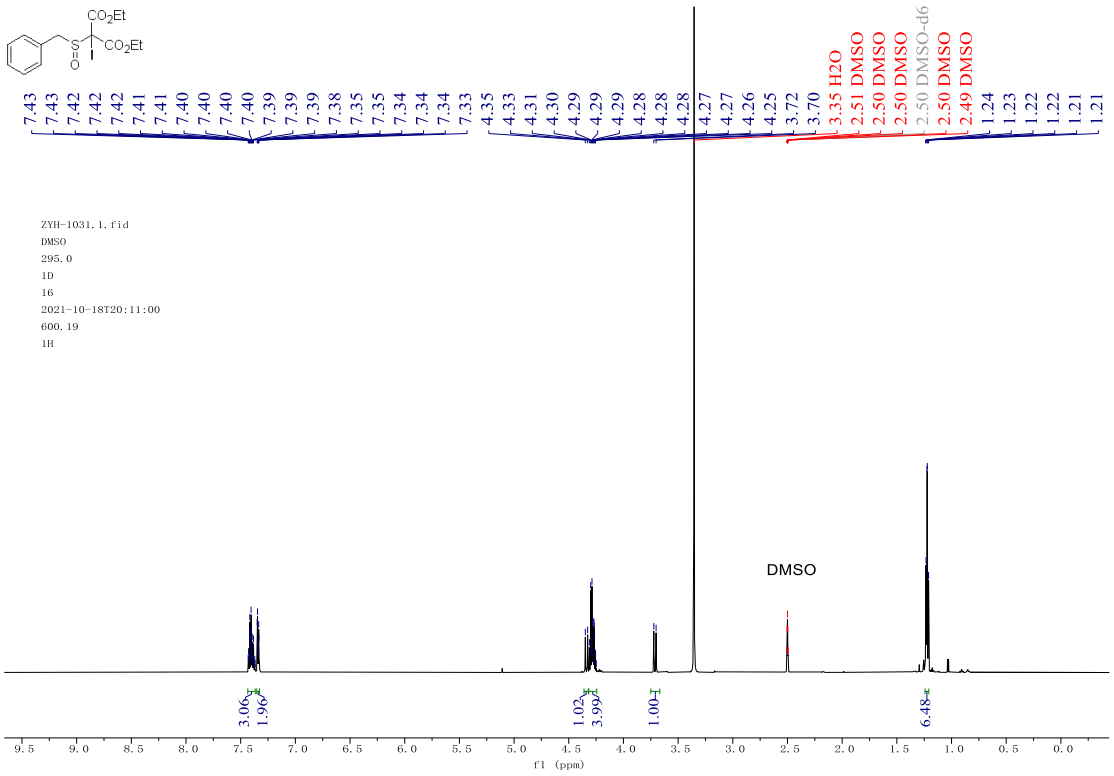


Figure S36. ^{1}H NMR spectrum of 3e in DMSO-d_6.

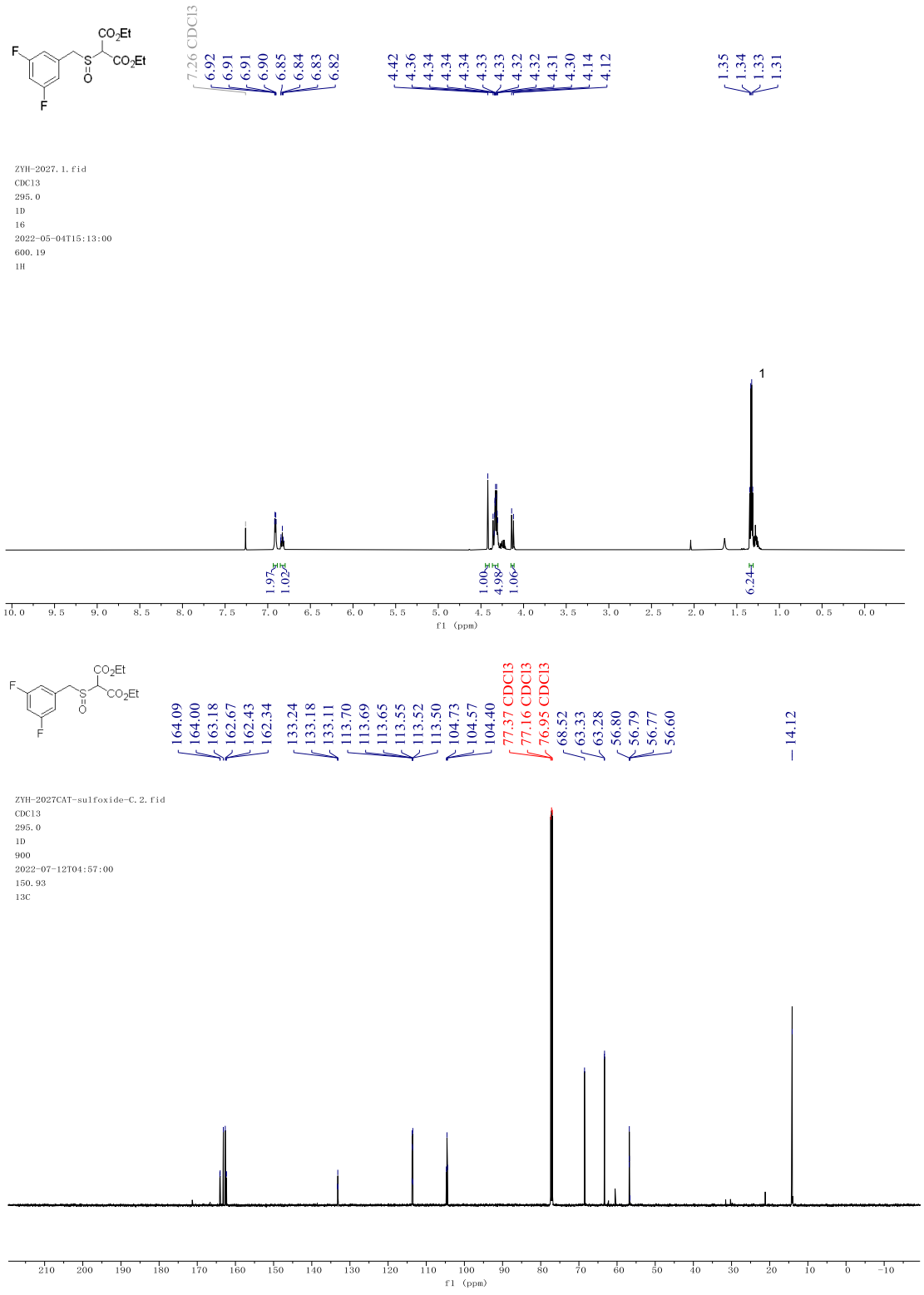


Figure S37. ^{1}H and ^{13}C NMR spectra of 2d in $CDCl_3$.

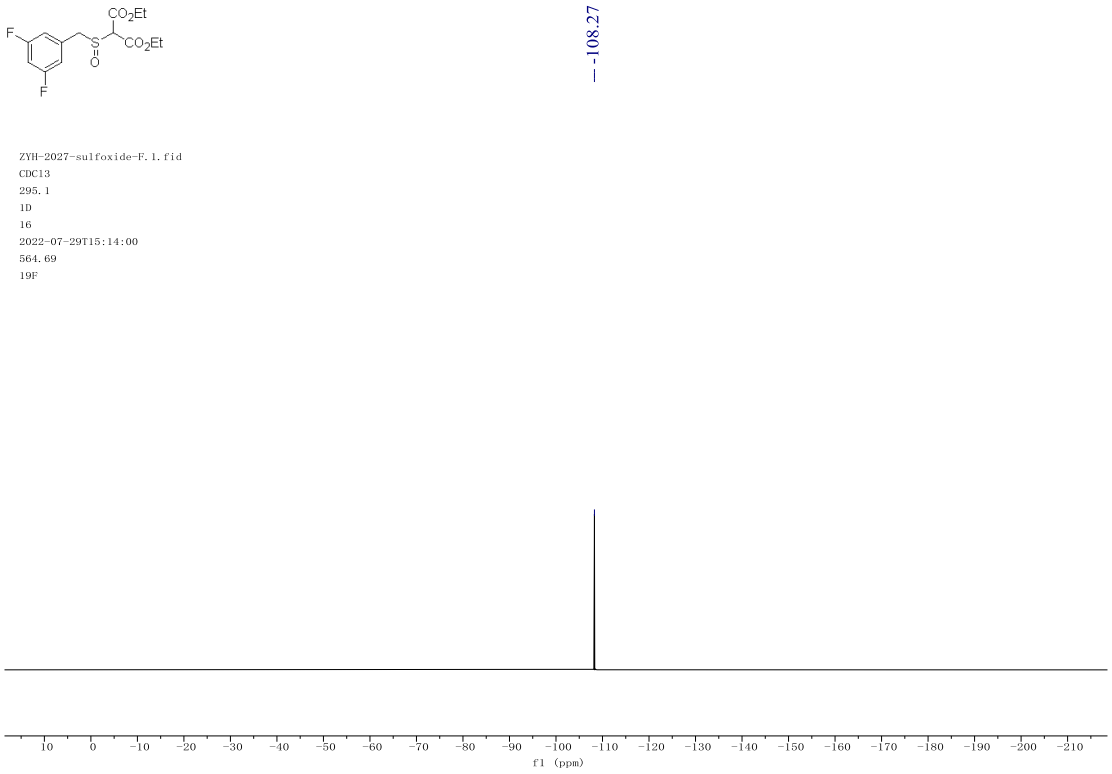


Figure S38. ^{19}F NMR spectrum of 2d in $CDCl_3$.

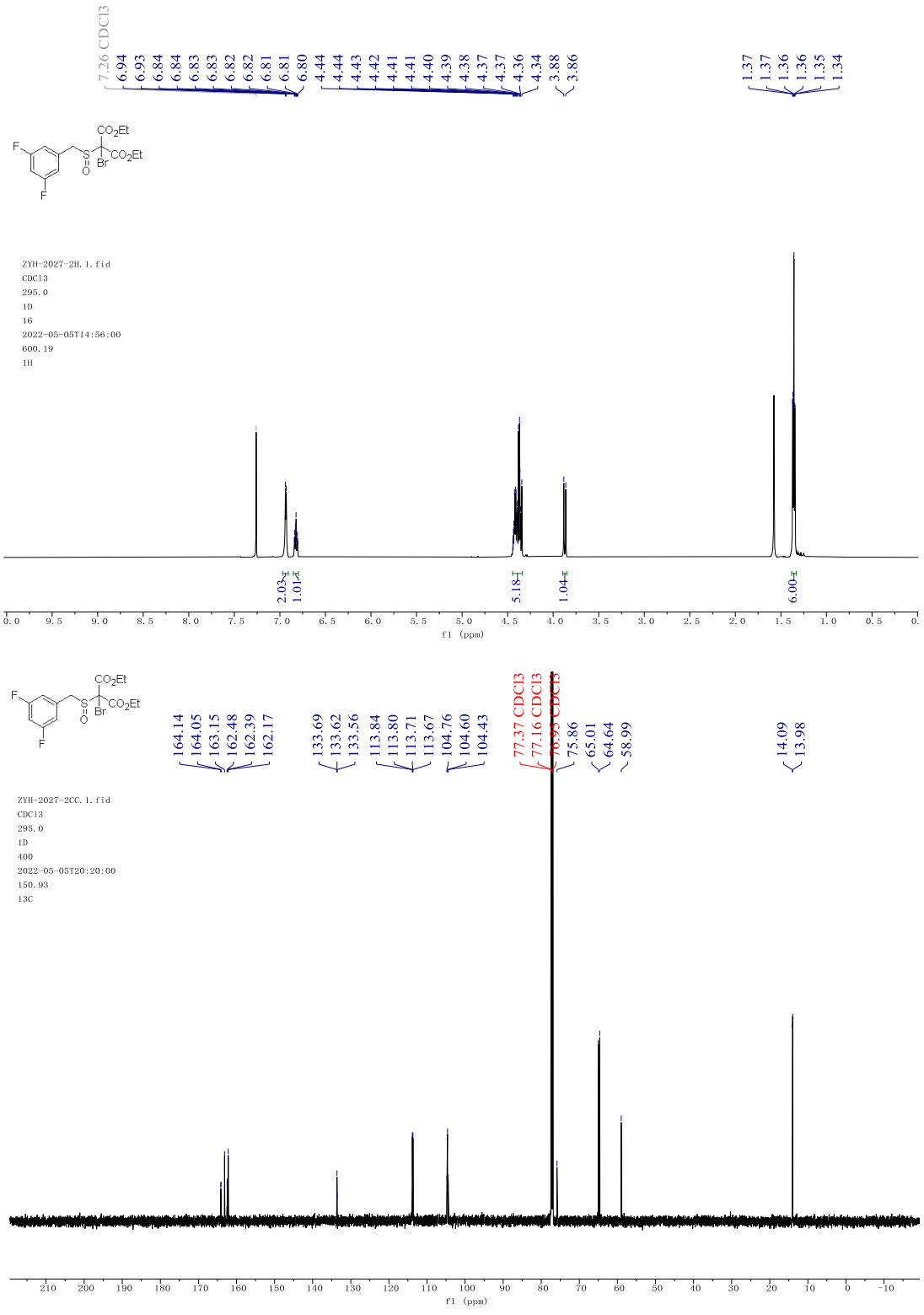


Figure S39. ^{1}H and ^{13}C NMR spectra of 3d in $CDCl_3$.

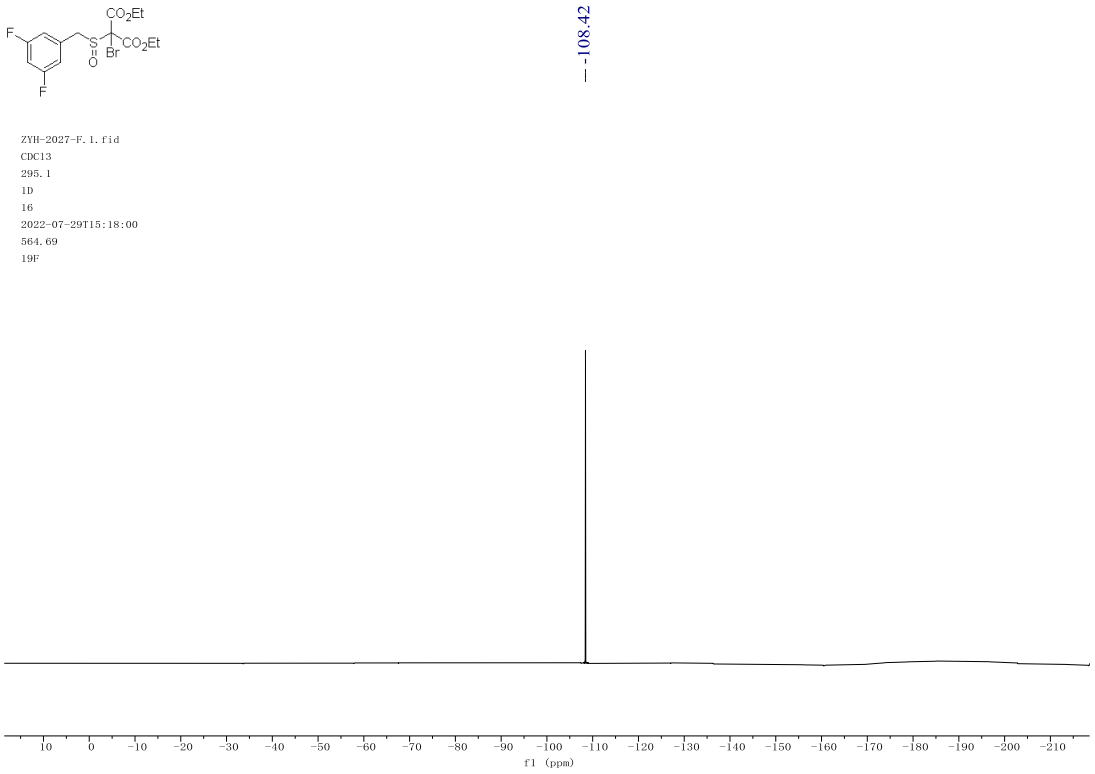


Figure S40. ^{19}F NMR spectrum of 3d in $CDCl_3$.

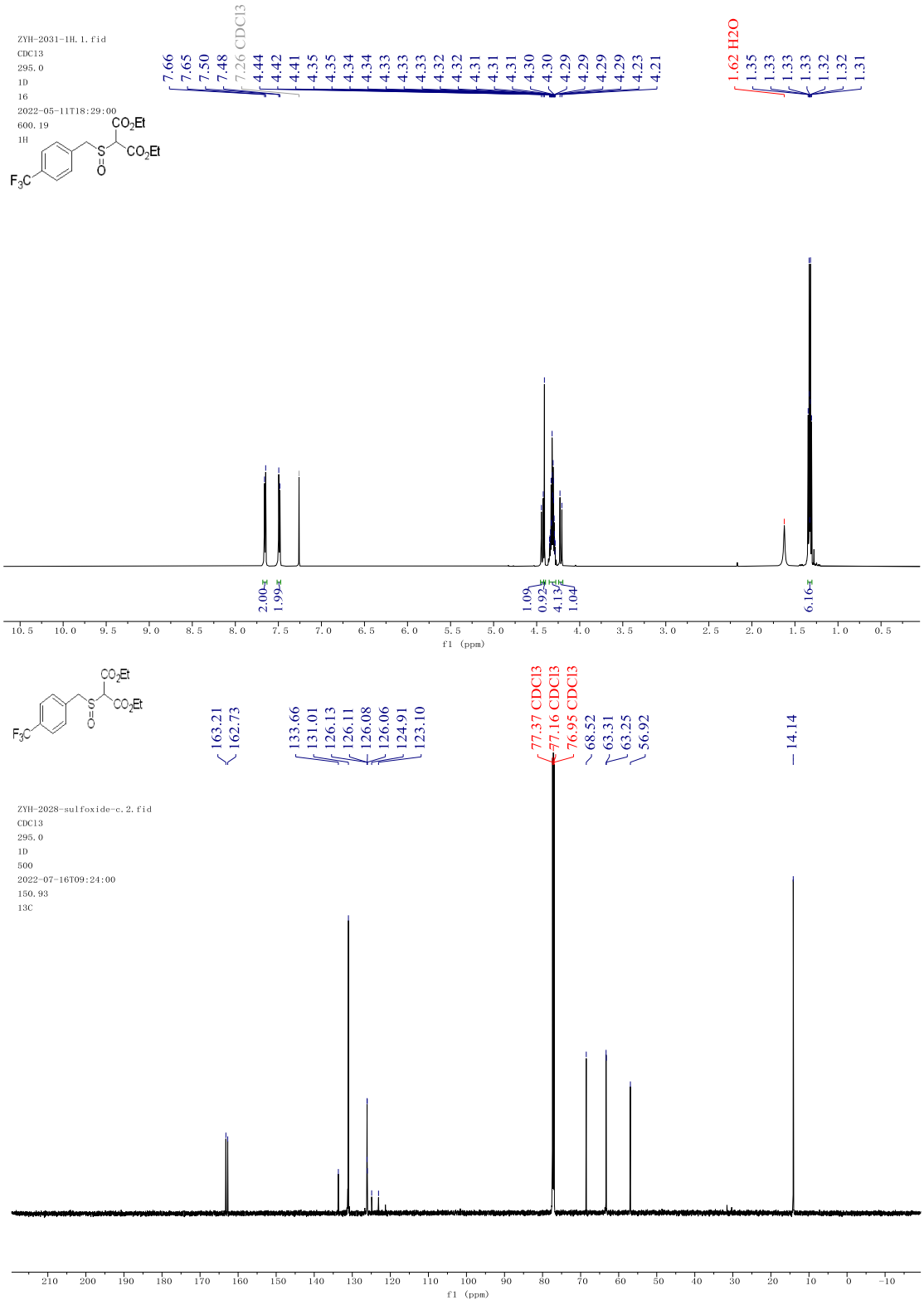


Figure S41. ^{1}H and ^{13}C NMR spectra of 2d' in $CDCl_3$.

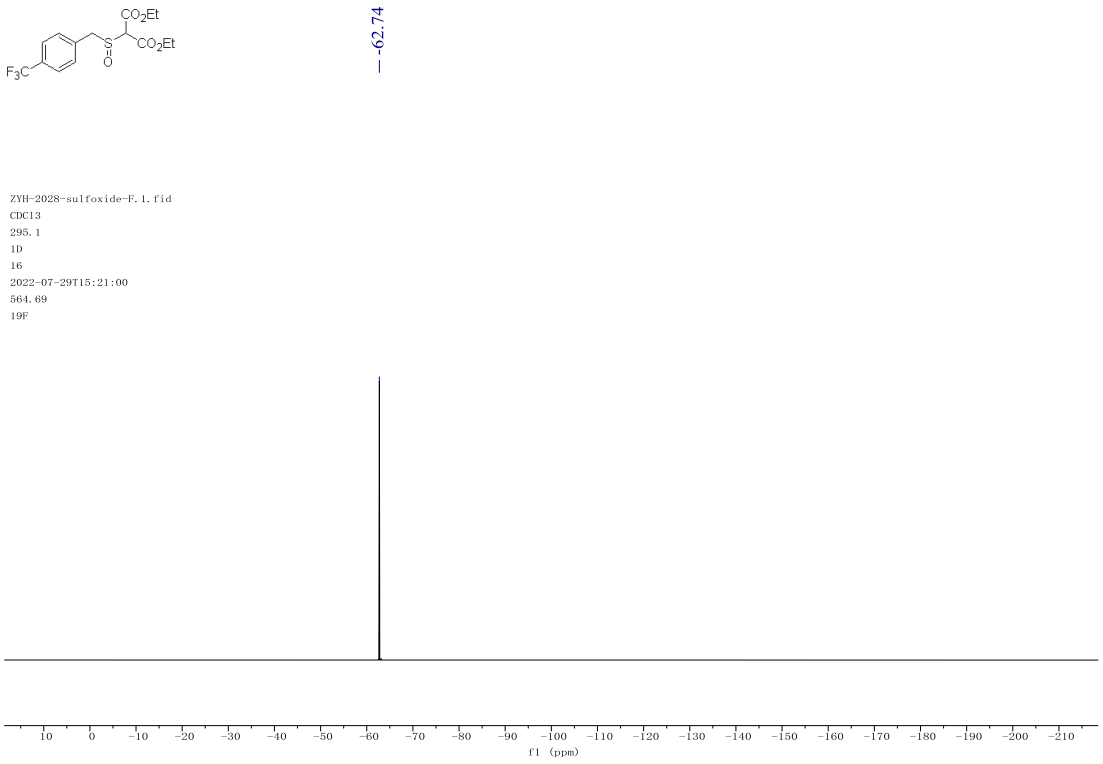


Figure S42. ^{19}F NMR spectrum of 2d' in $CDCl_3$.

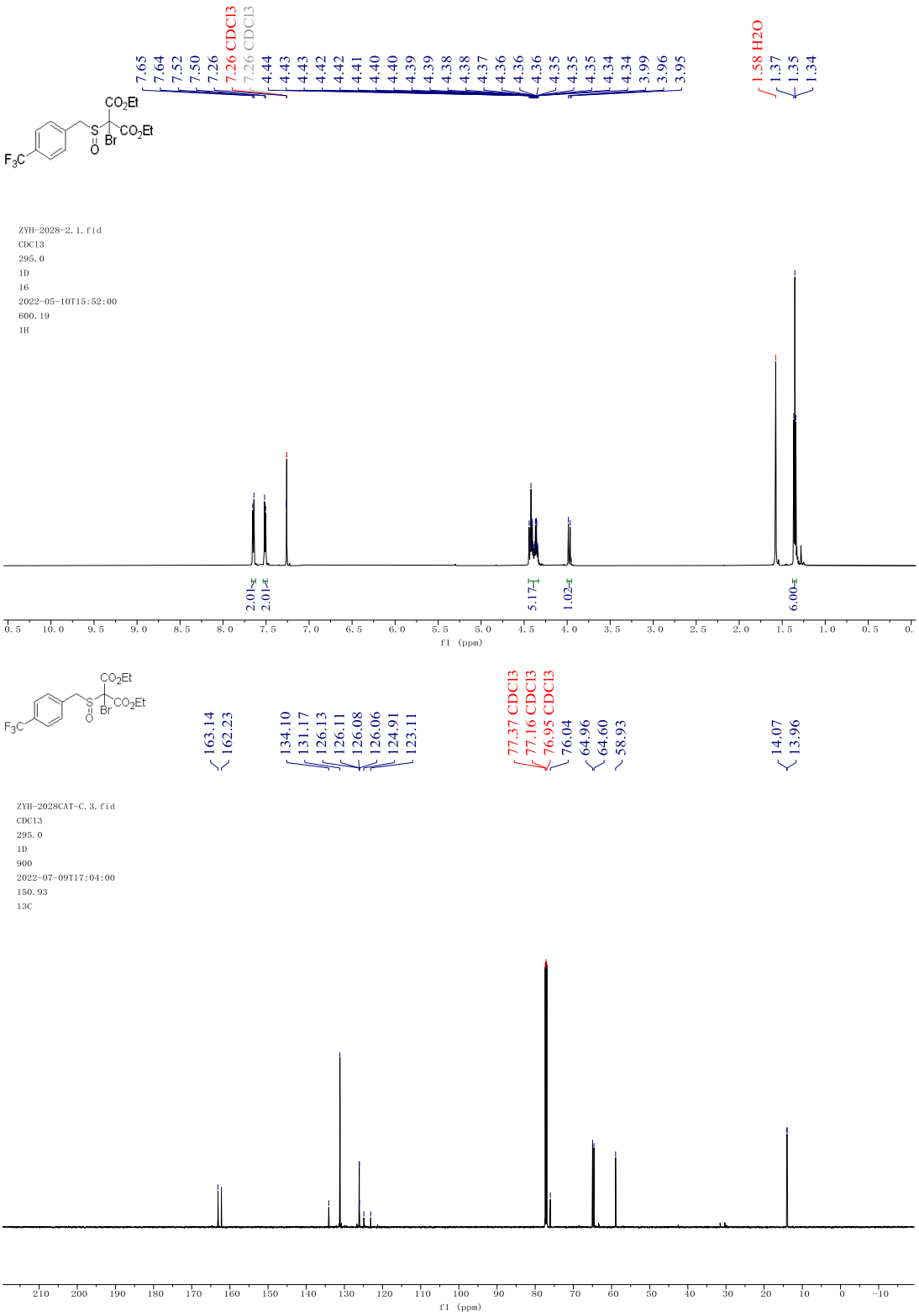


Figure S43. ^{1}H and ^{13}C NMR spectra of 3d' in $CDCl_3$.

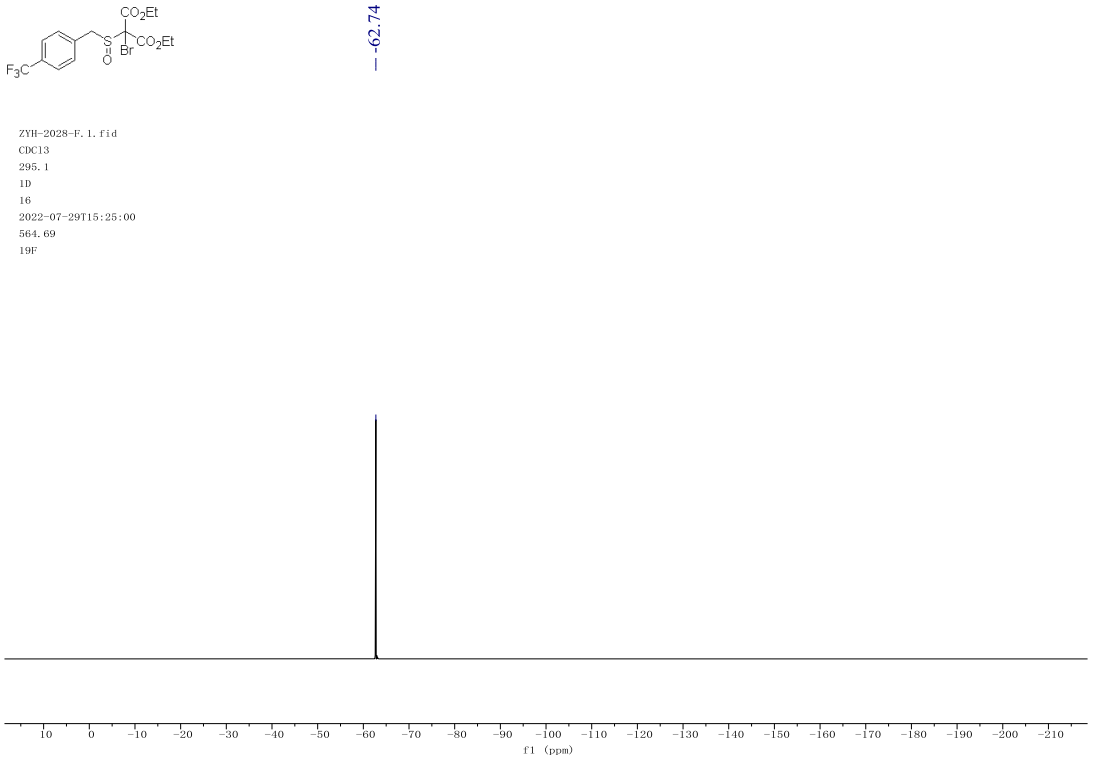


Figure S44. ^{19}F NMR spectrum of 3d' in $CDCl_3$.

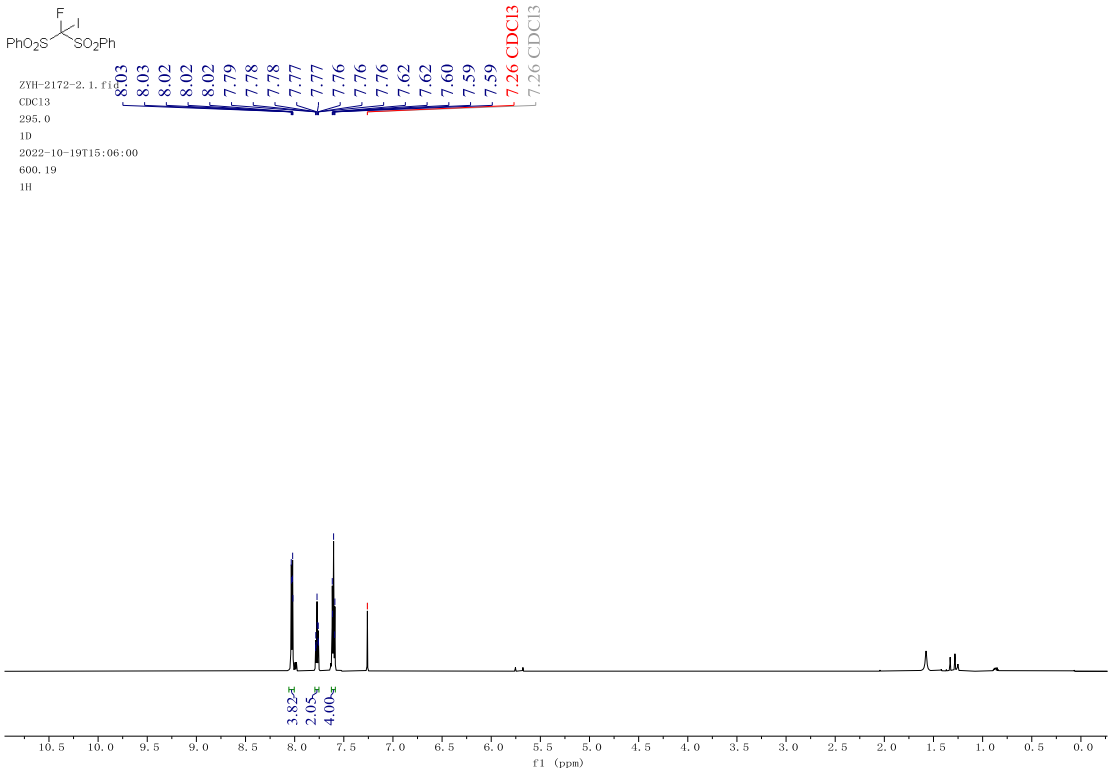


Figure S45. ^{1}H NMR spectrum of 3f in $CDCl_3$.

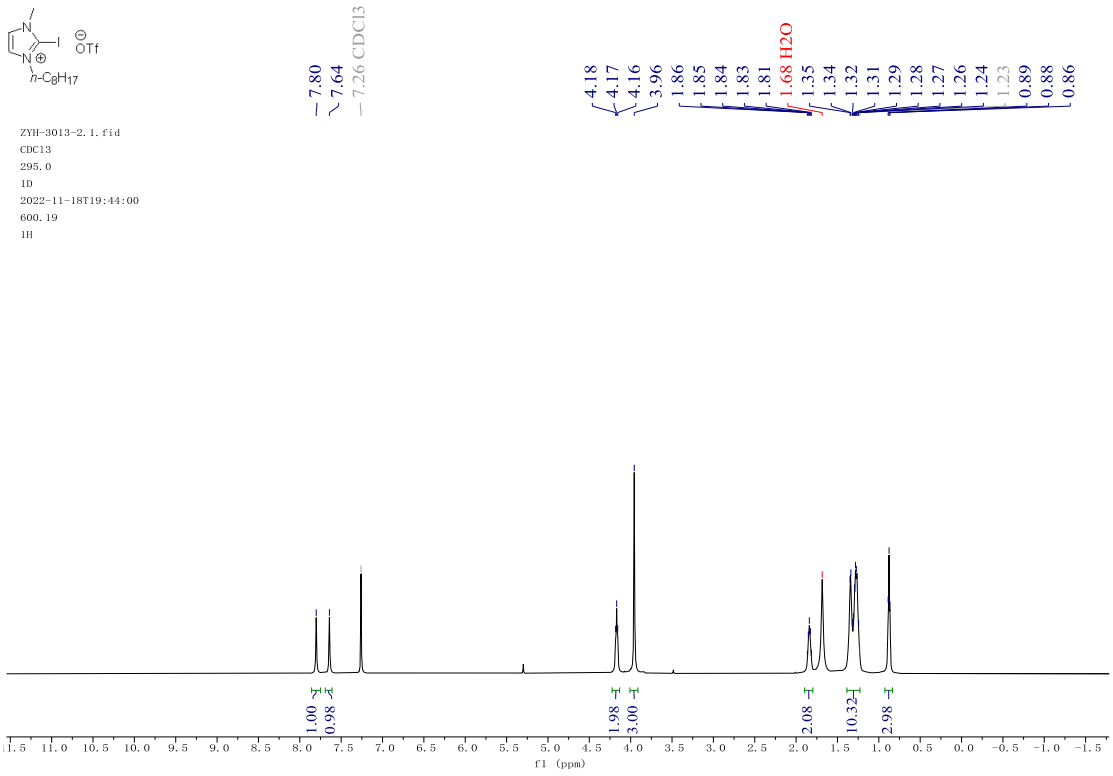


Figure S46. ^{1}H NMR spectrum of 3g in $CDCl_3$.

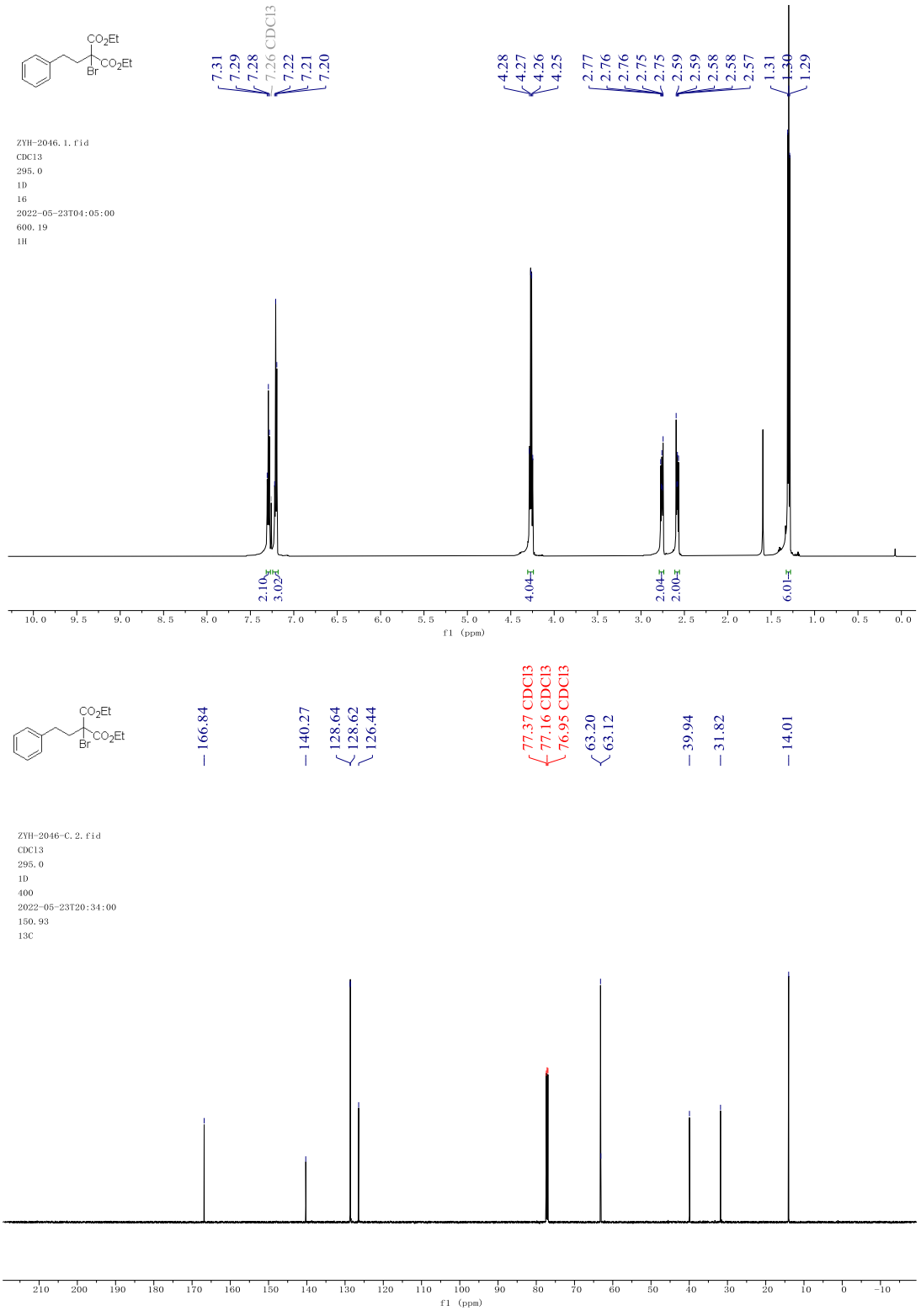


Figure S47. ^{1}H and ^{13}C NMR spectra of 3h in $CDCl_3$.

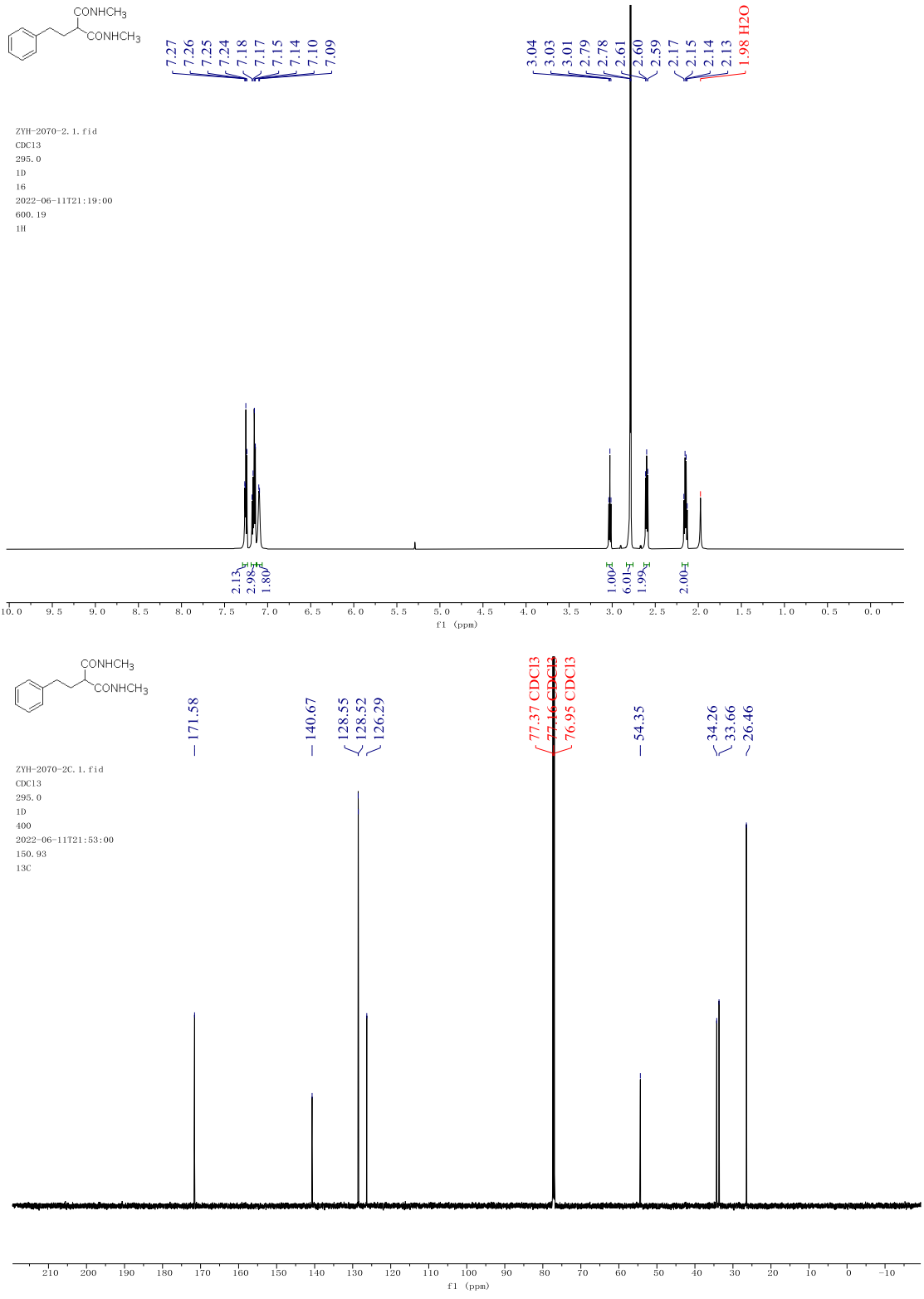


Figure S48. ^{1}H and ^{13}C NMR spectra of 2i in $CDCl_3$.

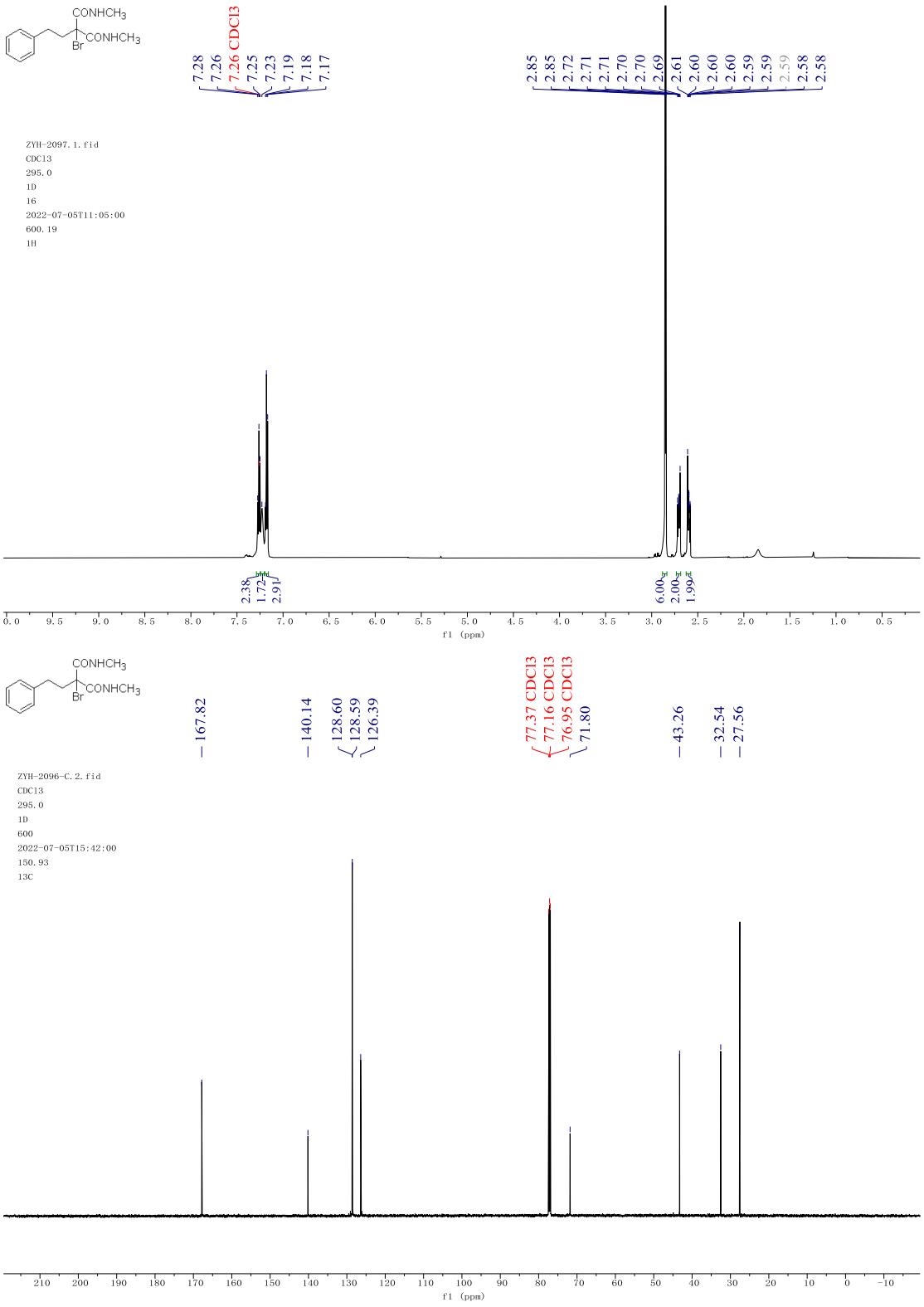


Figure S49. ^{1}H and ^{13}C NMR spectra of 3i in $CDCl_3$.

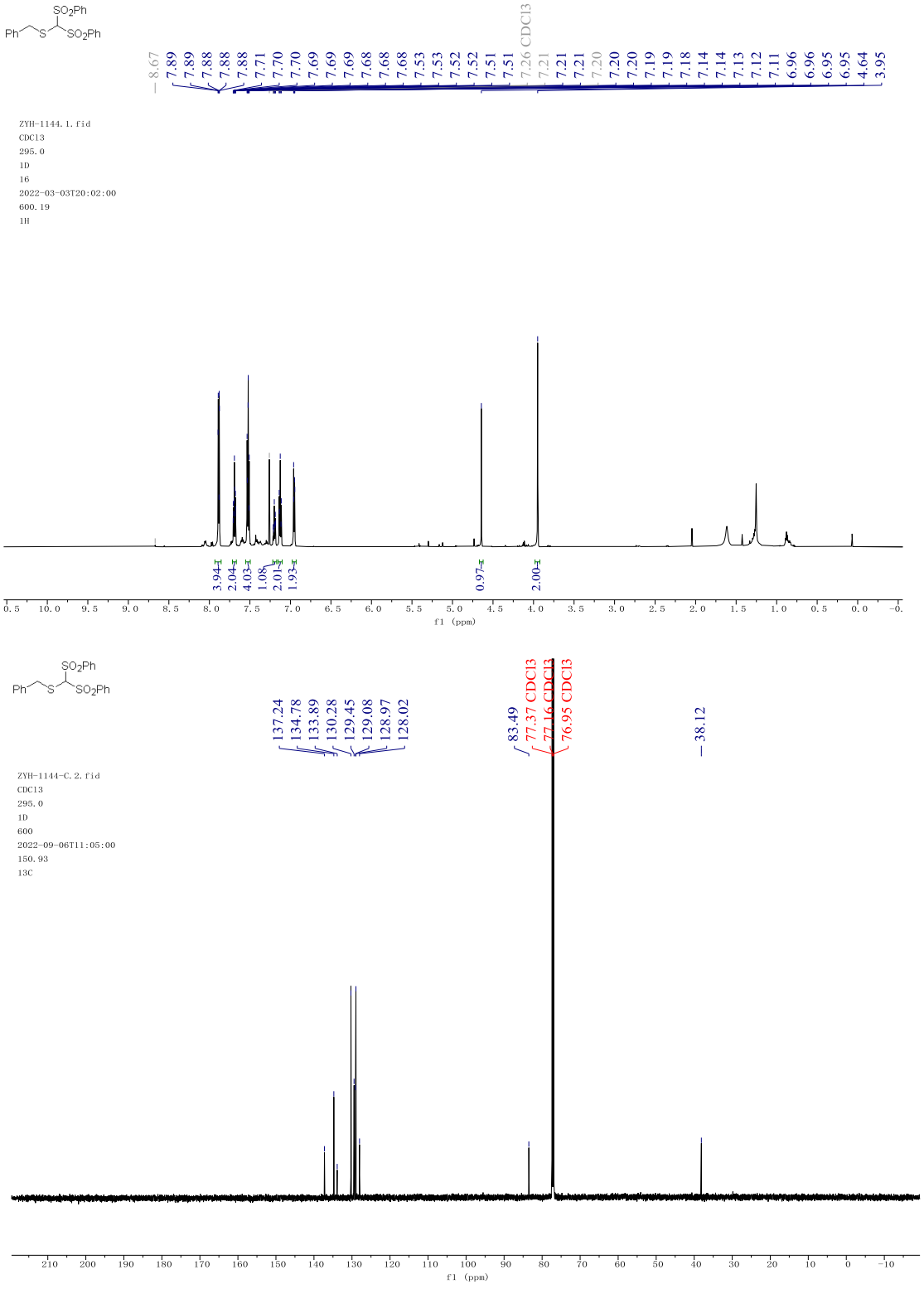


Figure S50. ^{1}H and ^{13}C NMR spectra of 1c in $CDCl_3$.

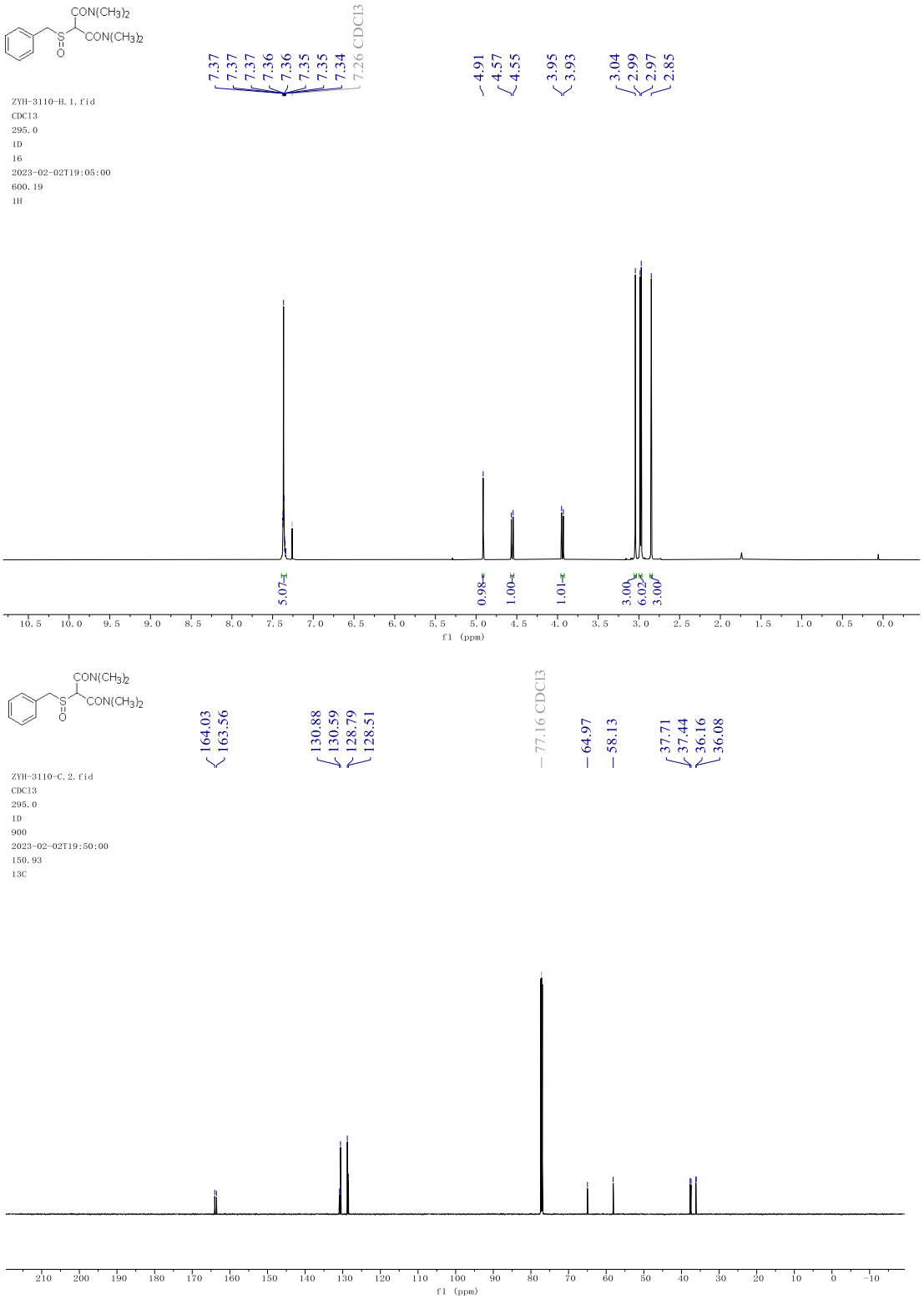


Figure S51. ^{1}H and ^{13}C NMR spectra of 2j in $CDCl_3$.

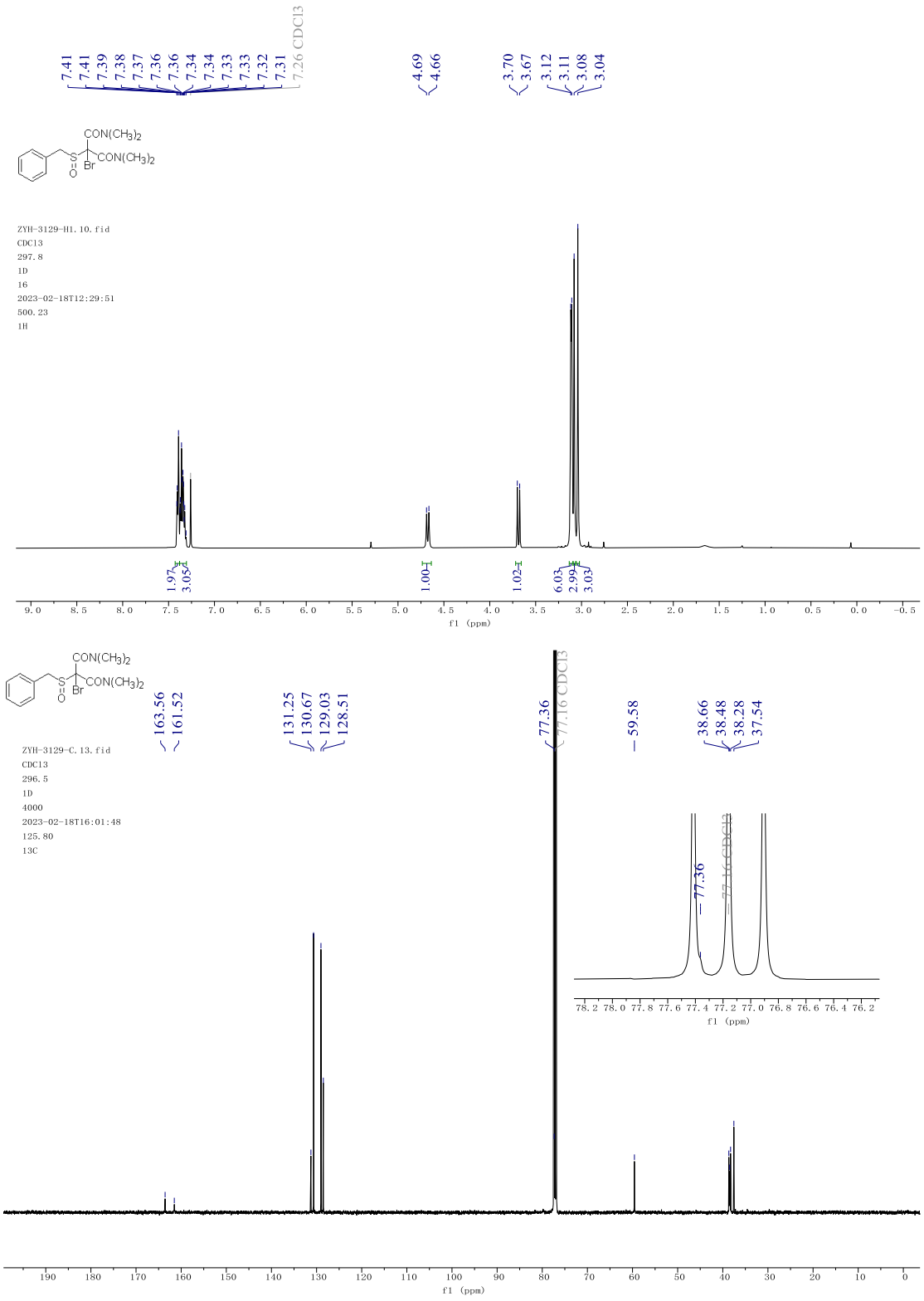


Figure S52. ^{1}H and ^{13}C NMR spectra of 3j in $CDCl_3$

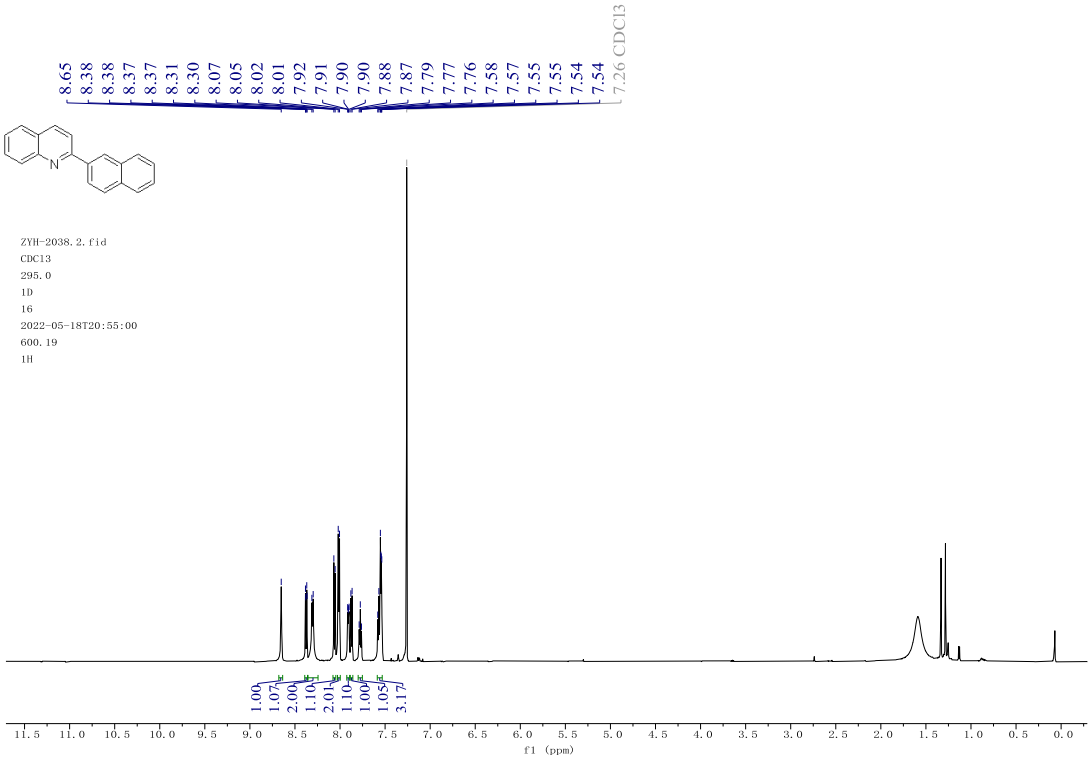


Figure S53. ^{1}H NMR spectrum of 4b in $CDCl_3$.

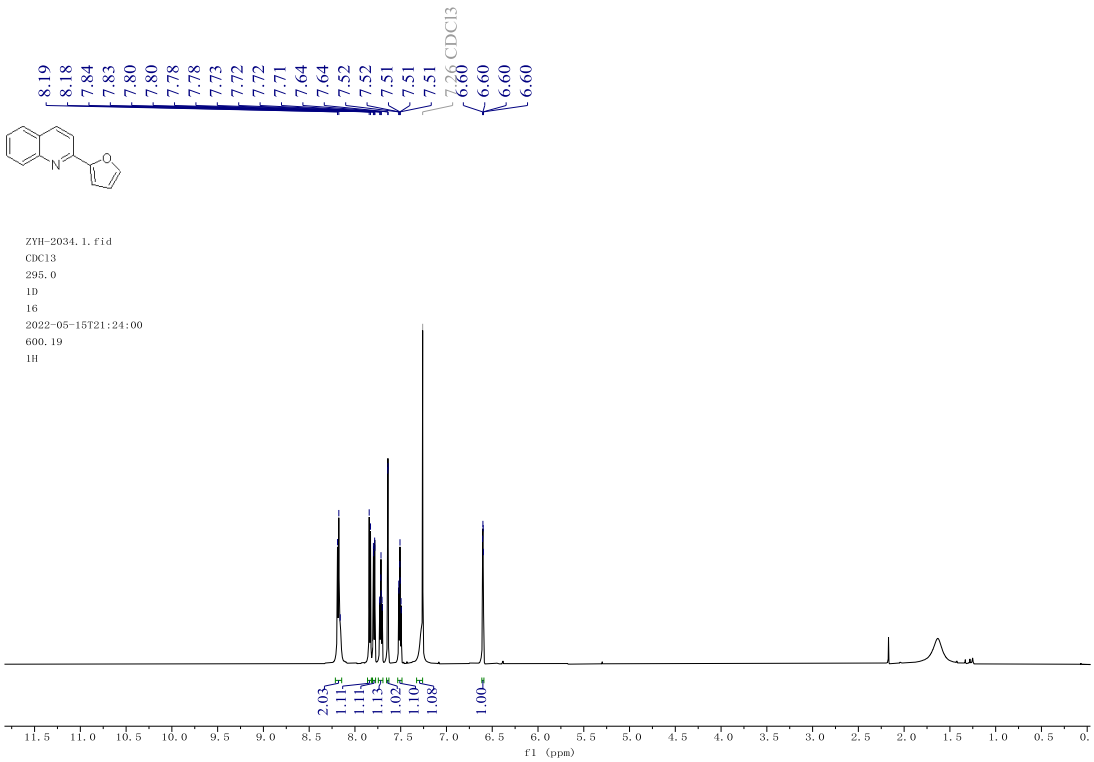


Figure S54. ^{1}H NMR spectrum of 4c in $CDCl_3$.

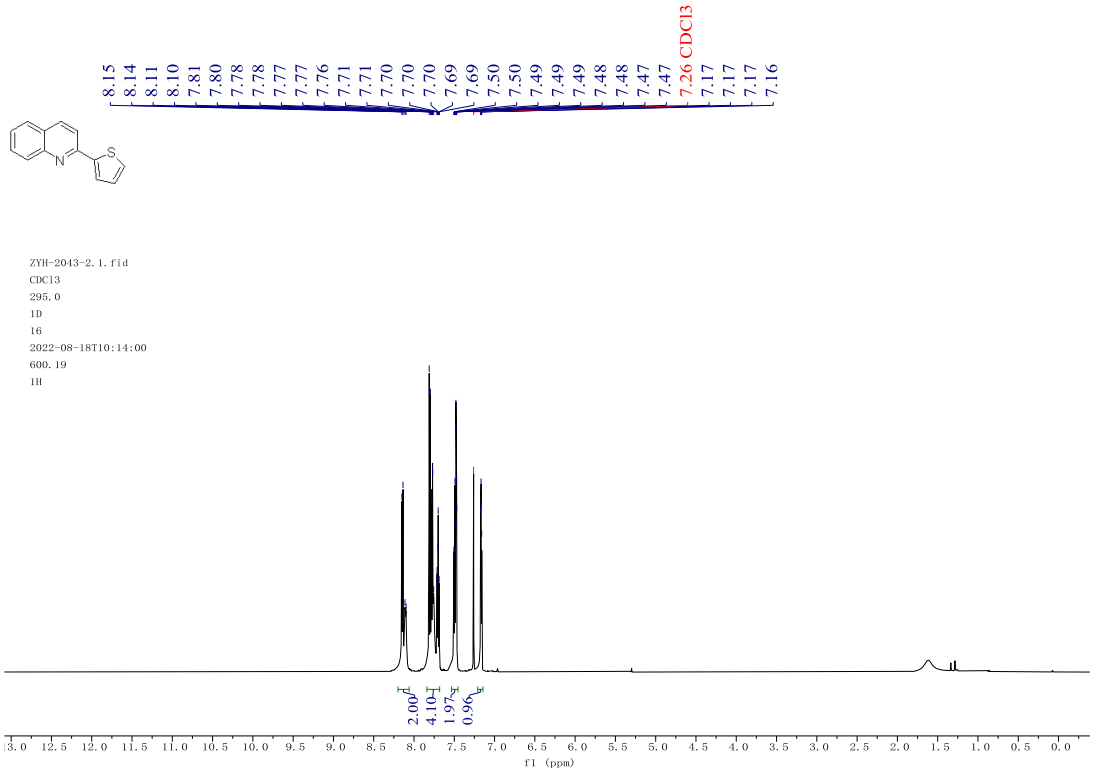


Figure S55. ^{1}H NMR spectrum of 4d in $CDCl_3$.

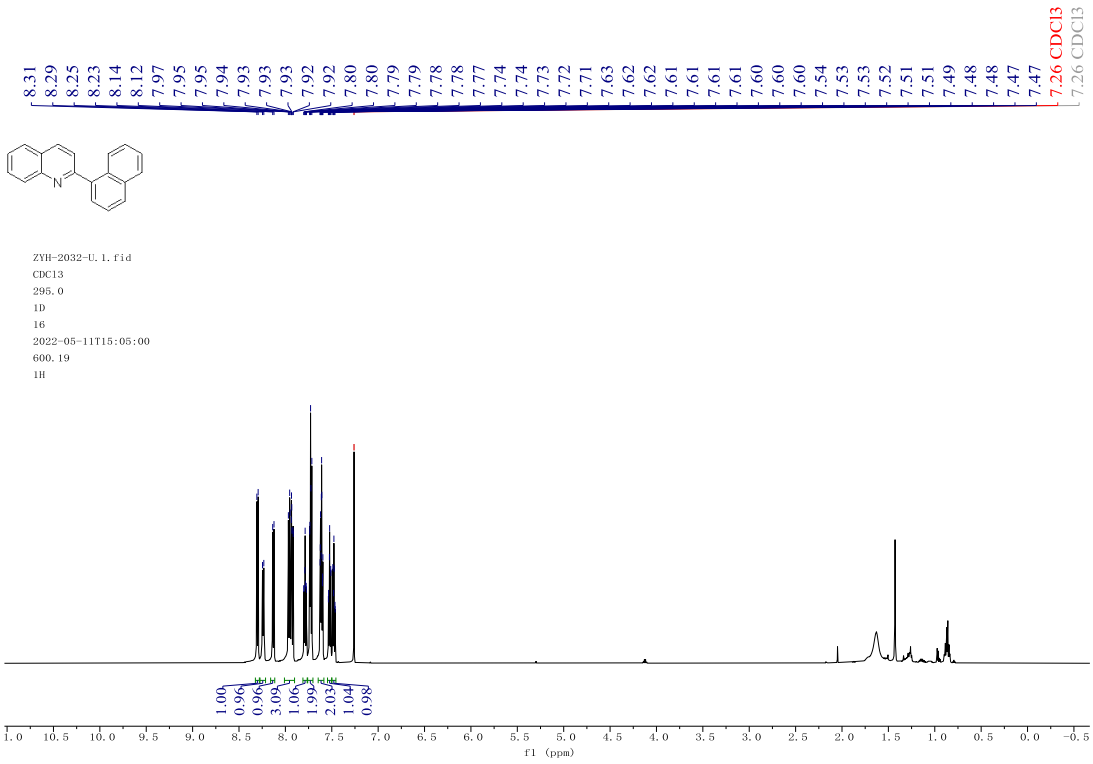


Figure S56. ^{1}H NMR spectrum of 4e in $CDCl_3$.

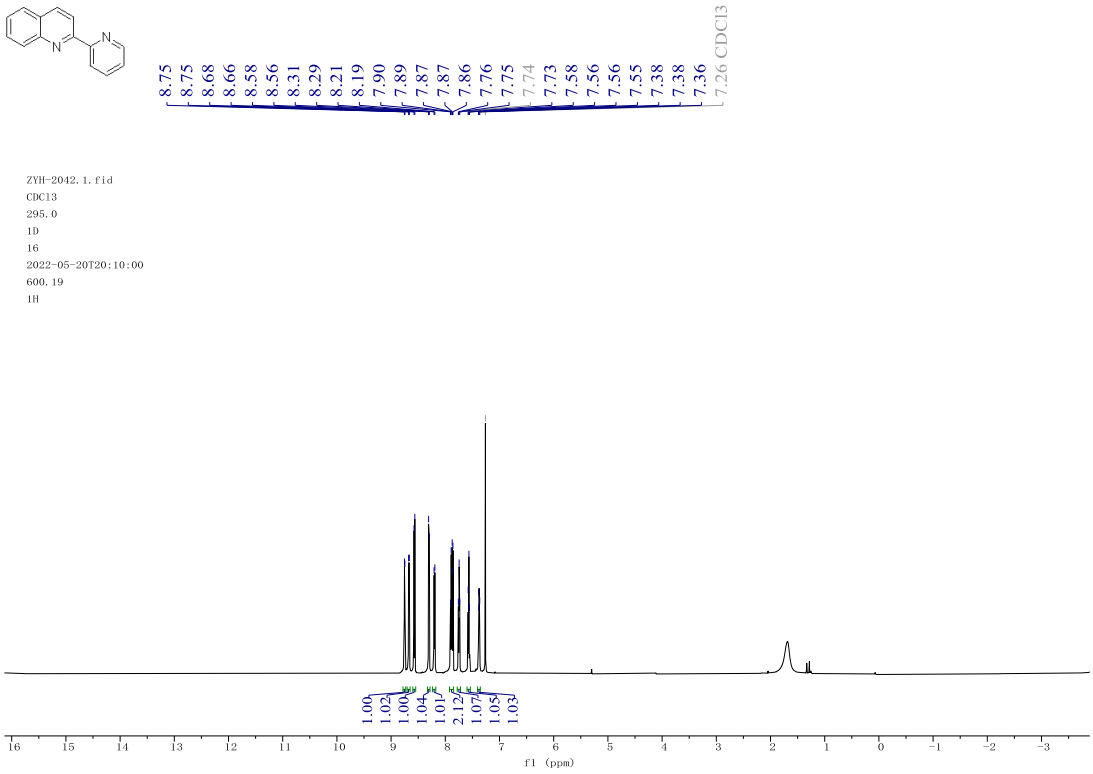


Figure S57. ^{1}H NMR spectrum of 4f in $CDCl_3$.

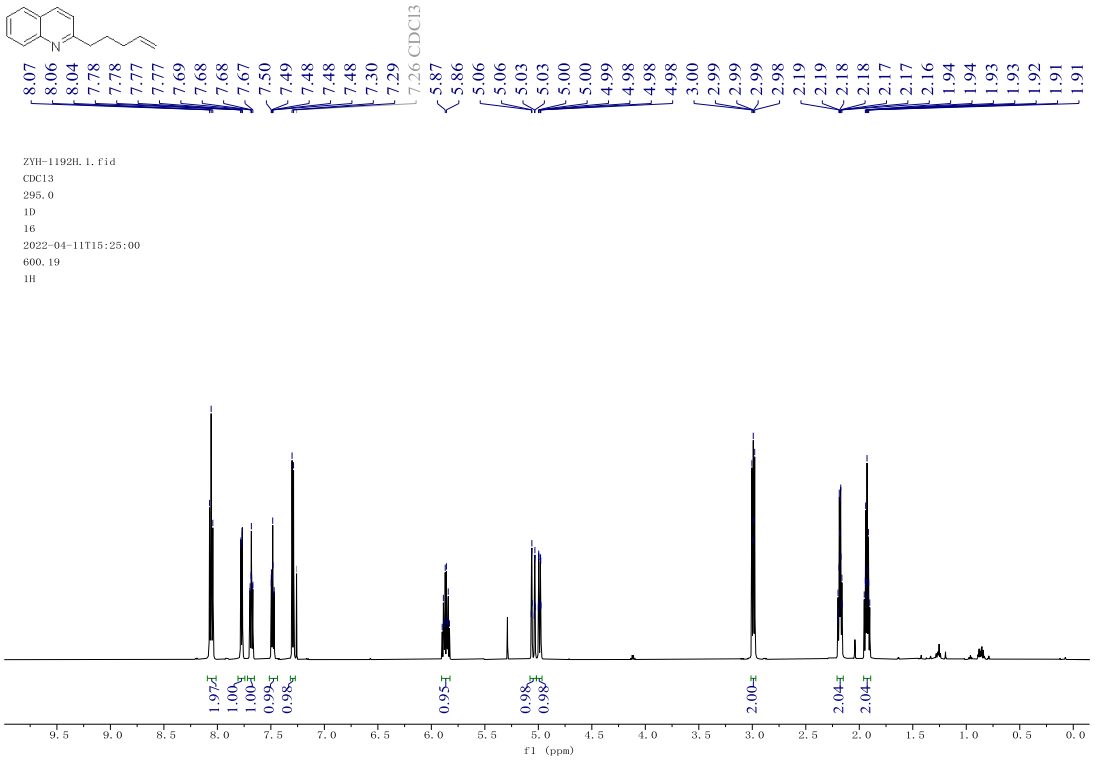


Figure S58. ^{1}H NMR spectrum of 4i in $CDCl_3$.

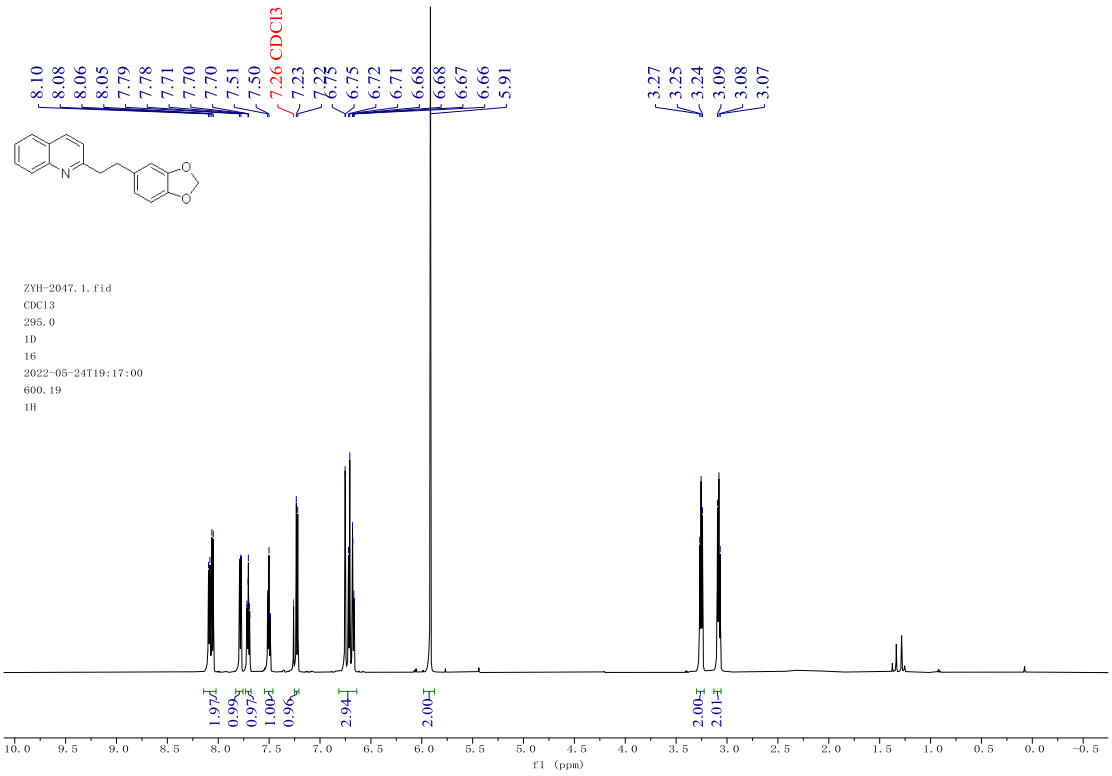


Figure S59. ^{1}H NMR spectrum of 4j in $CDCl_3$.

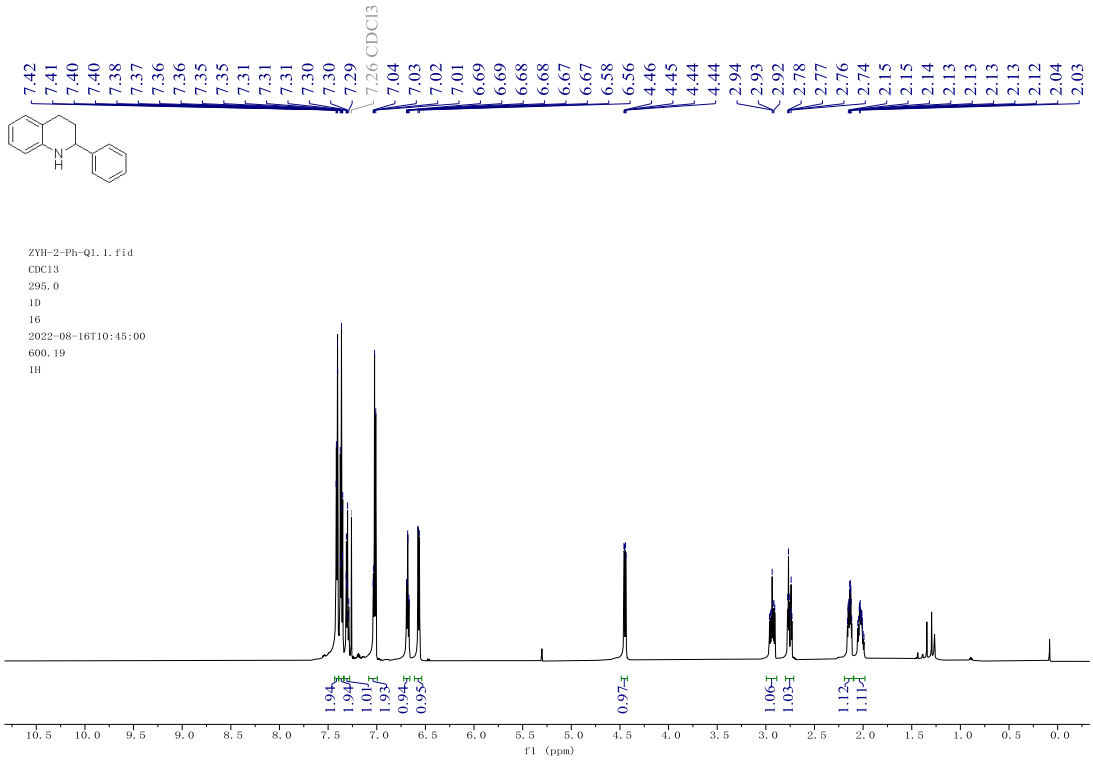


Figure S60. ^{1}H NMR spectrum of 5a in $CDCl_3$.

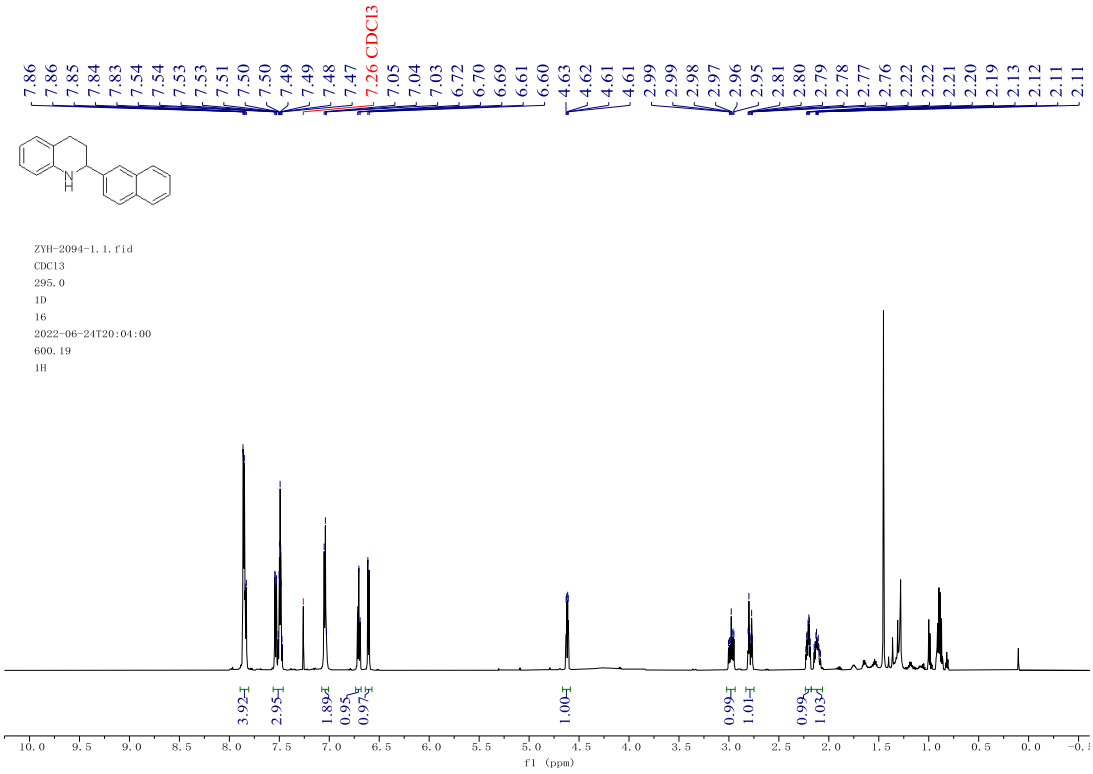


Figure S61. ^{1}H NMR spectrum of 5b in $CDCl_3$.

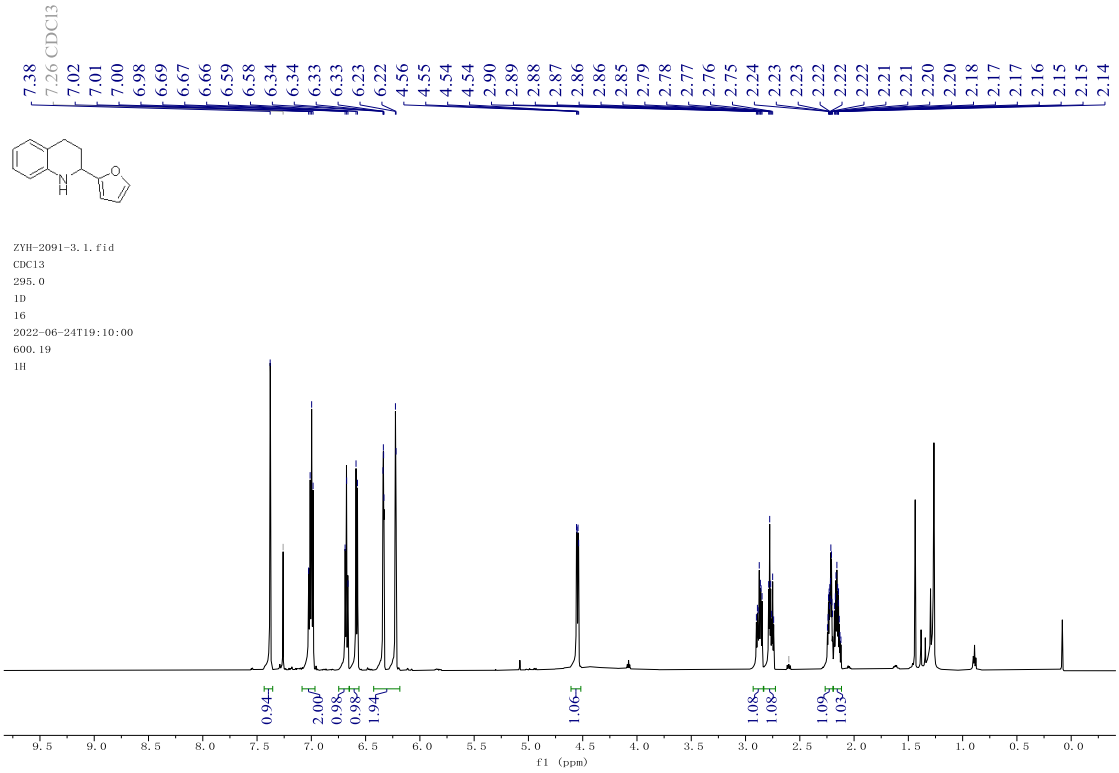


Figure S62. ^{1}H NMR spectrum of 5c in $CDCl_3$.

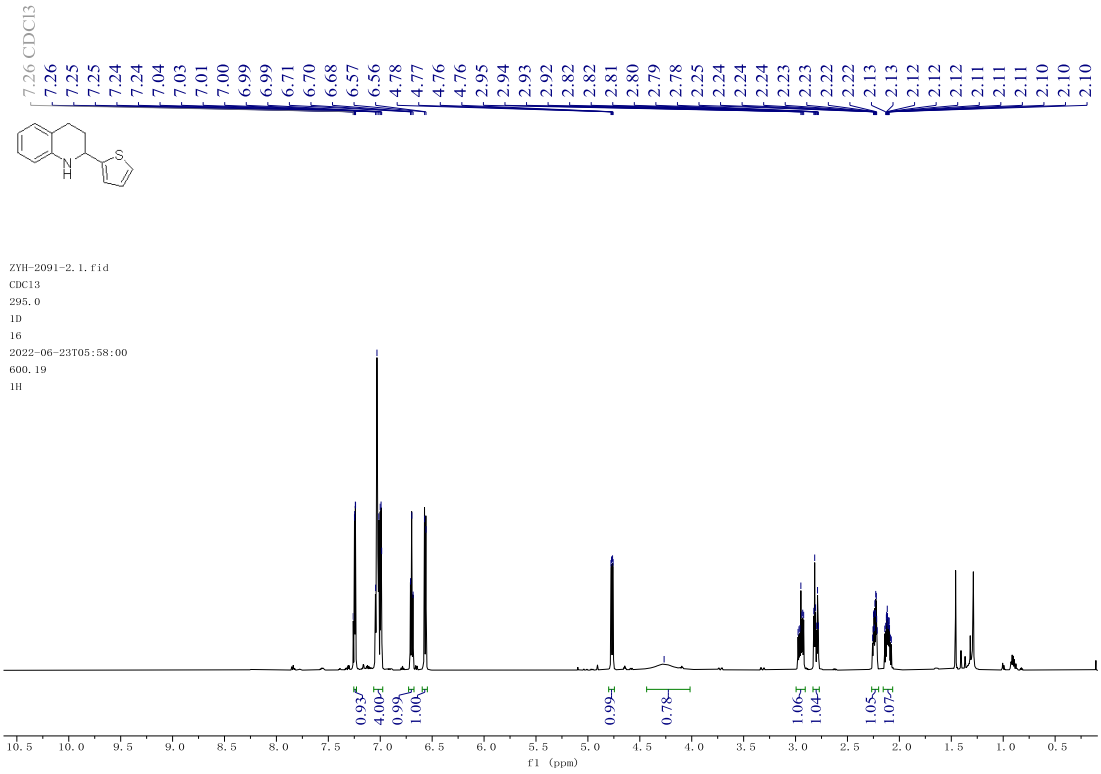


Figure S63. ^{1}H NMR spectrum of 5d in $CDCl_3$.

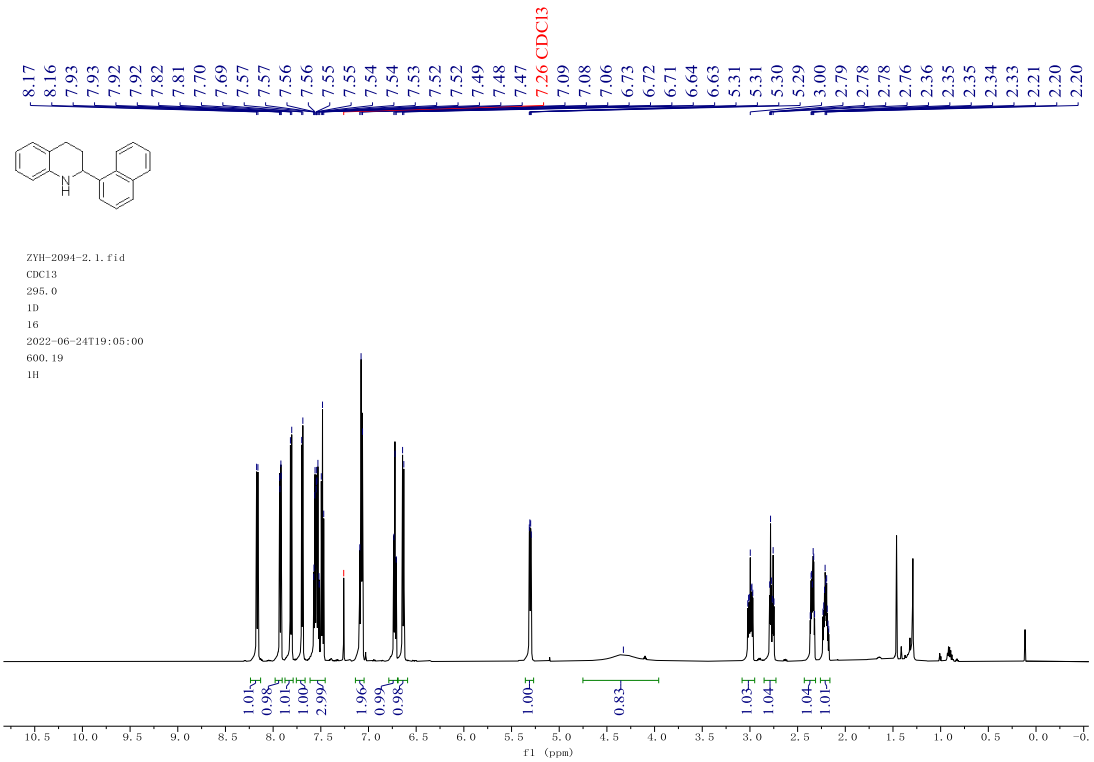


Figure S64. ^{1}H NMR spectrum of 5e in $CDCl_3$.

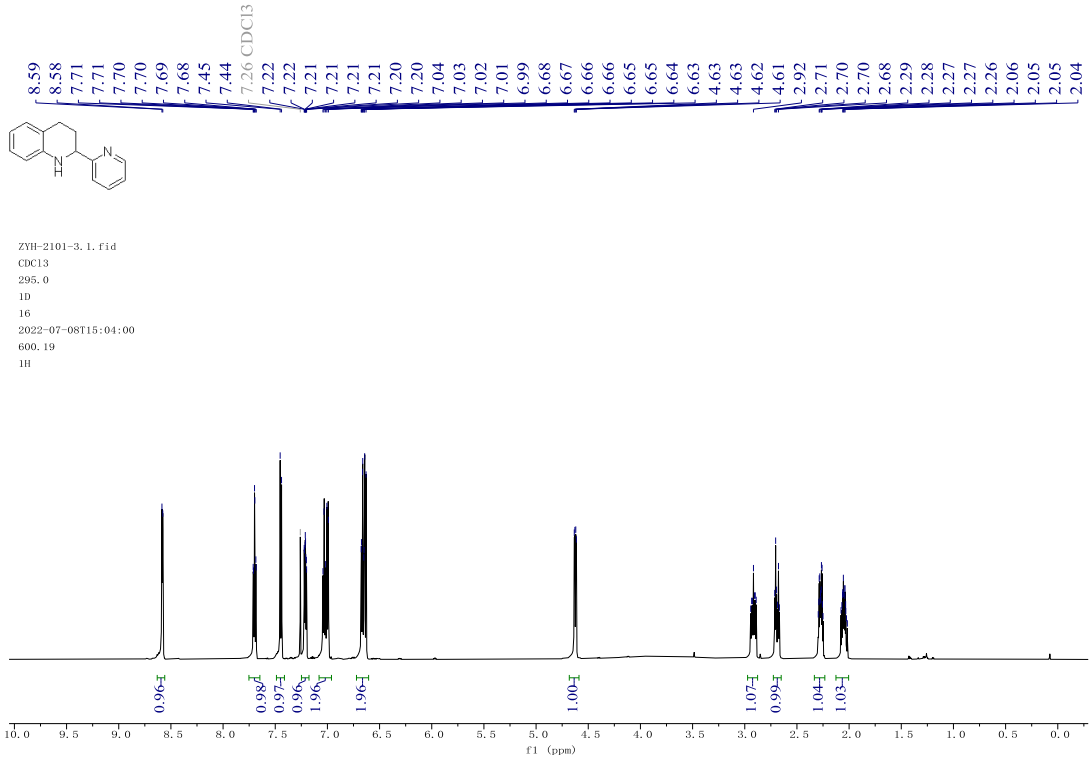


Figure S65. ^{1}H NMR spectrum of 5f in $CDCl_3$.

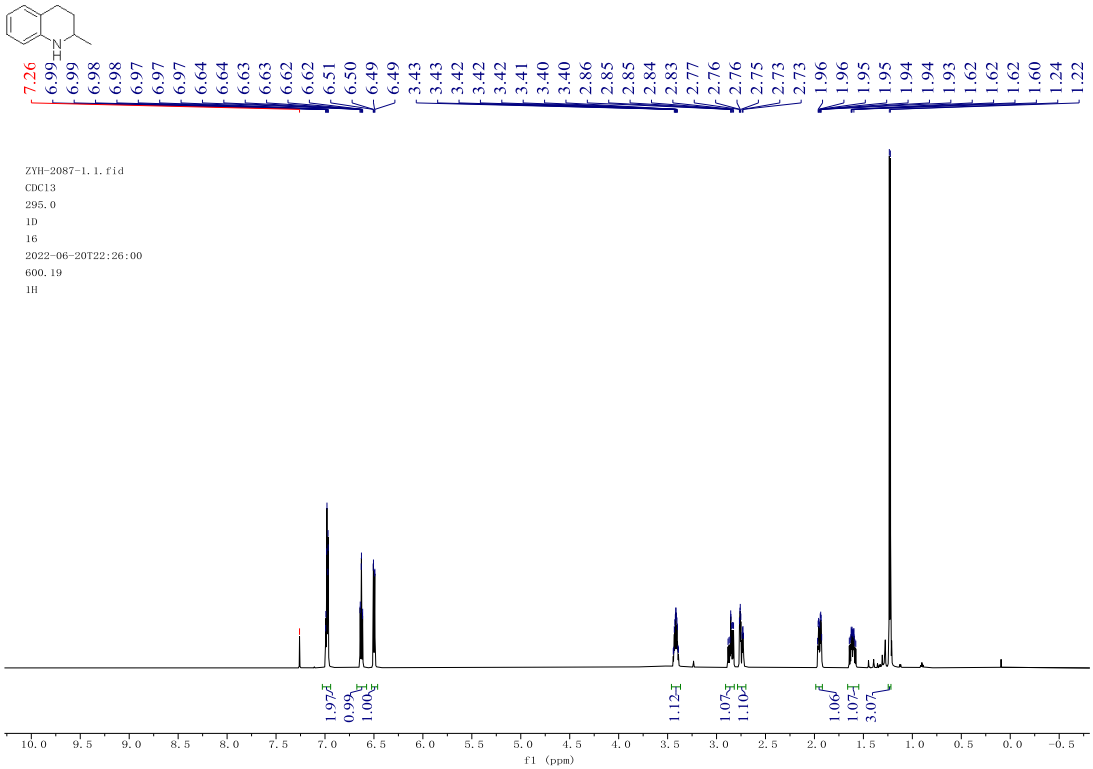


Figure S66. ^{1}H NMR spectrum of 5g in $CDCl_3$.

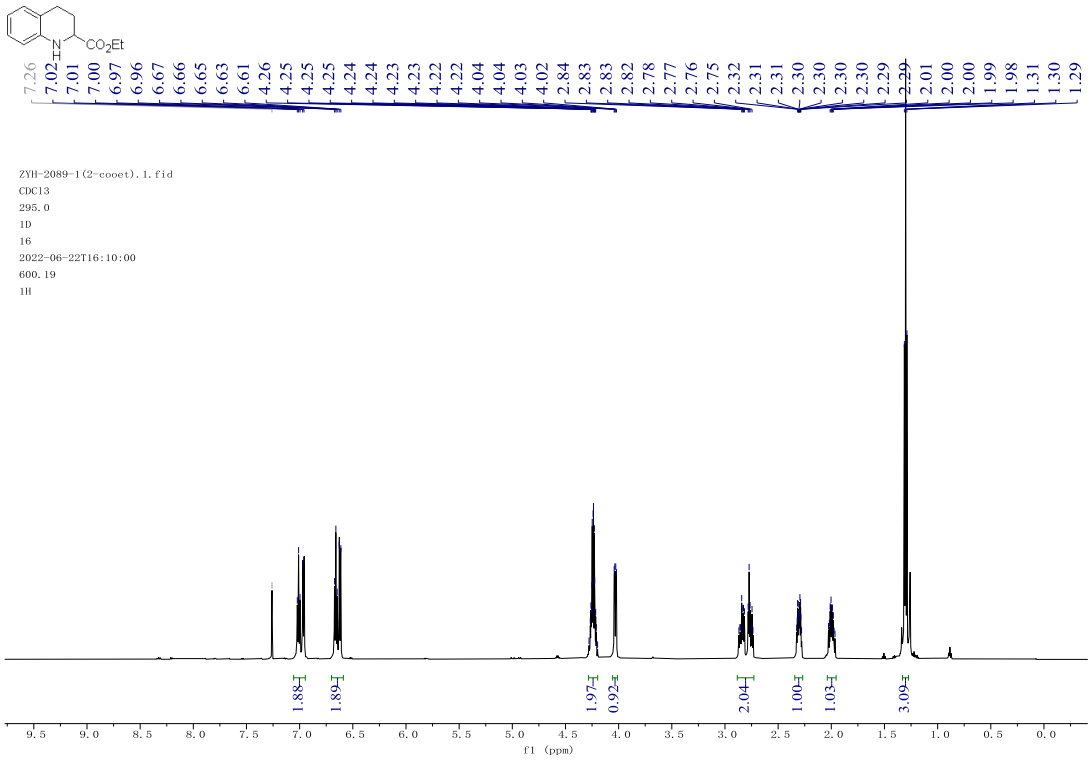


Figure S67. ^{1}H NMR spectrum of 5h in $CDCl_3$.

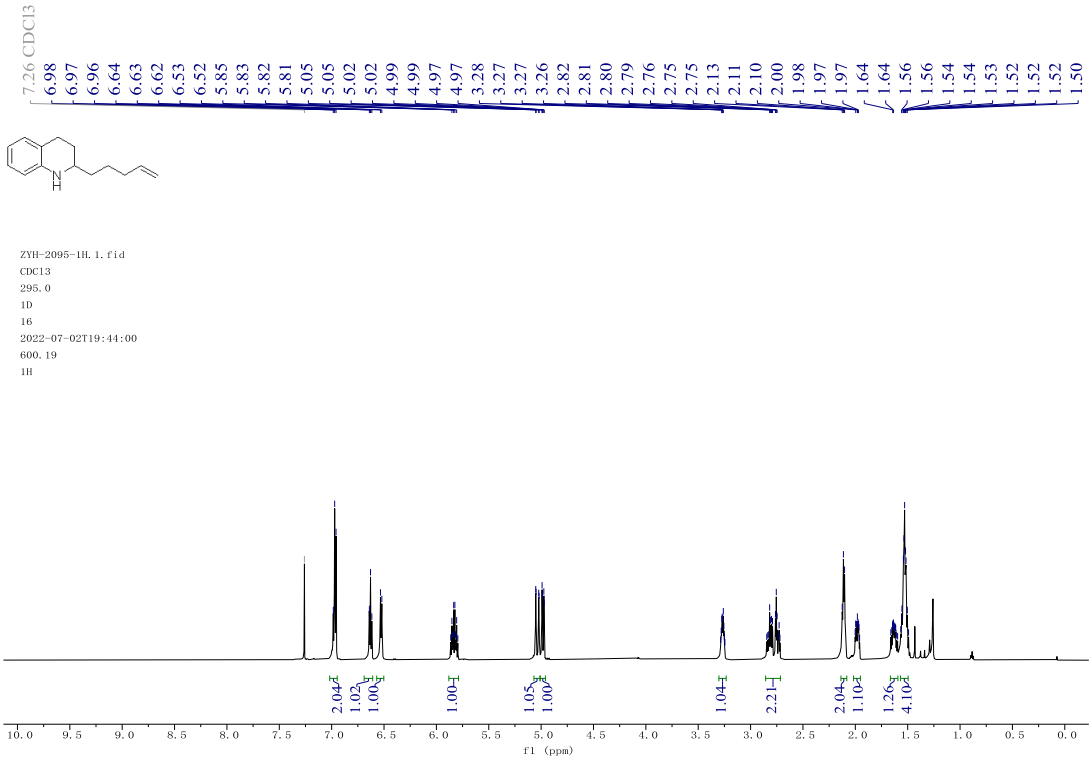


Figure S68. ^{1}H NMR spectrum of 5i in $CDCl_3$.

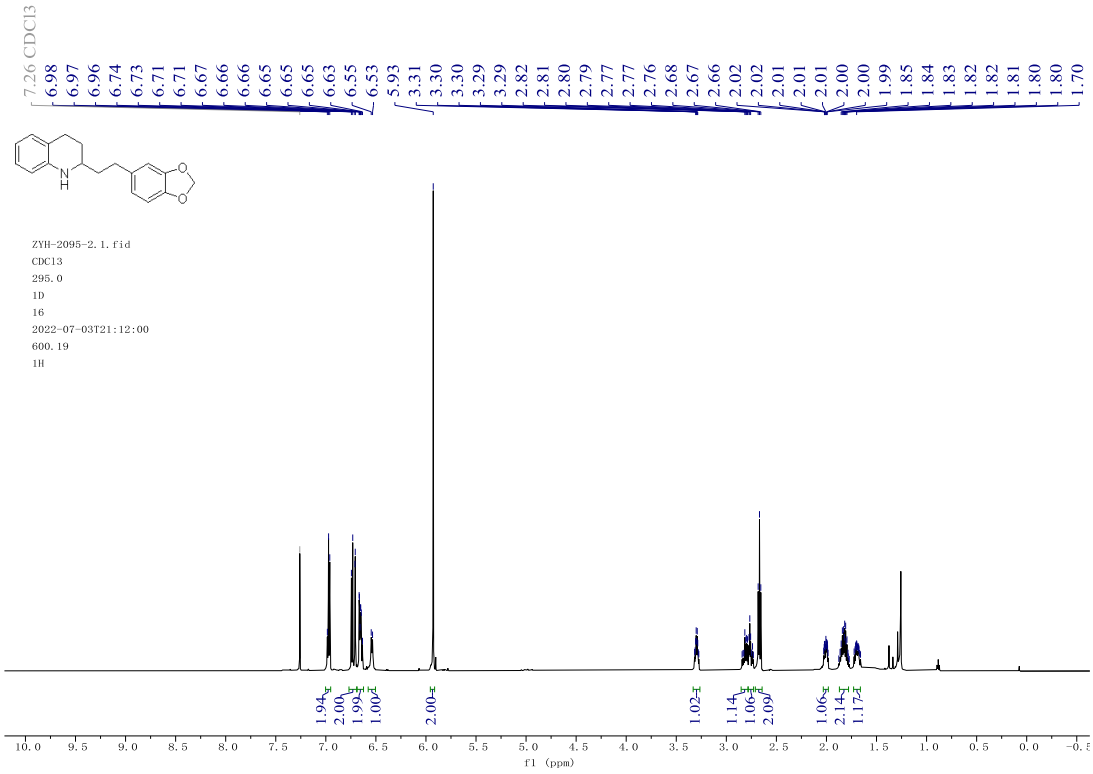


Figure S69. ^{1}H NMR spectrum of 5j in $CDCl_3$.

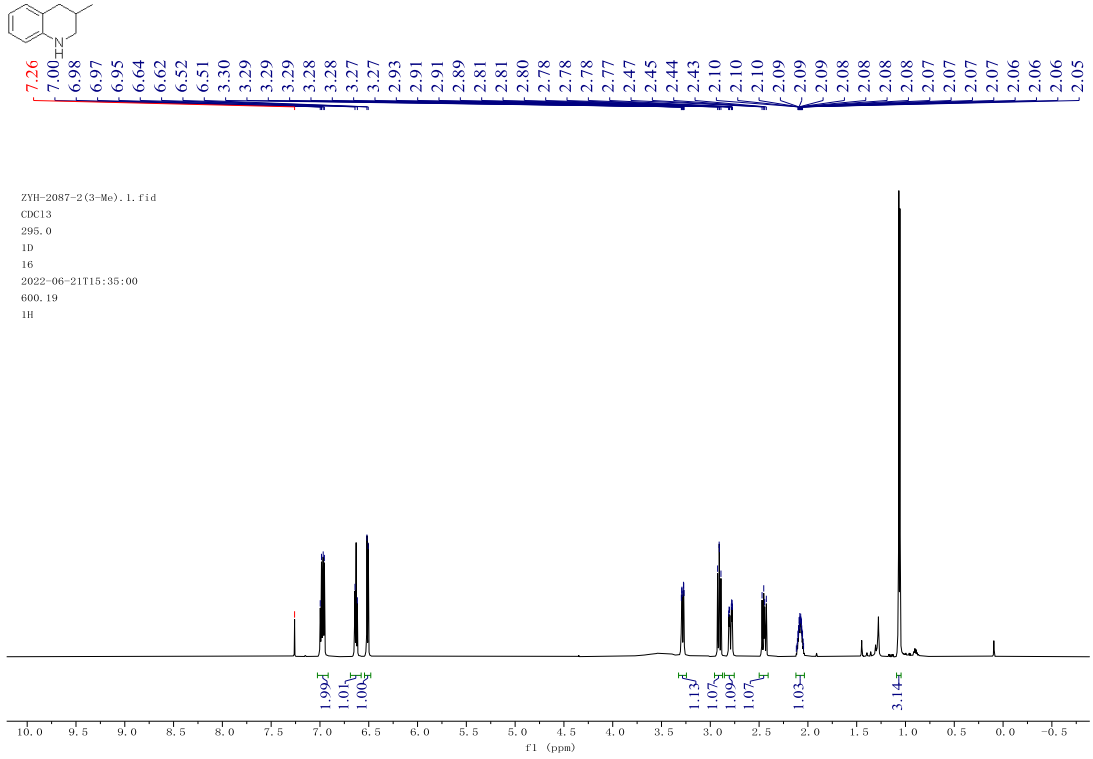


Figure S70. ^{1}H NMR spectrum of 5k in $CDCl_3$.

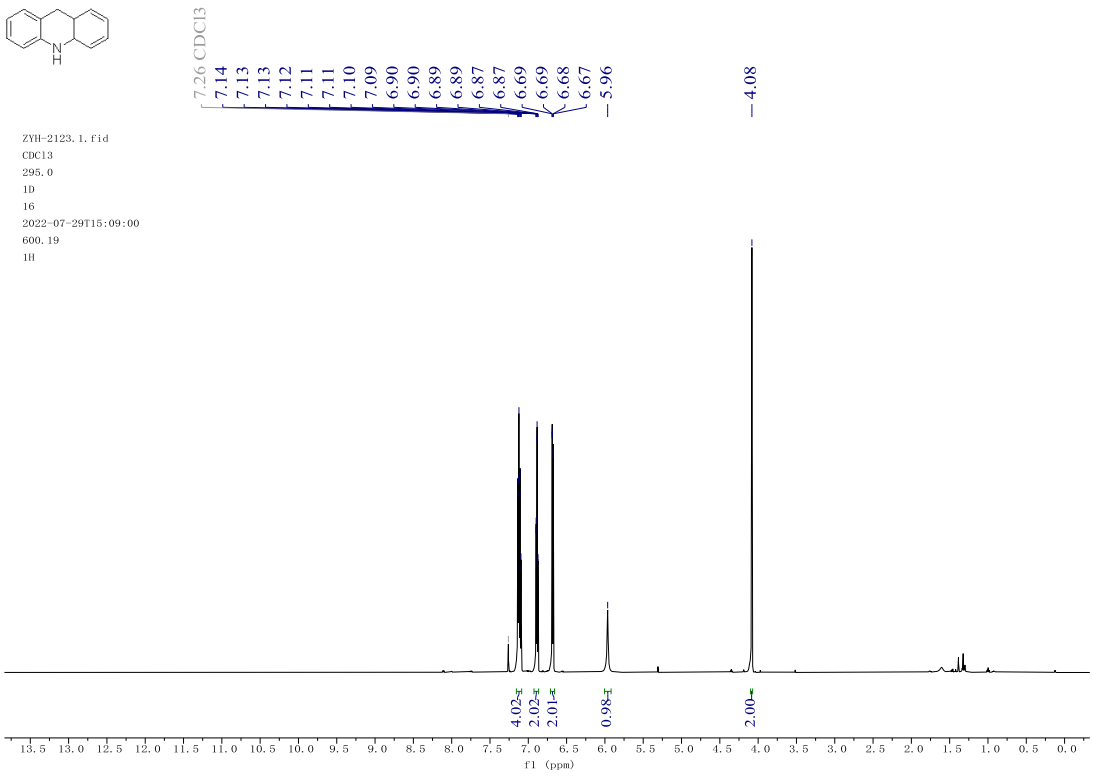


Figure S71. ^{1}H NMR spectrum of 5l in $CDCl_3$.

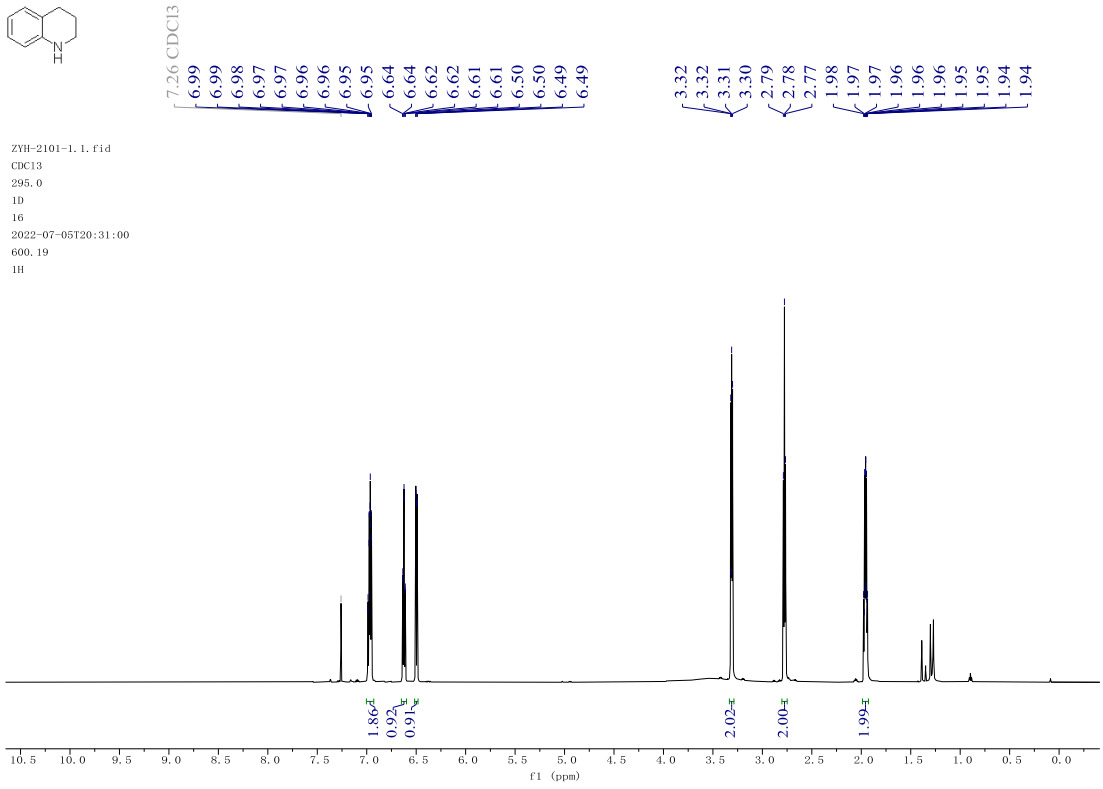


Figure S72. ^{1}H NMR spectrum of 5m in $CDCl_3$.

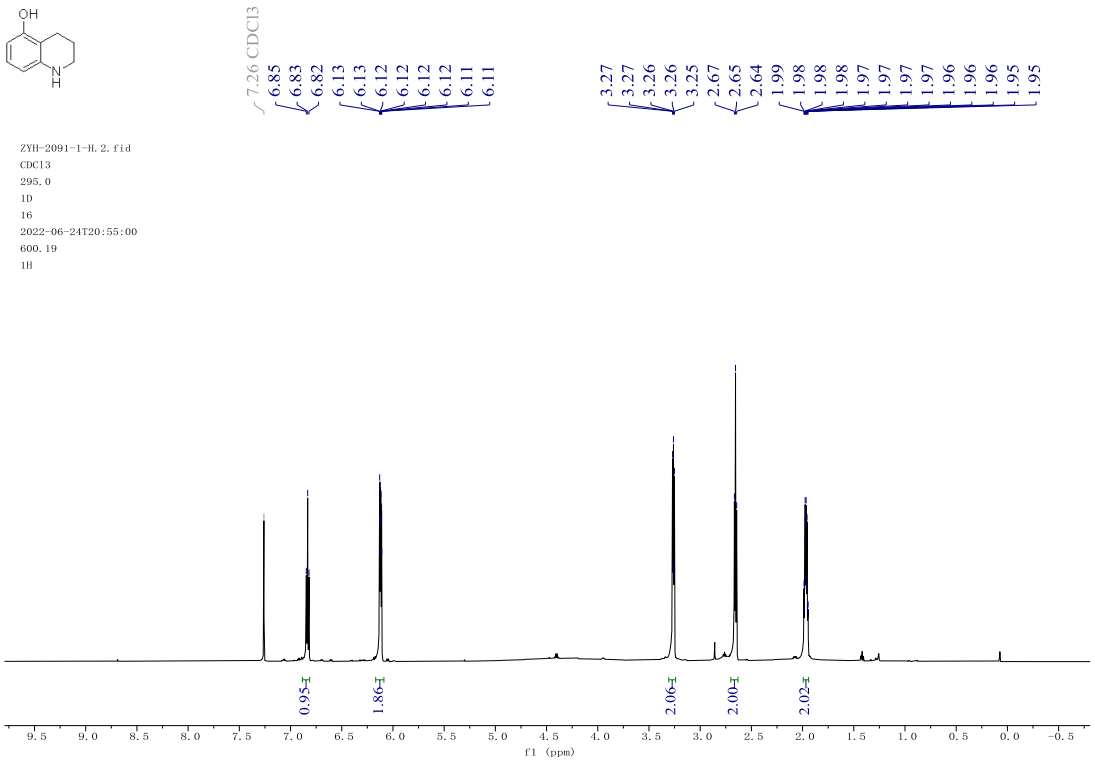


Figure S73. ^{1}H NMR spectrum of 5n in $CDCl_3$.

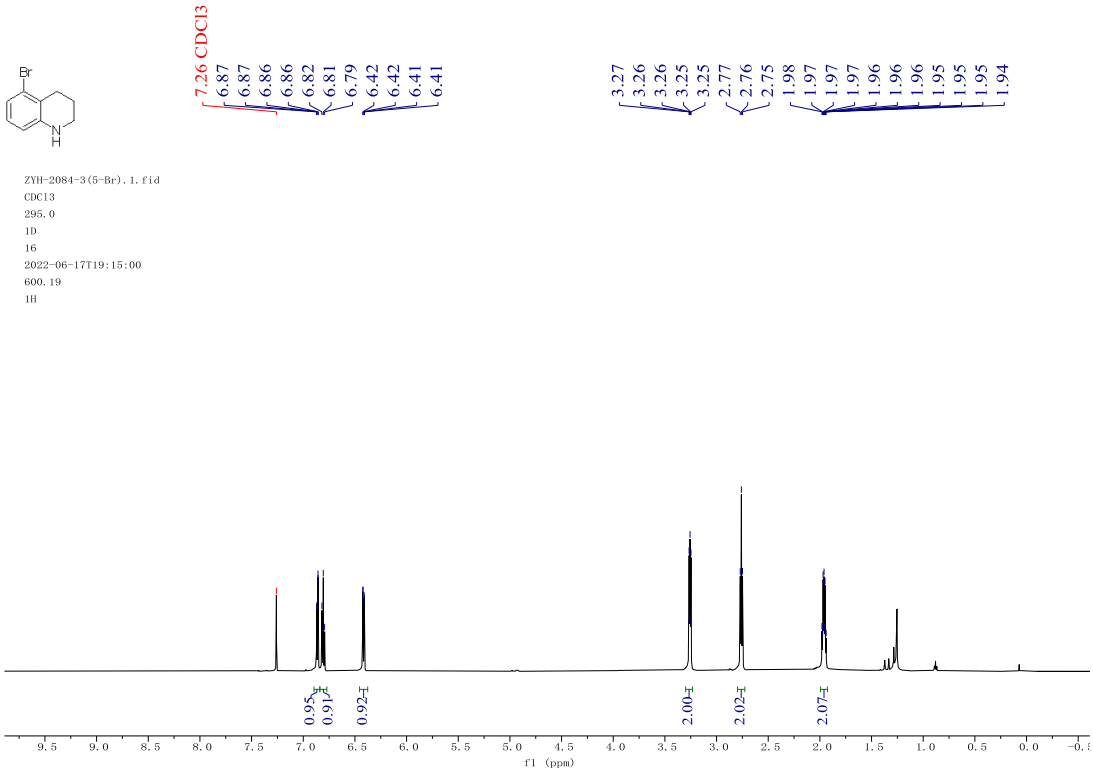


Figure S74. ^{1}H NMR spectrum of 5o in $CDCl_3$.

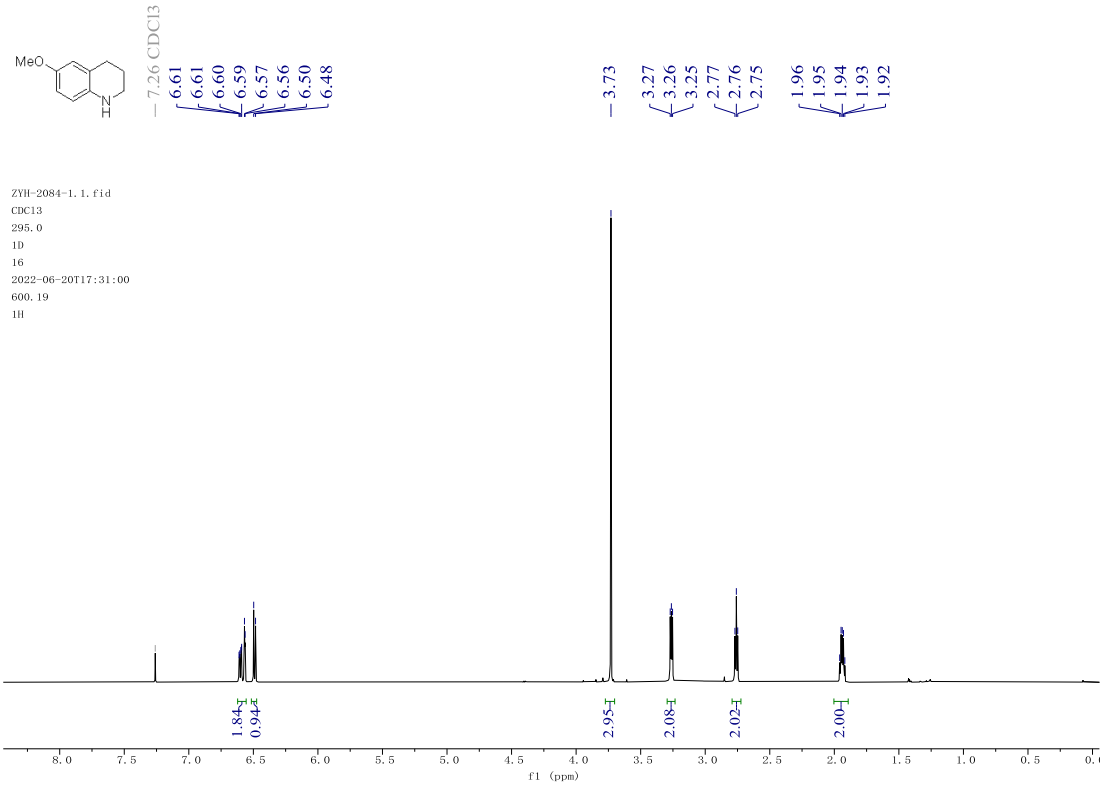


Figure S75. ^{1}H NMR spectrum of 5p in $CDCl_3$.

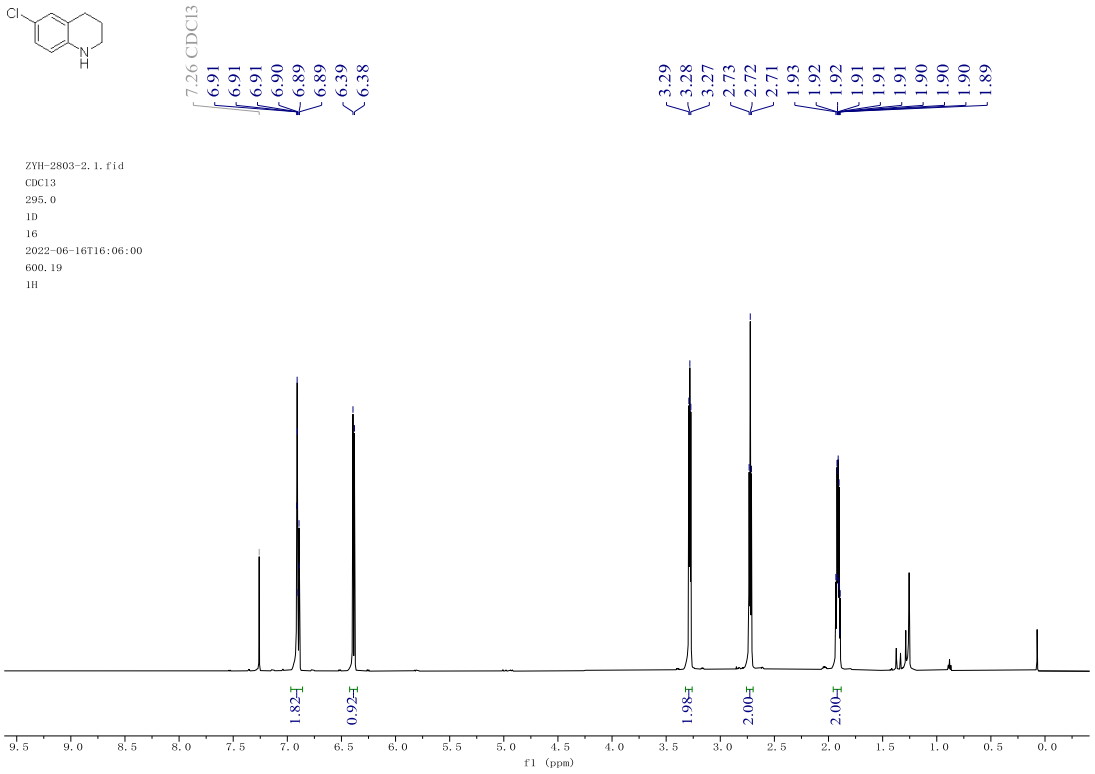


Figure S76. ^{1}H NMR spectrum of 5q in $CDCl_3$.

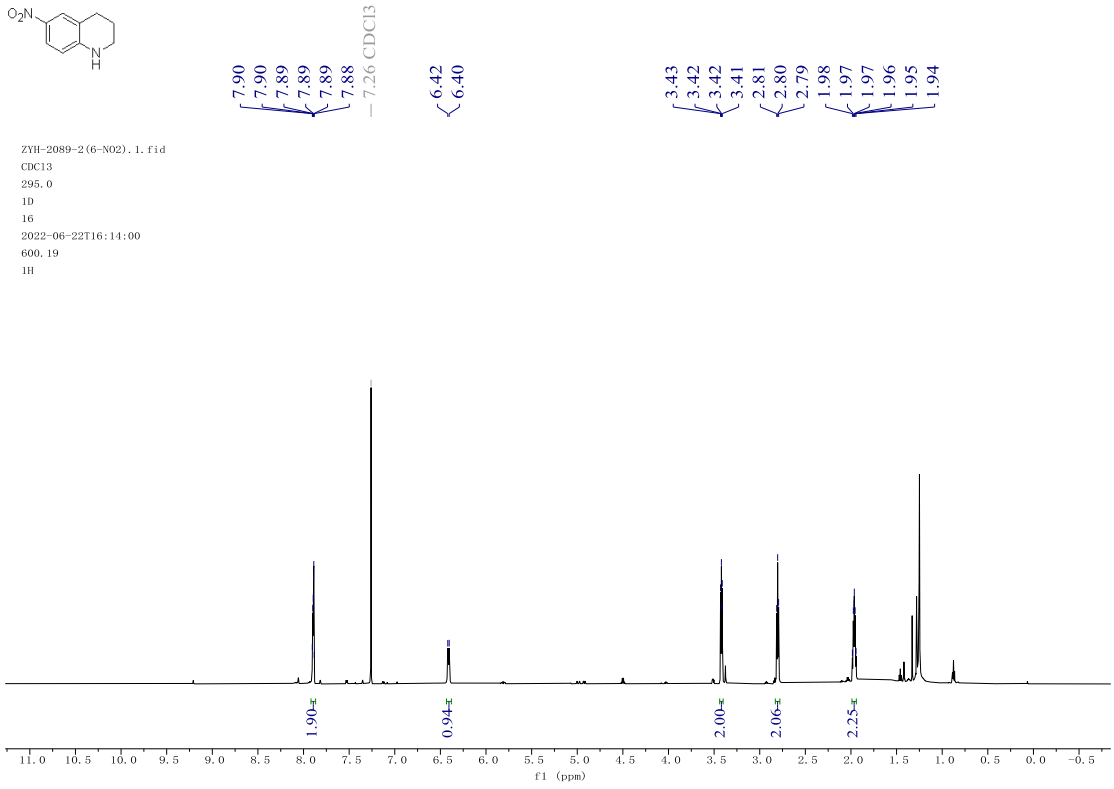


Figure S77. ^{1}H NMR spectrum of 5r in $CDCl_3$.

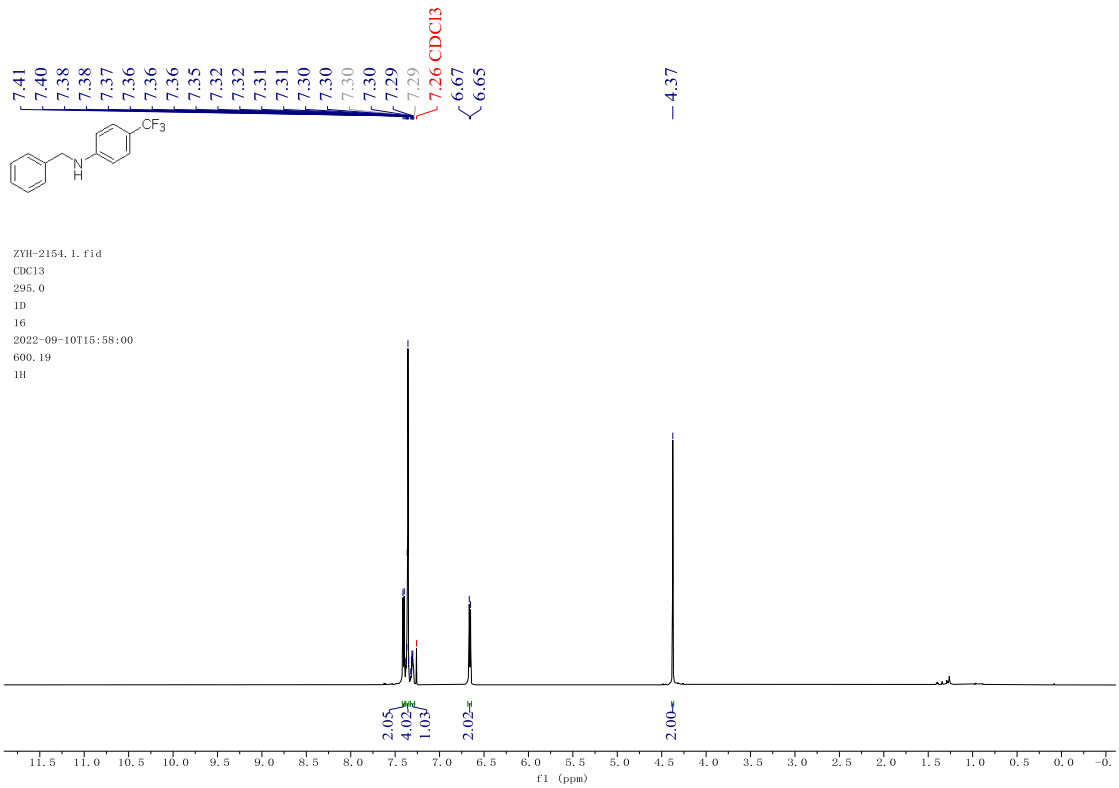


Figure S78. ^{1}H NMR spectrum of 5s in $CDCl_3$.

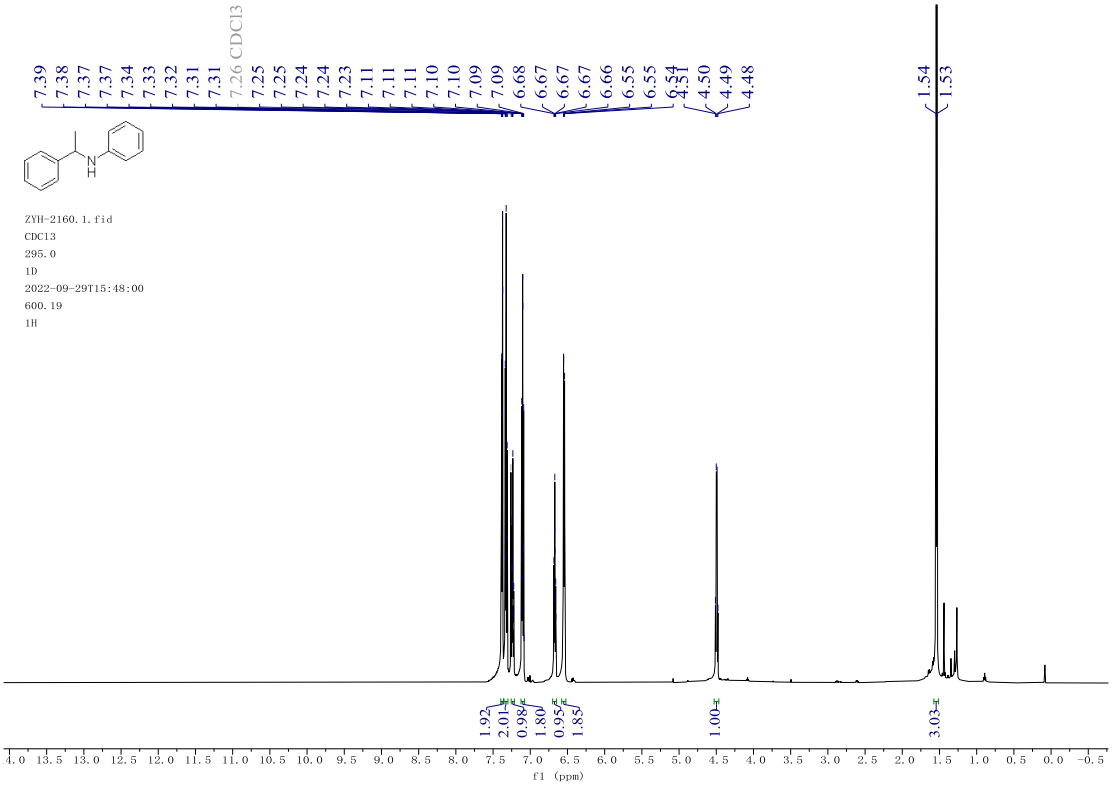


Figure S79. ^{1}H NMR spectrum of 5t in $CDCl_3$.

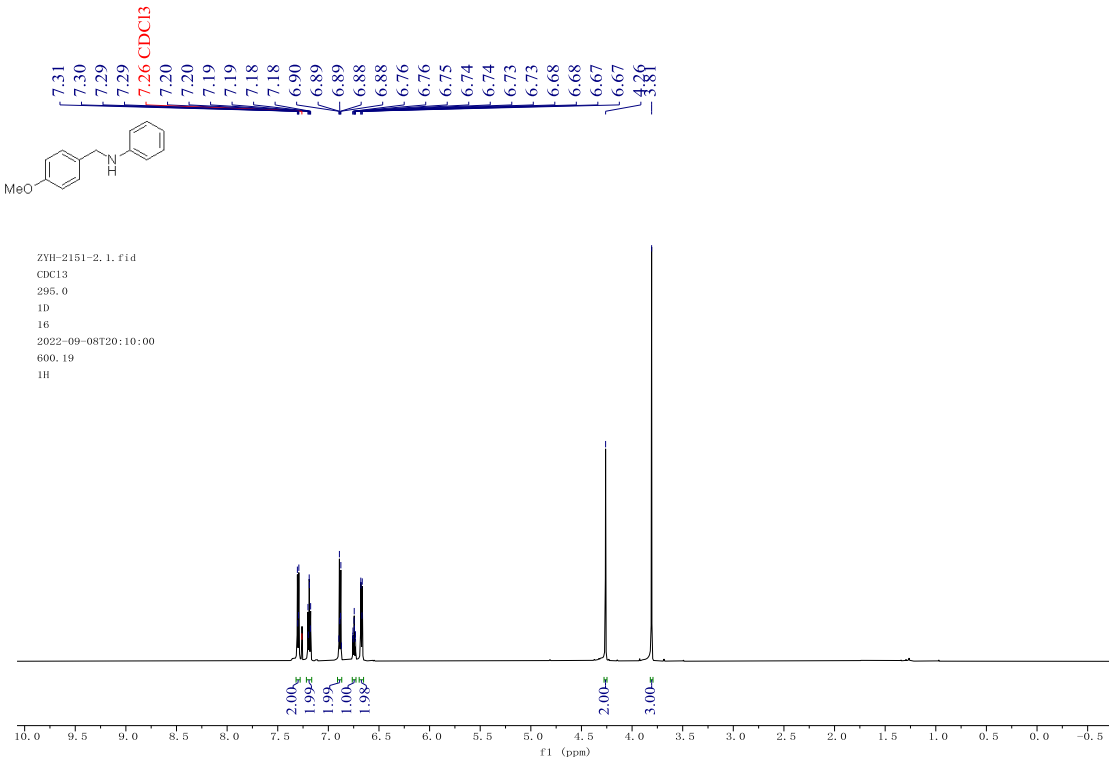


Figure S80. ^{1}H NMR spectrum of 5u in $CDCl_3$.

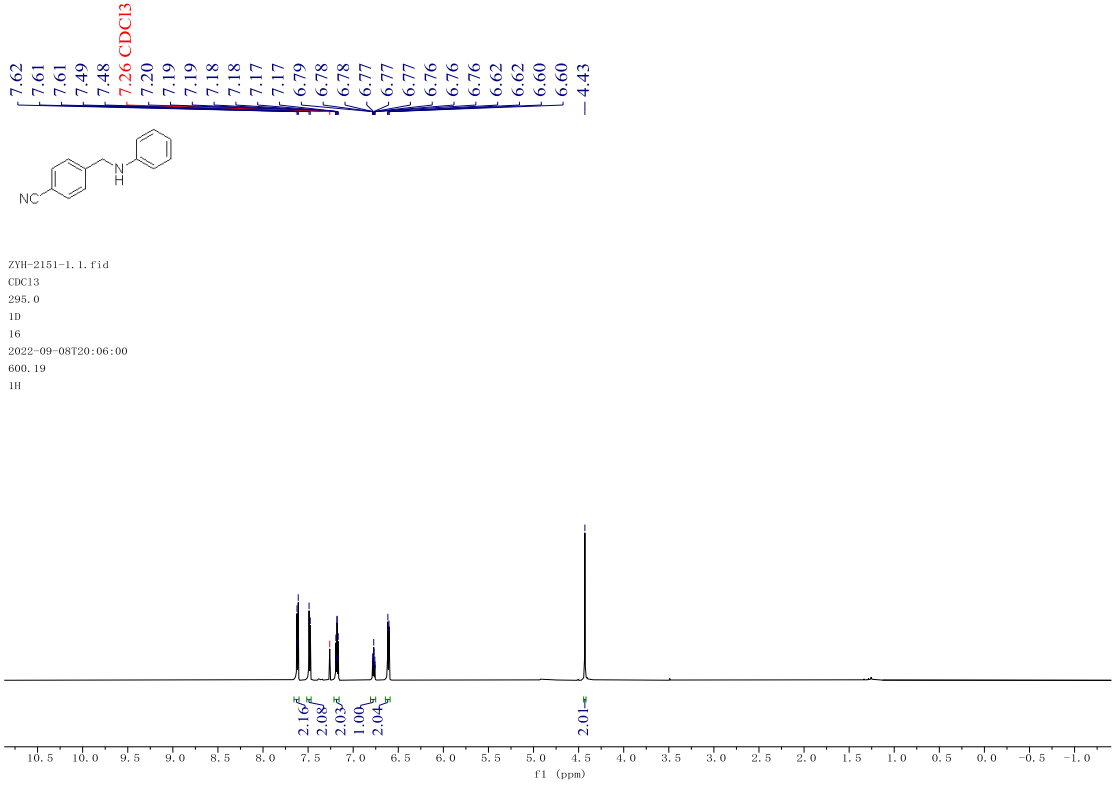


Figure S81. ^{1}H NMR spectrum of 5v in $CDCl_3$.

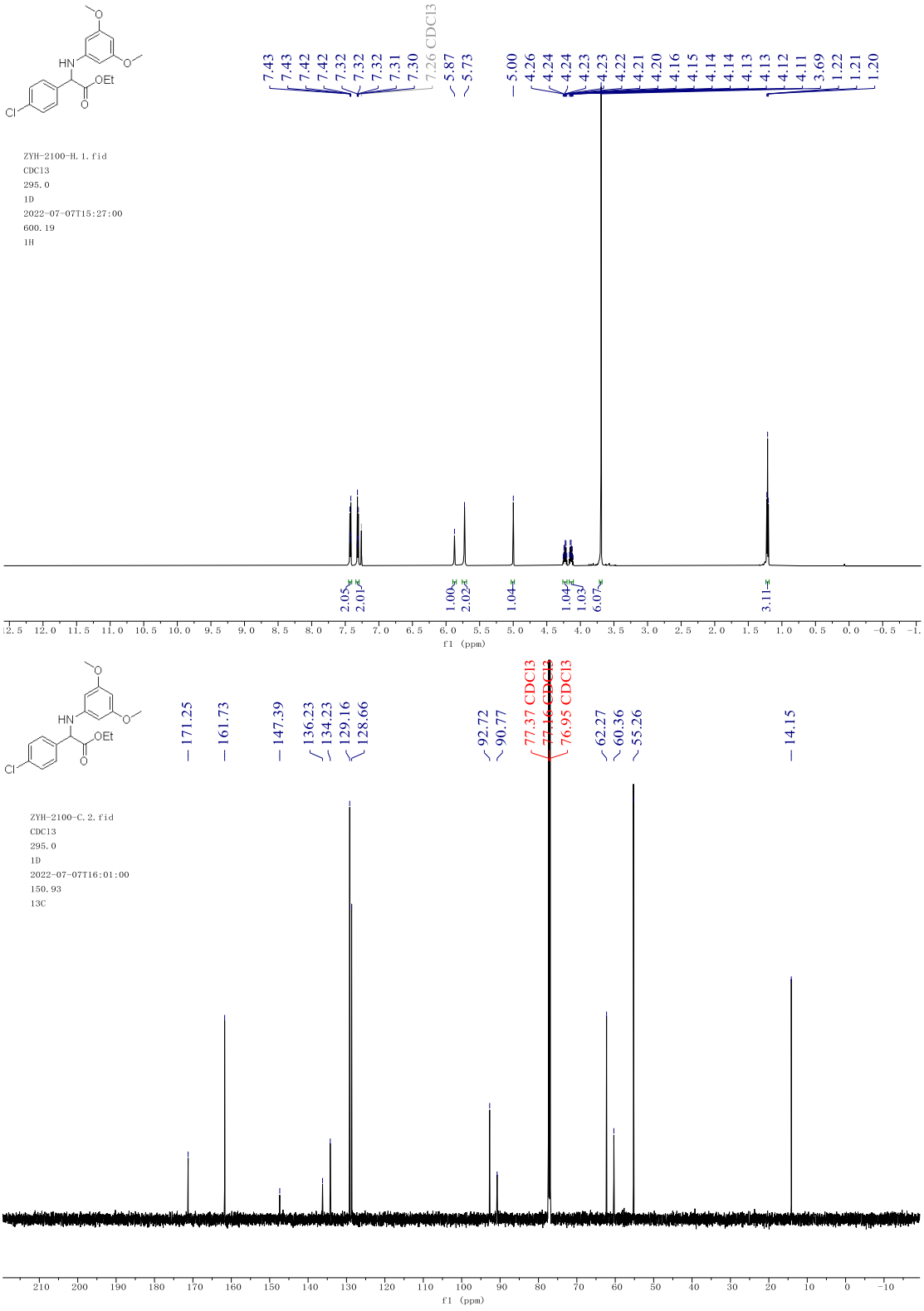


Figure S82. ^{1}H and ^{13}C NMR spectrum of 5w in $CDCl_3$.

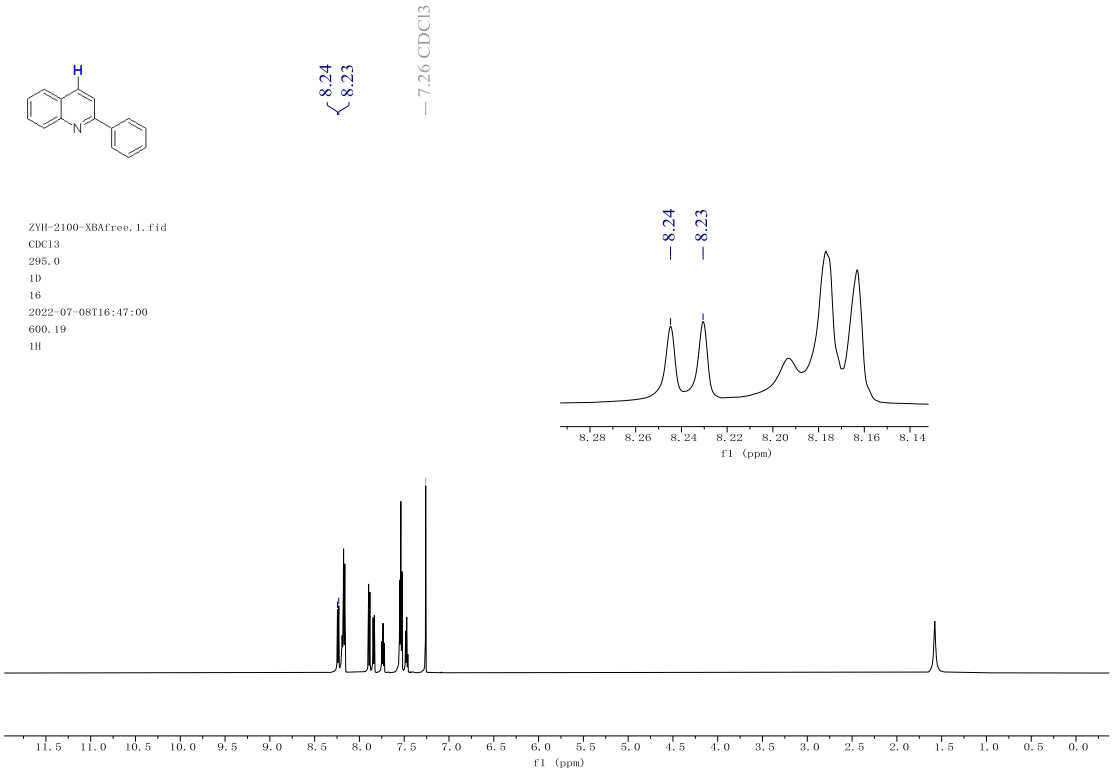


Figure S83. ^{1}H NMR spectrum of 4a in $CDCl_3$.

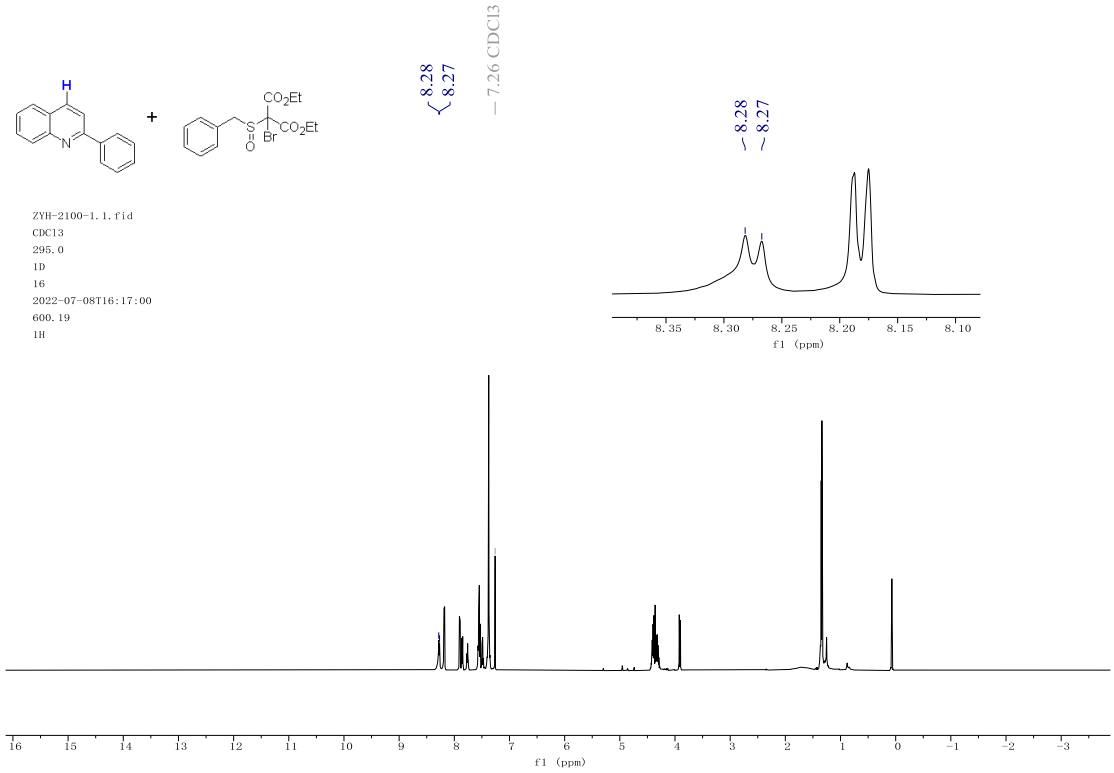


Figure S84. ^{1}H NMR spectrum of 4a•3a (1:1).

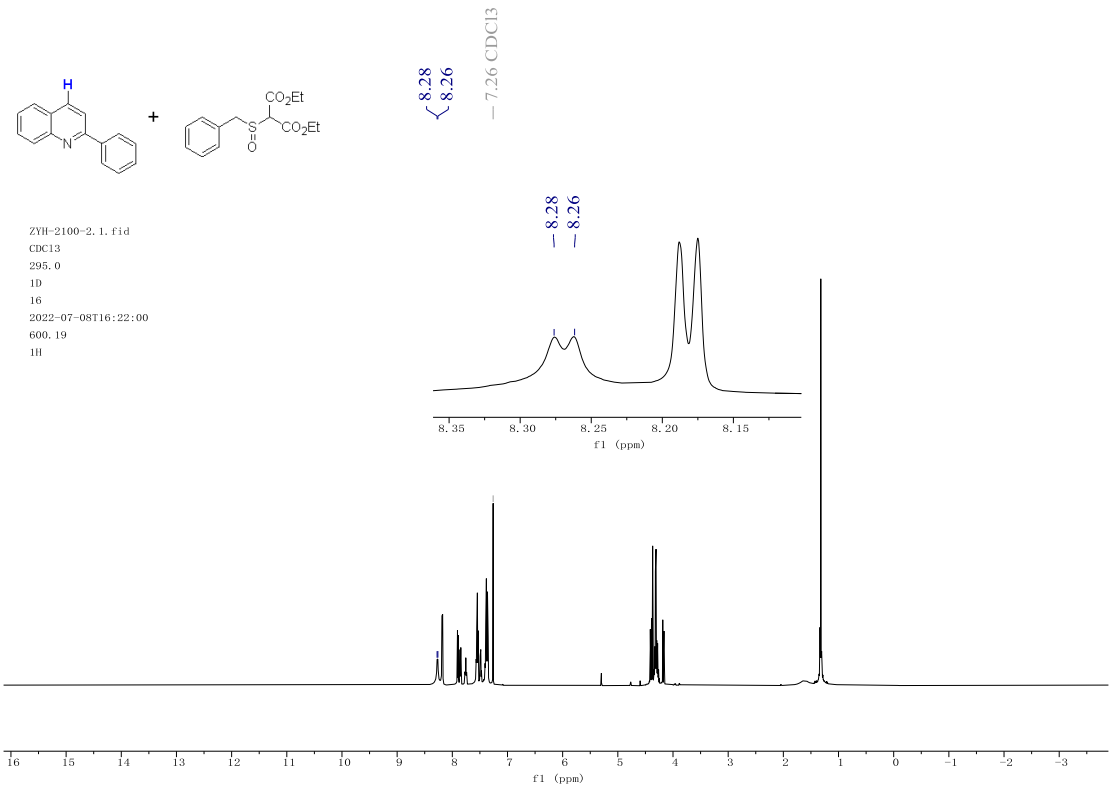


Figure S85. ^{1}H NMR spectrum of 4a•2a (1:1).

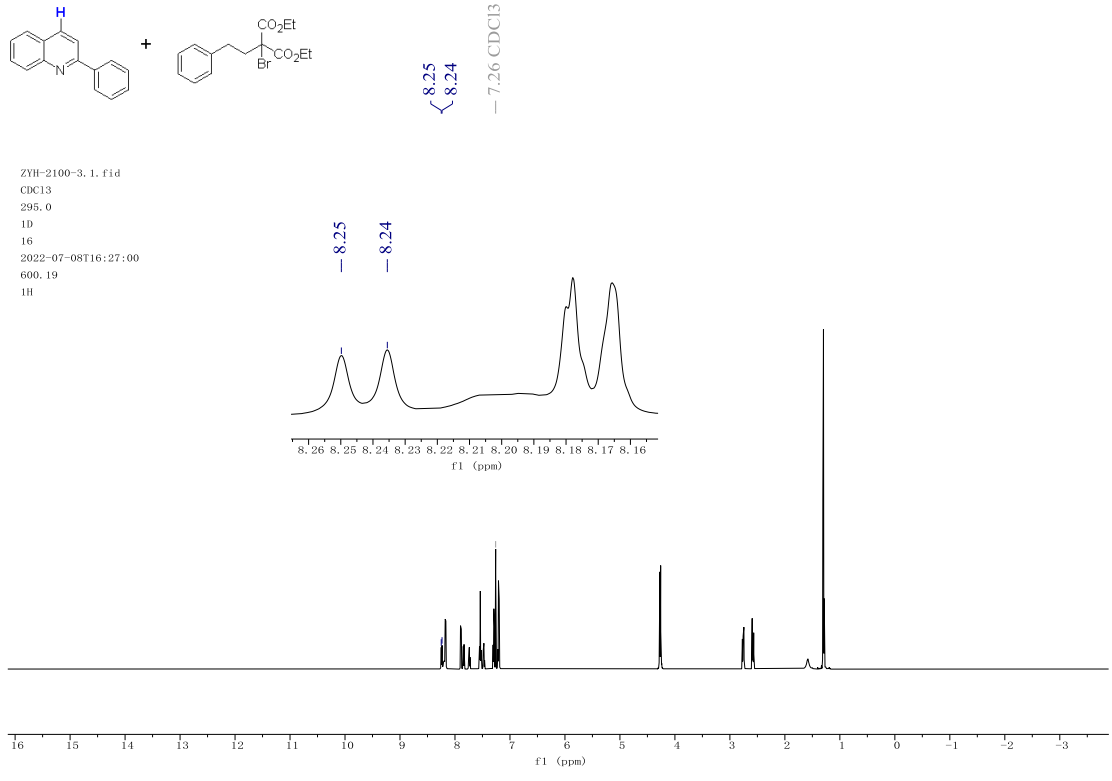


Figure S86. ^{1}H NMR spectrum of 4a•3h (1:1).

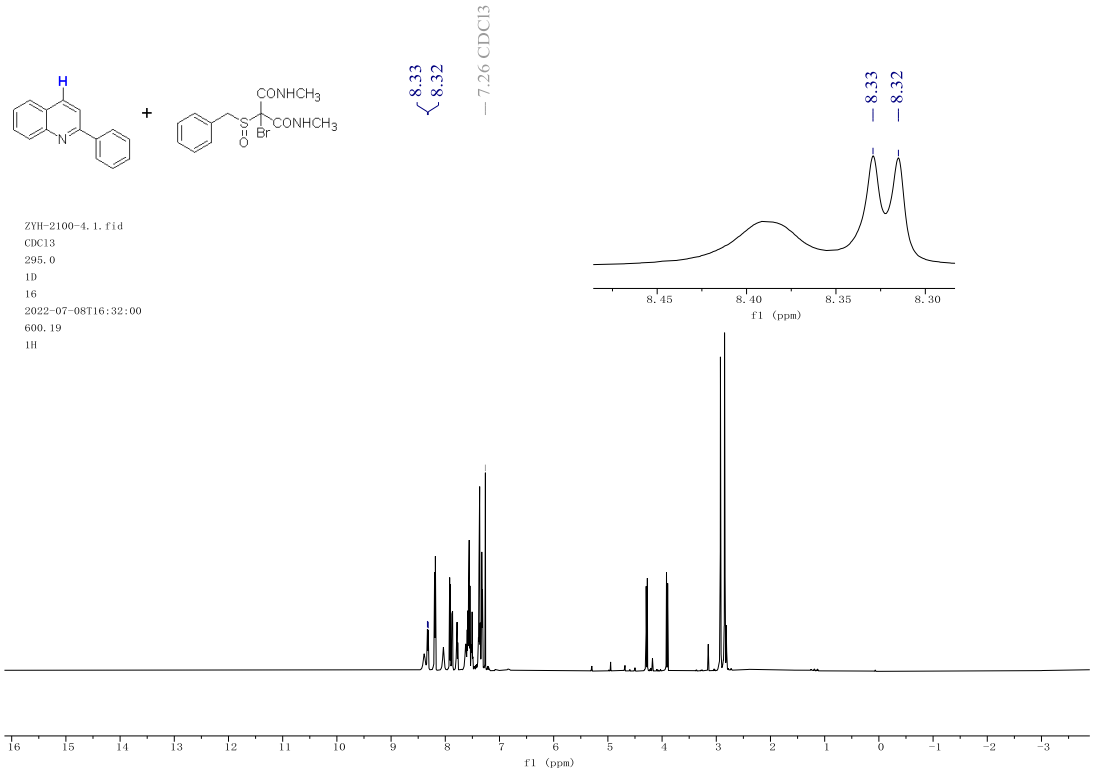


Figure S87. ^{1}H NMR spectrum of 4a•3b (1:1).

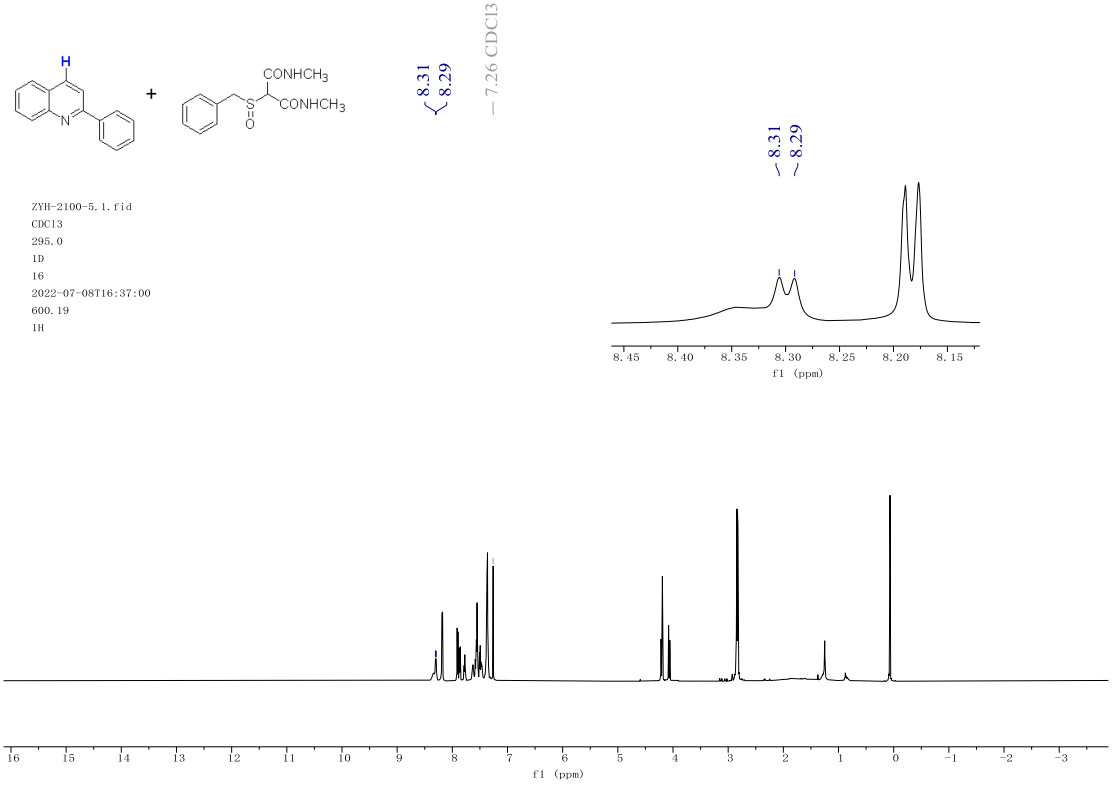


Figure S88. ^{1}H NMR spectrum of 4a•2b (1:1).

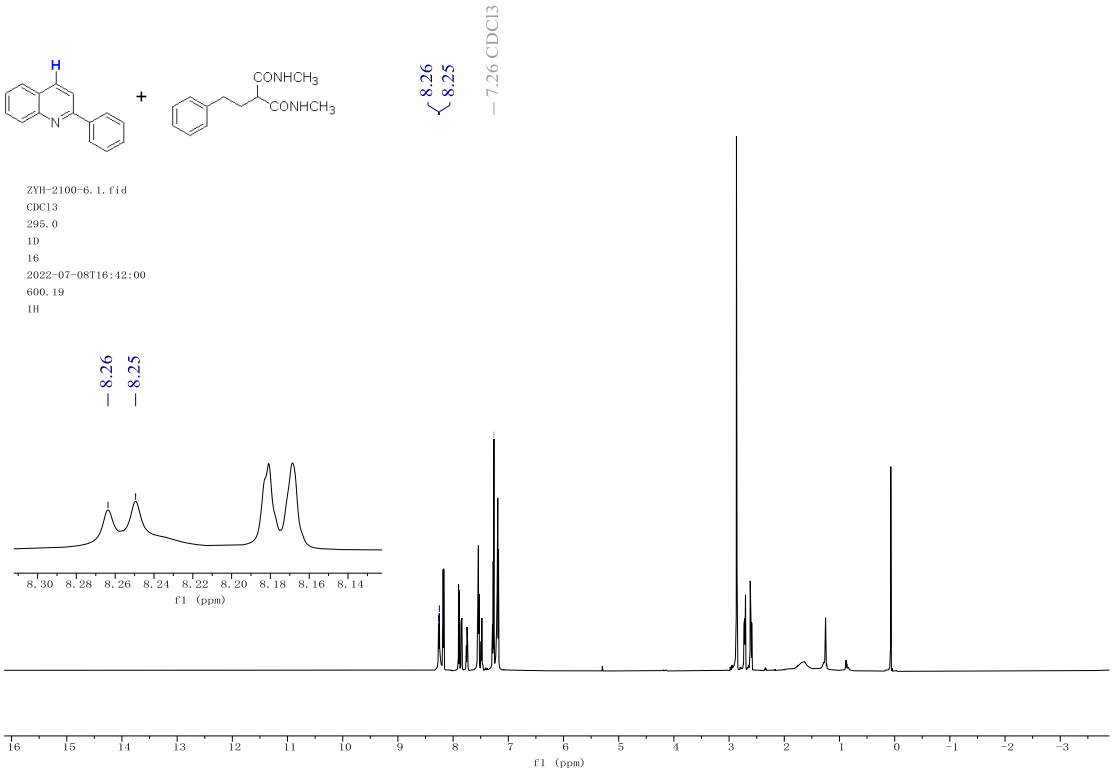


Figure S89. ^{1}H NMR spectrum of 4a•3i (1:1).

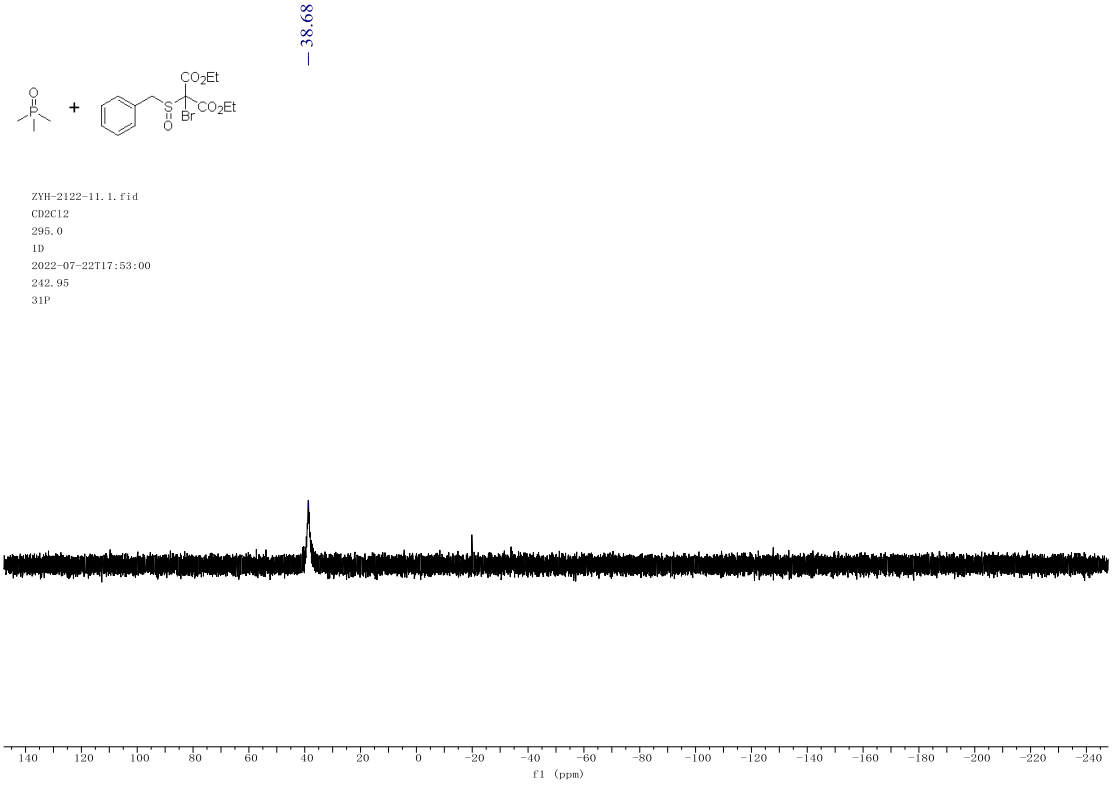


Figure S90. ^{31}P NMR spectrum of TMPO•3a (1:5).

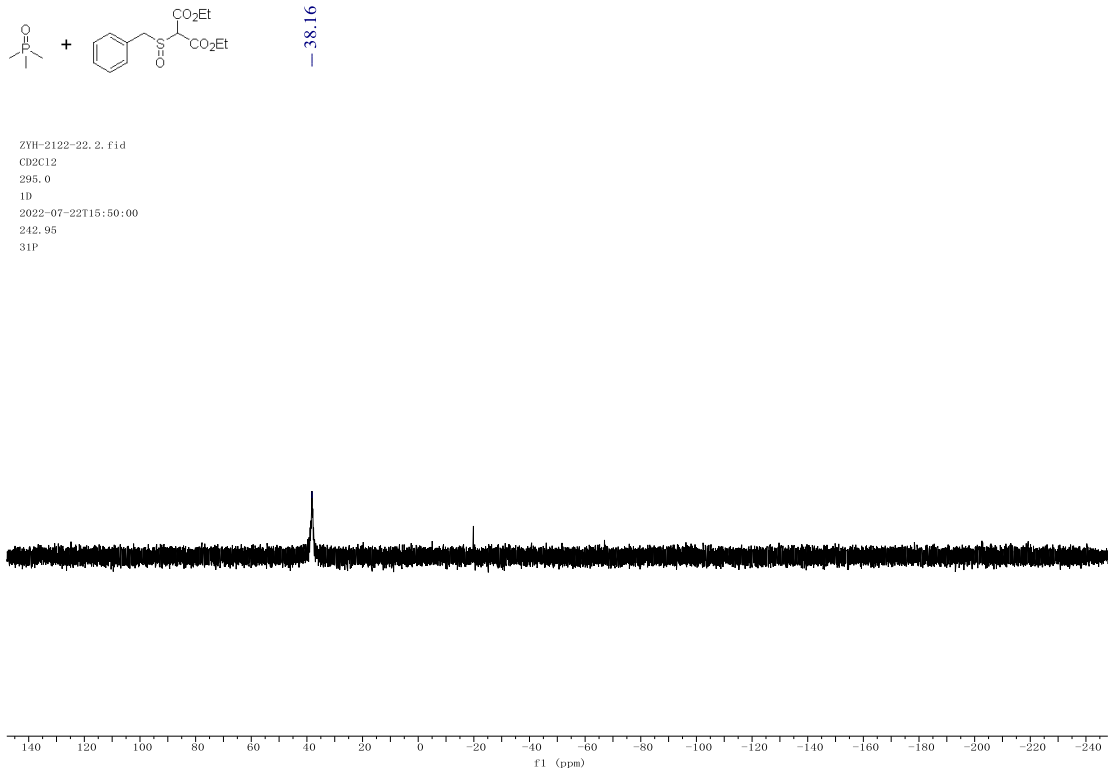


Figure S91. ^{31}P NMR spectrum of TMPO•2a (1:5).

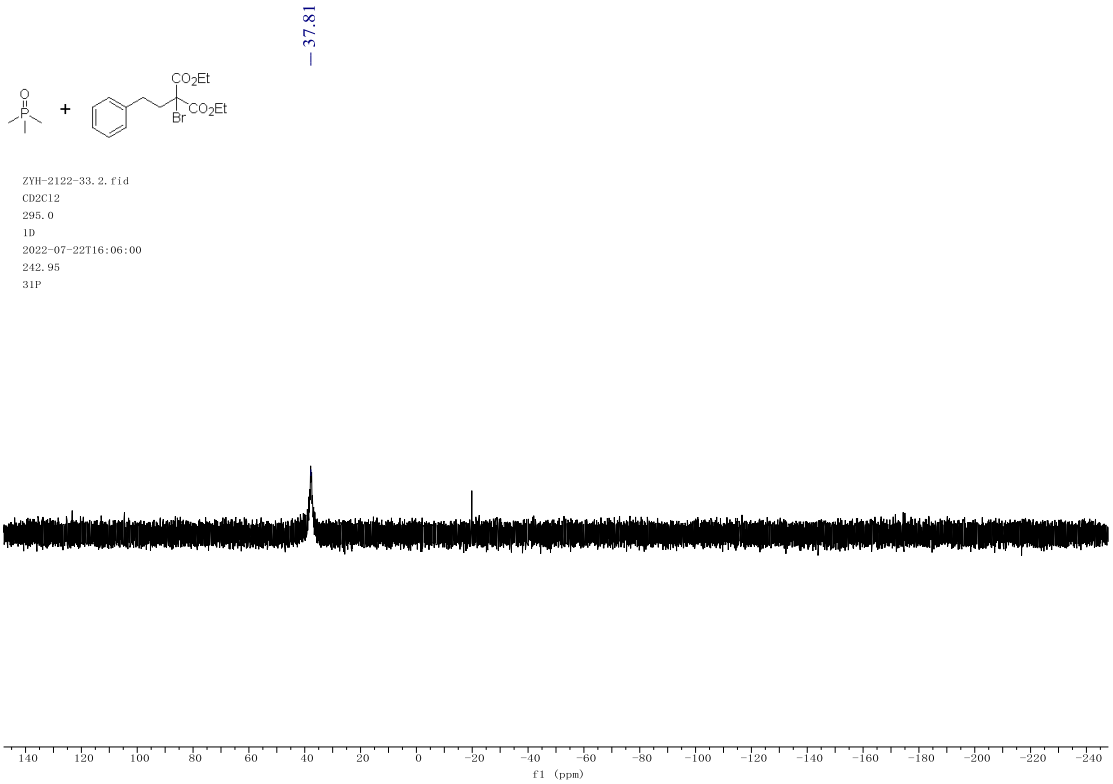


Figure S92. ^{31}P NMR spectrum of TMPO•3h (1:5).

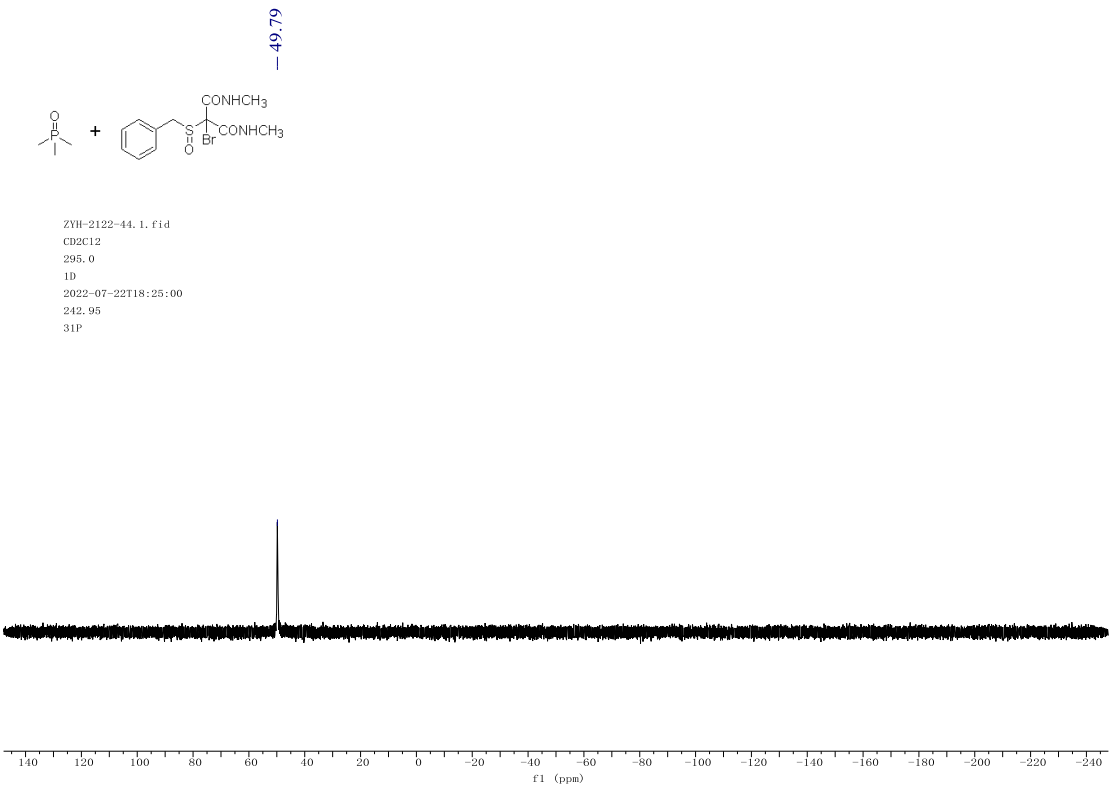


Figure S93. ^{31}P NMR spectrum of TMPO•3b (1:5).

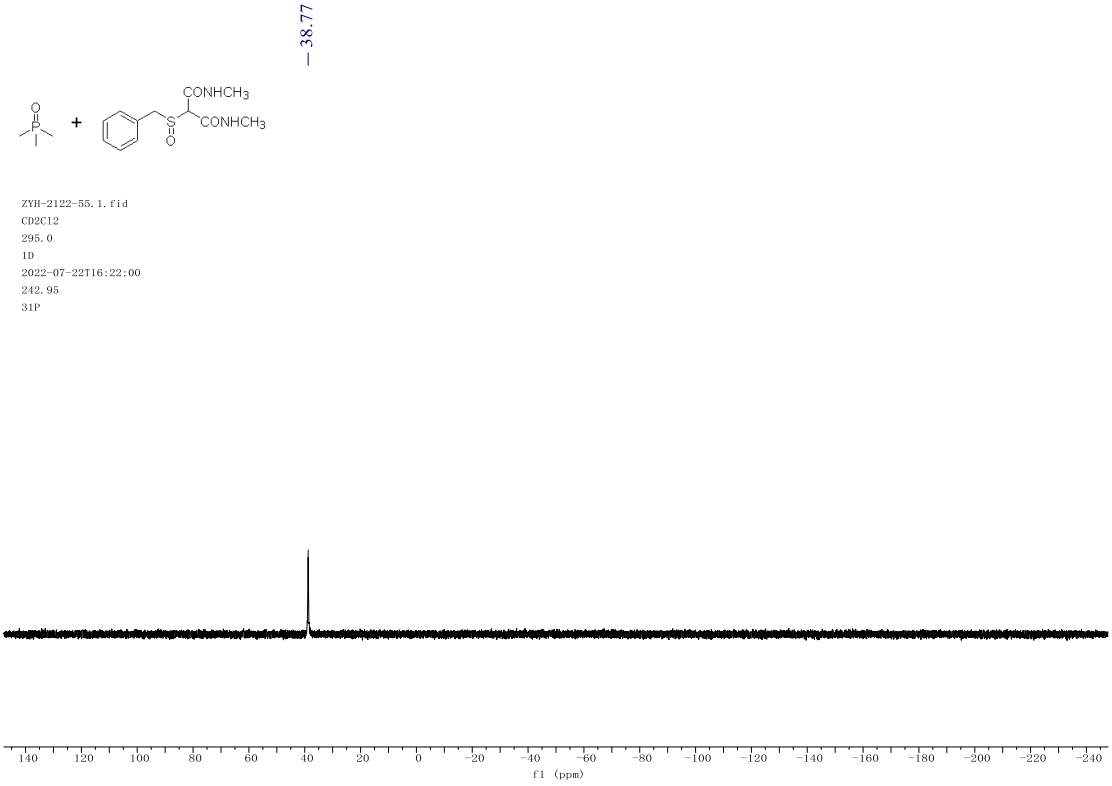


Figure S94. ^{31}P NMR spectrum of TMPO•2b (1:5).

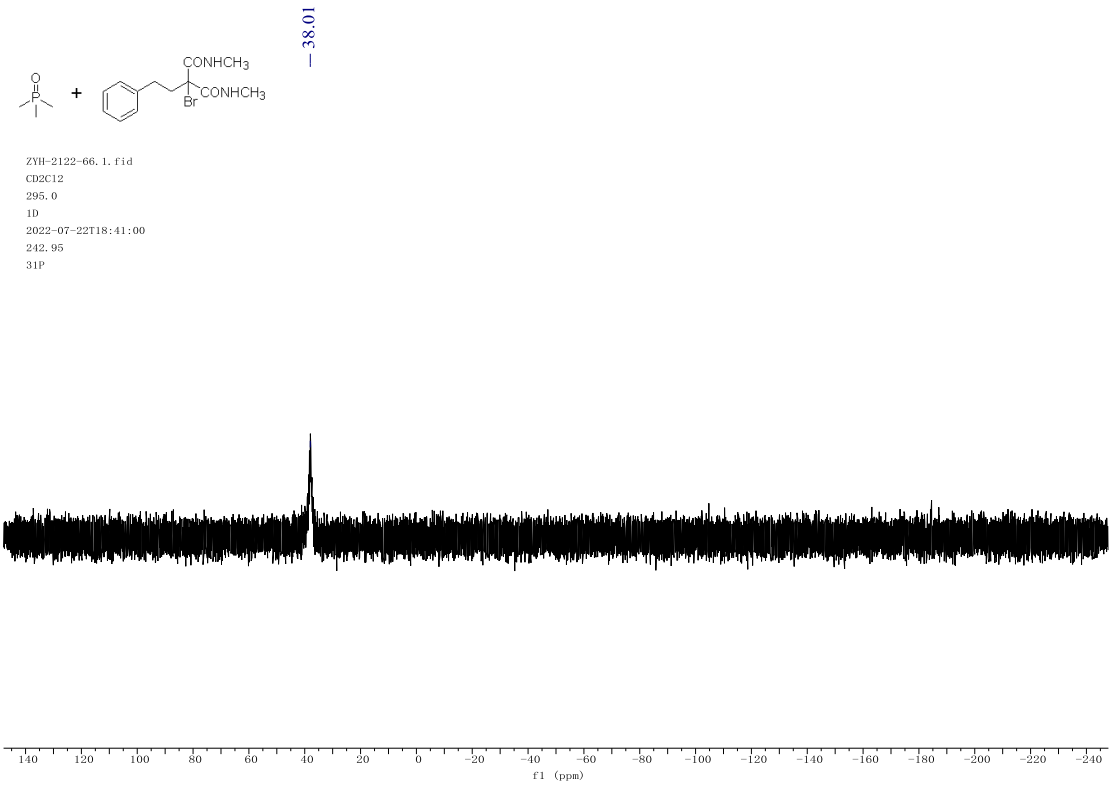


Figure S95. ^{31}P NMR spectrum of TMPO•3i (1:5).

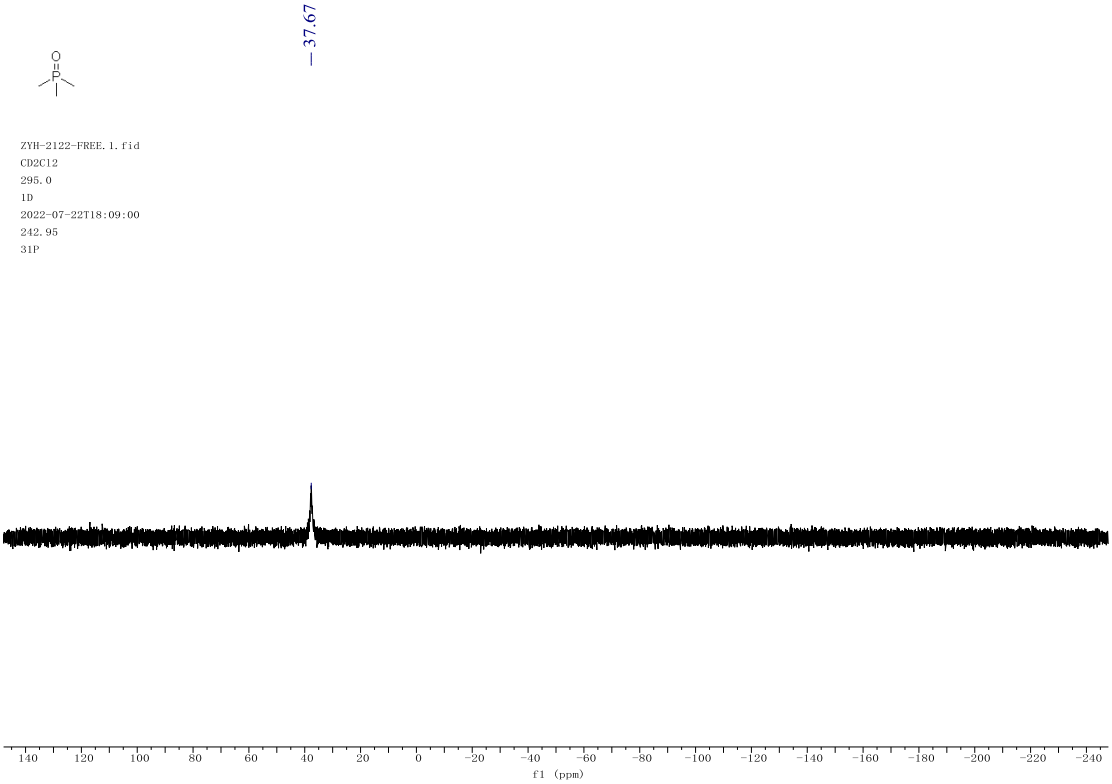


Figure S96. ^{31}P NMR spectrum of TMPO.

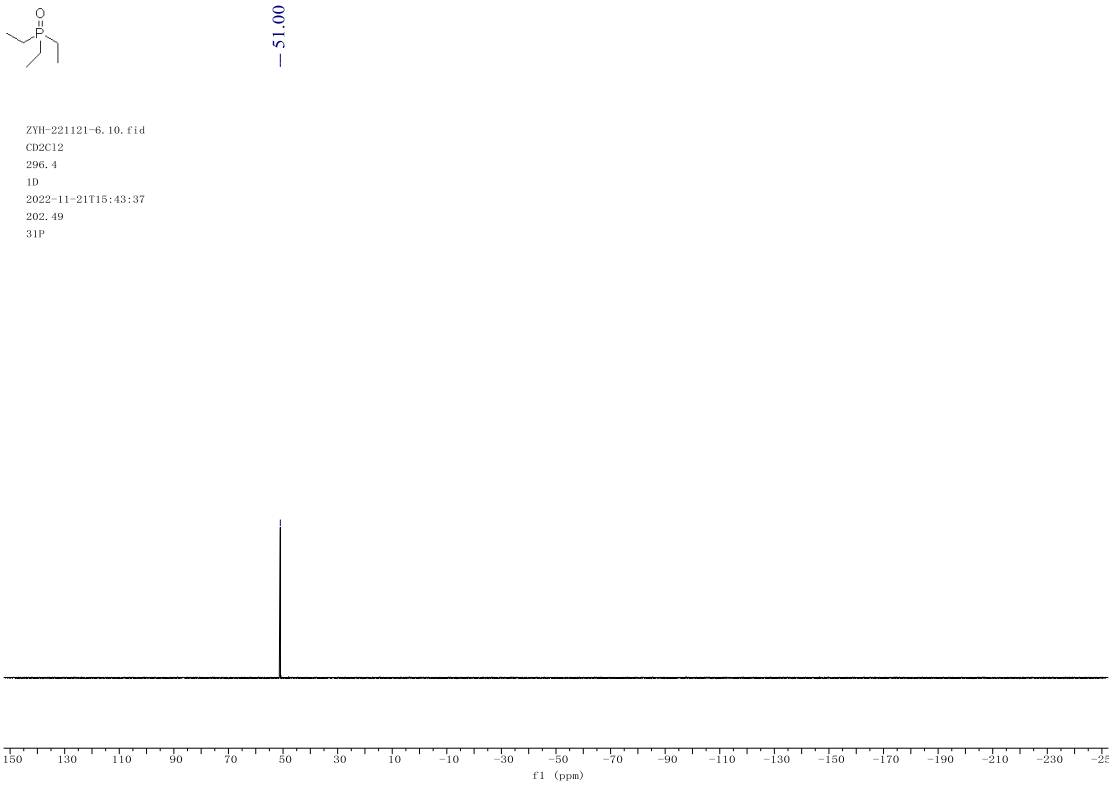


Figure S97. ^{31}P NMR spectrum of TEPO.

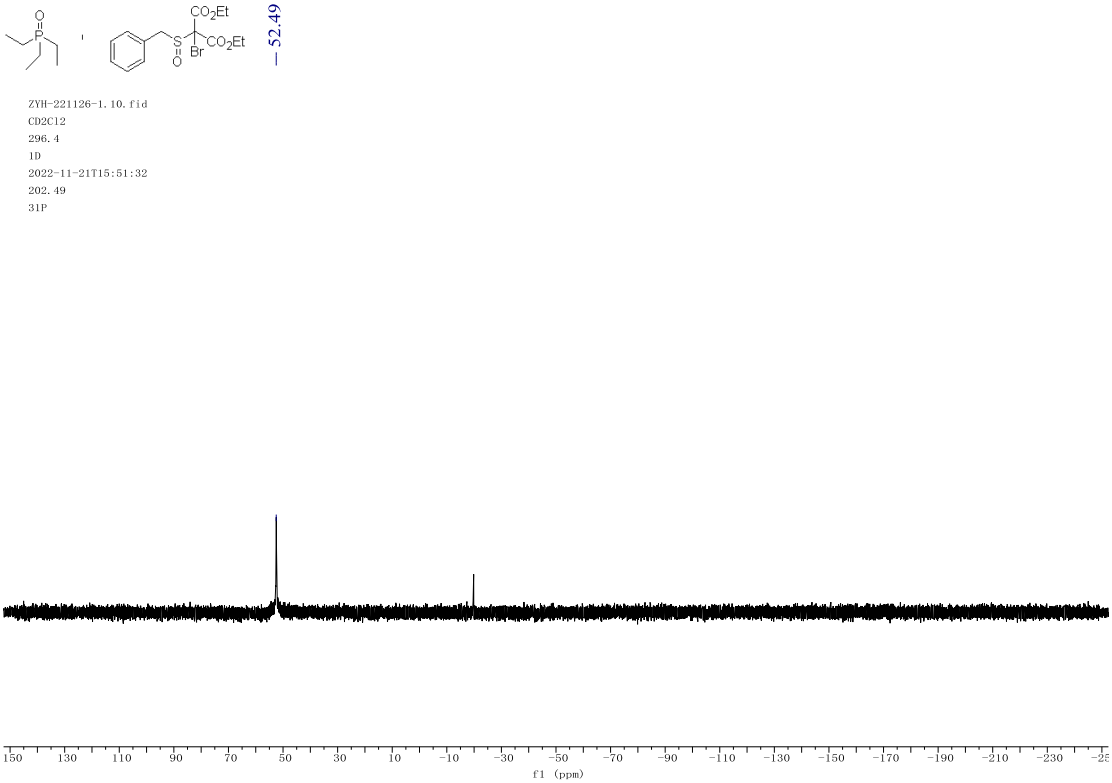


Figure S98. ^{31}P NMR spectrum of TEPO•3a (1:10).

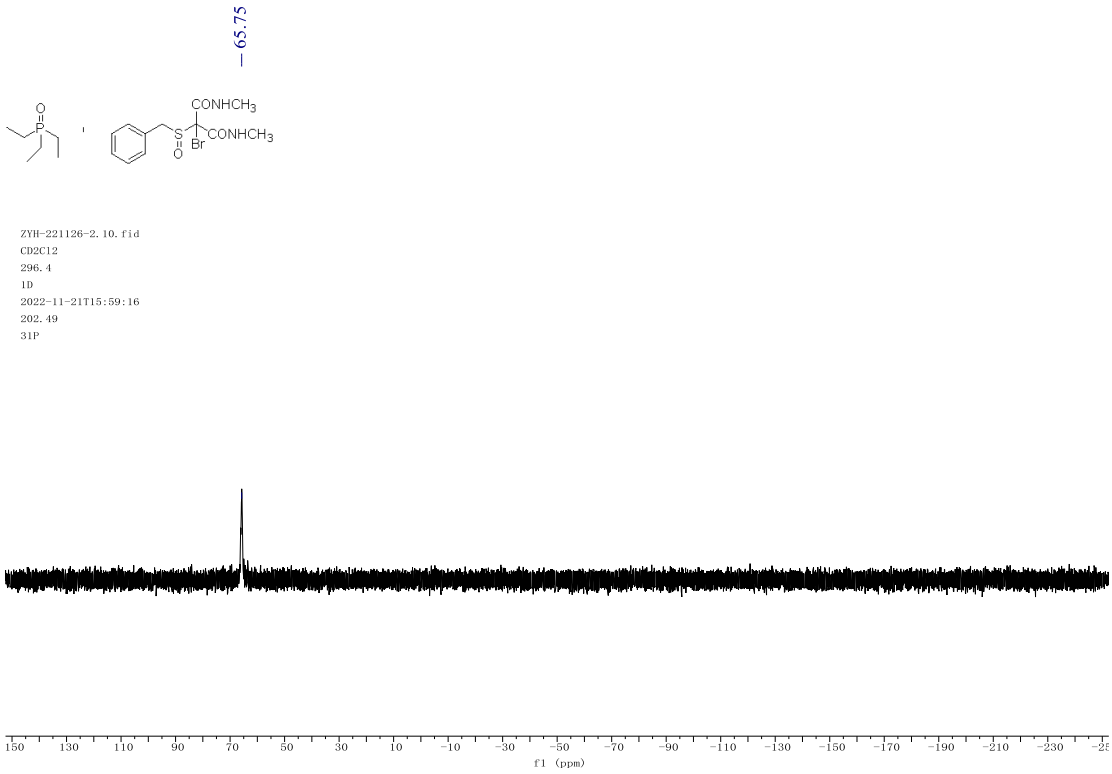


Figure S99. ^{31}P NMR spectrum of TEPO•3b (1:10).

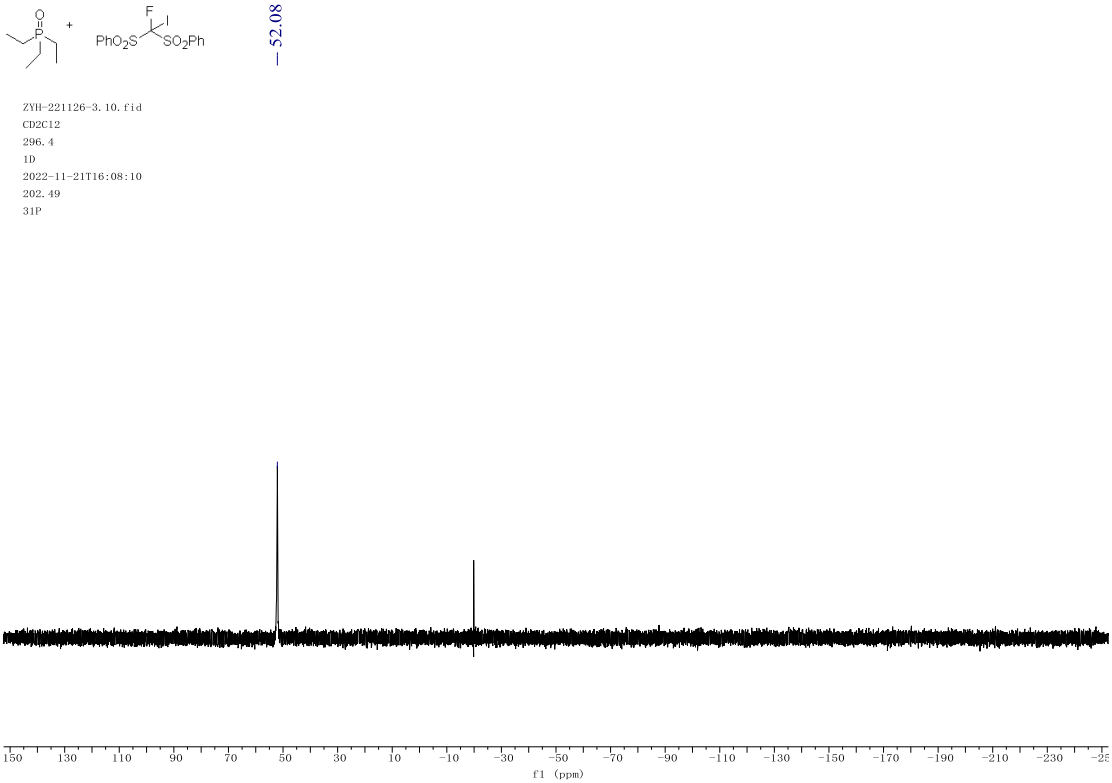


Figure S100. ^{31}P NMR spectrum of TEPO•3f (1:10).

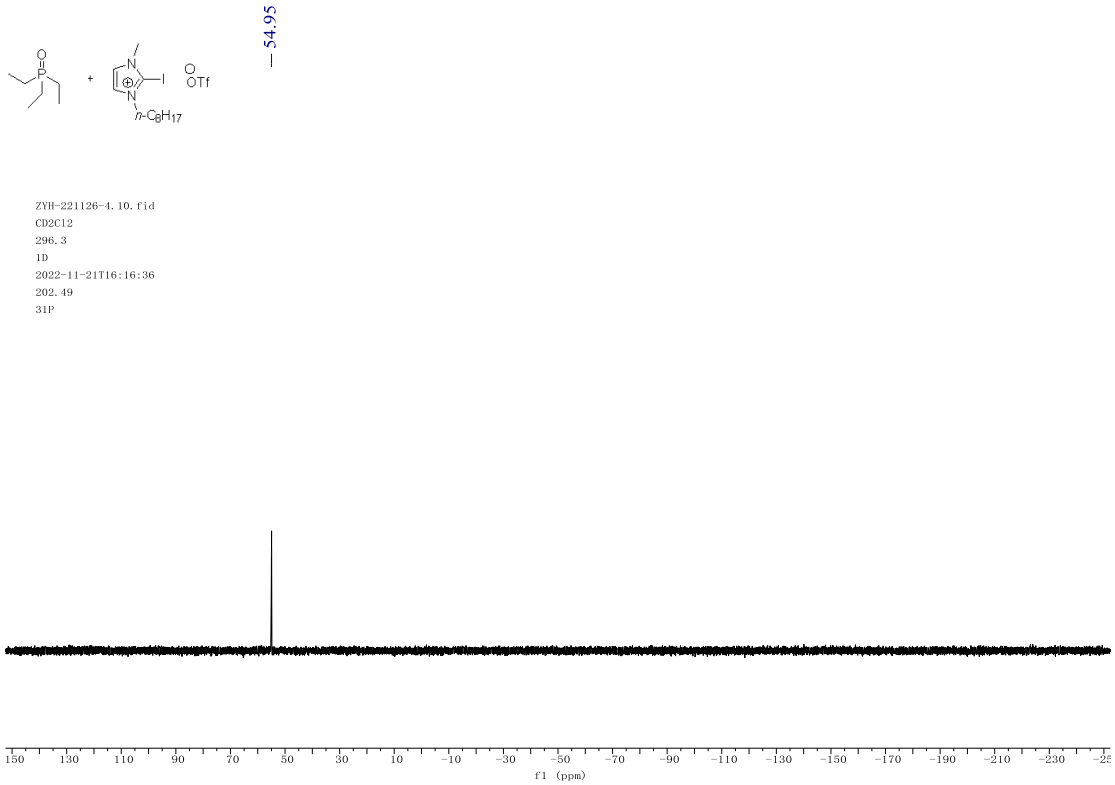


Figure S101. ^{31}P NMR spectrum of TEPO•3g (1:10).

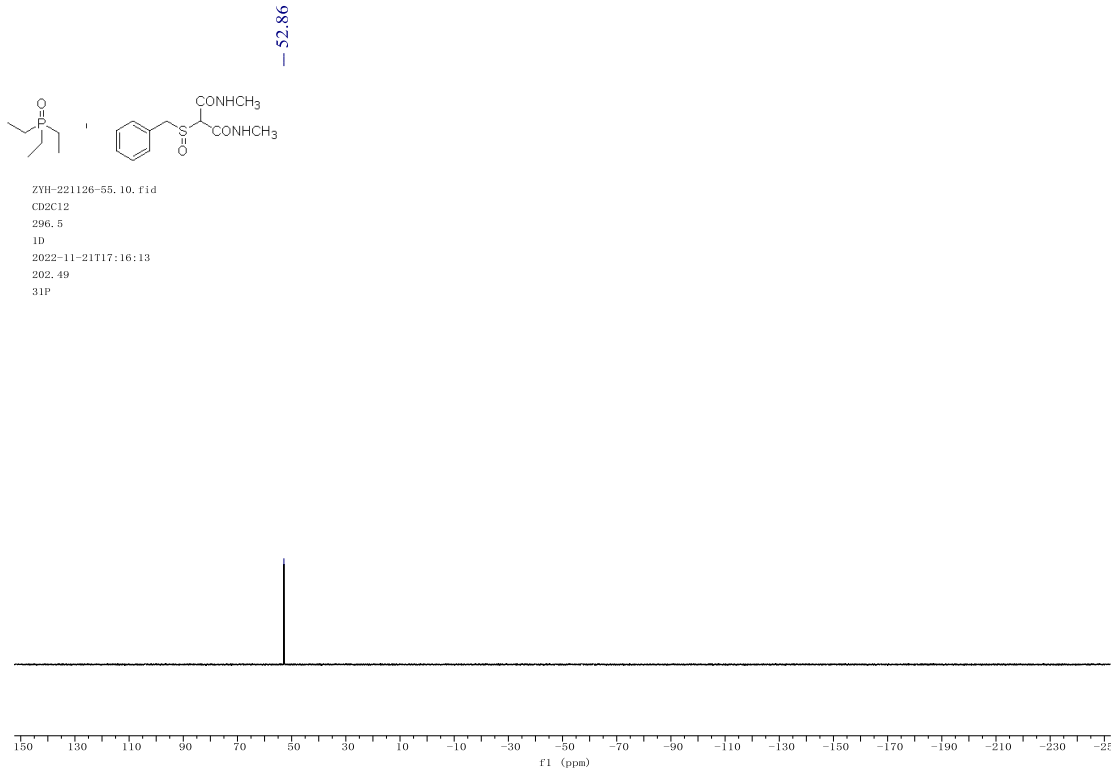


Figure S102. ^{31}P NMR spectrum of TEPO•2b (1:10).